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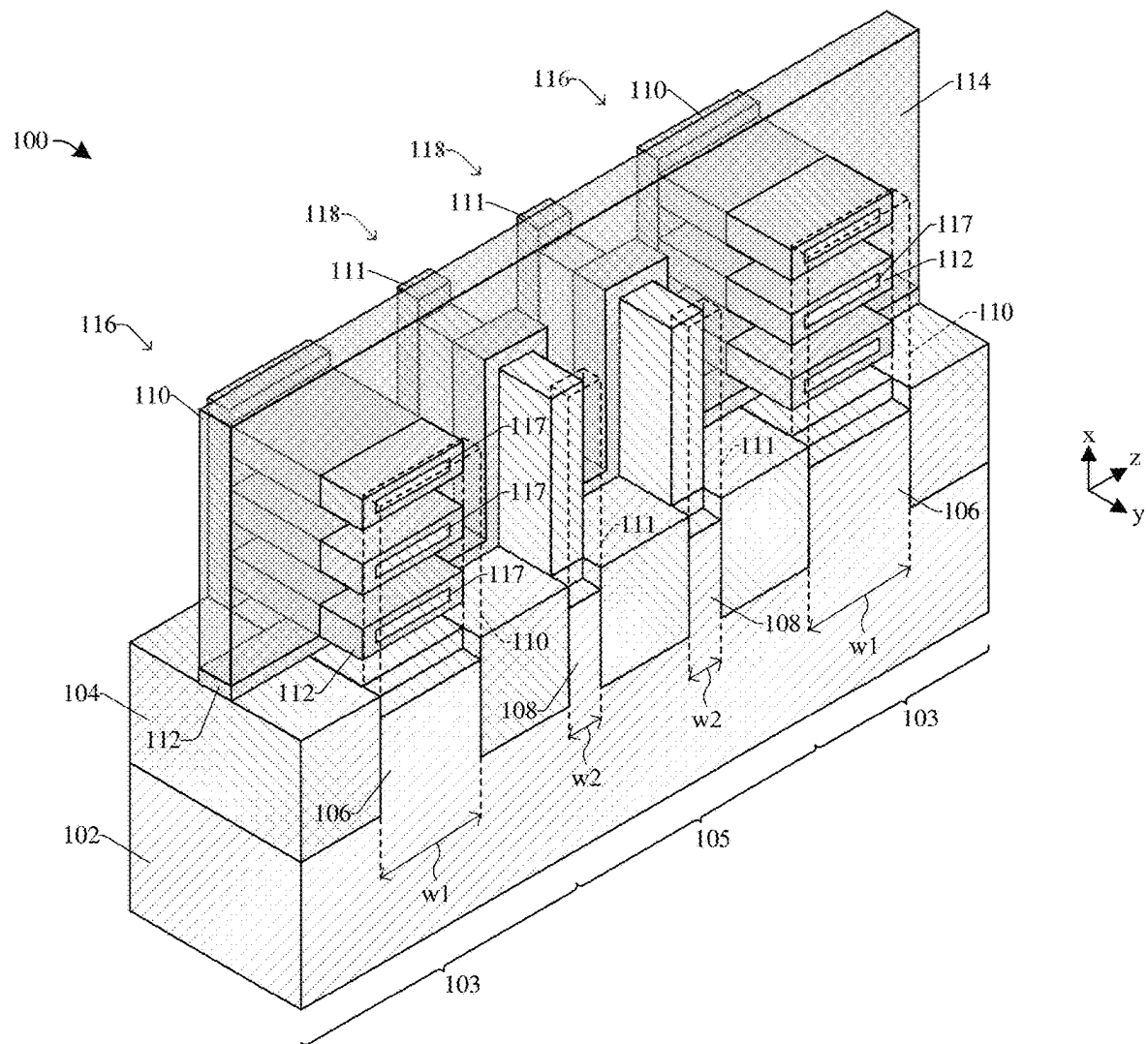
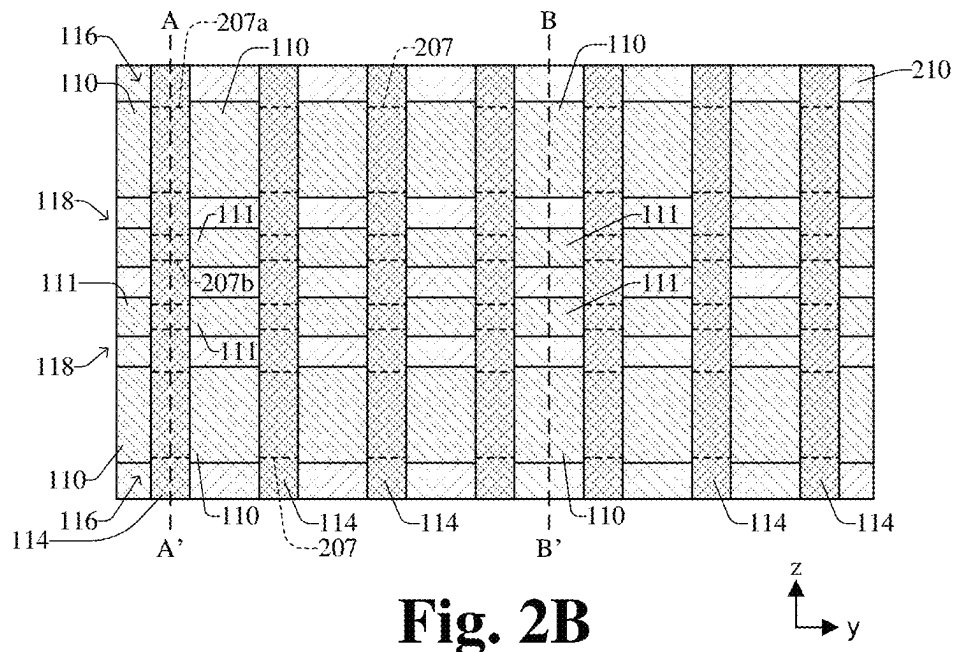
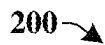
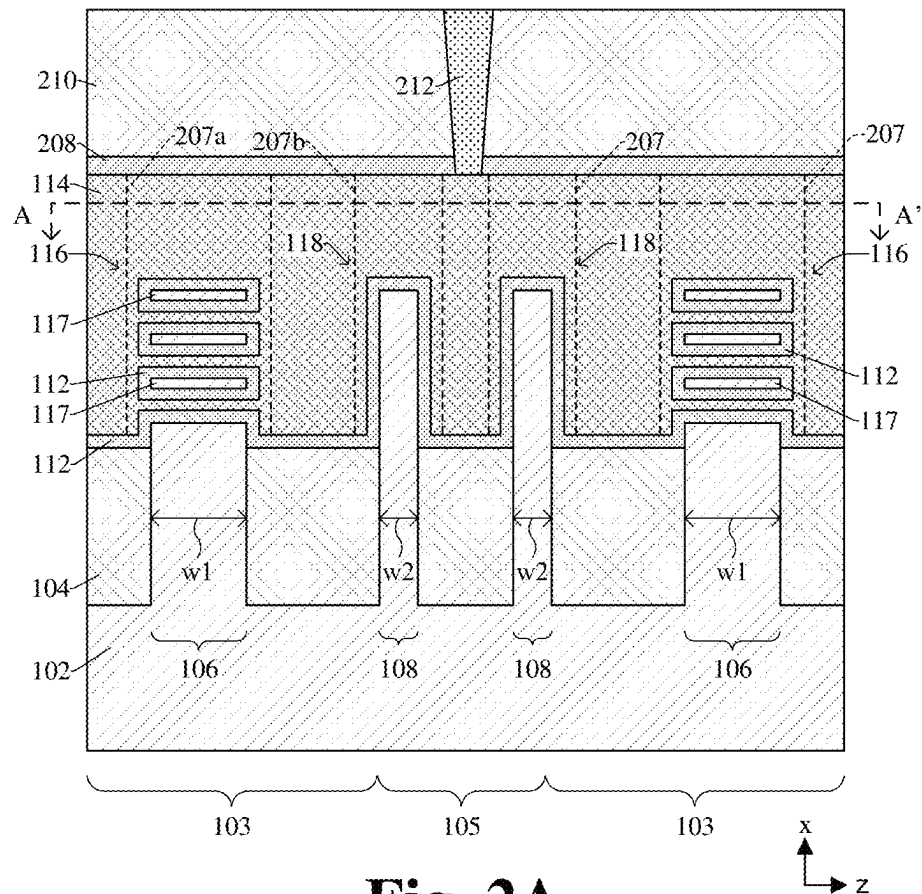
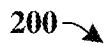
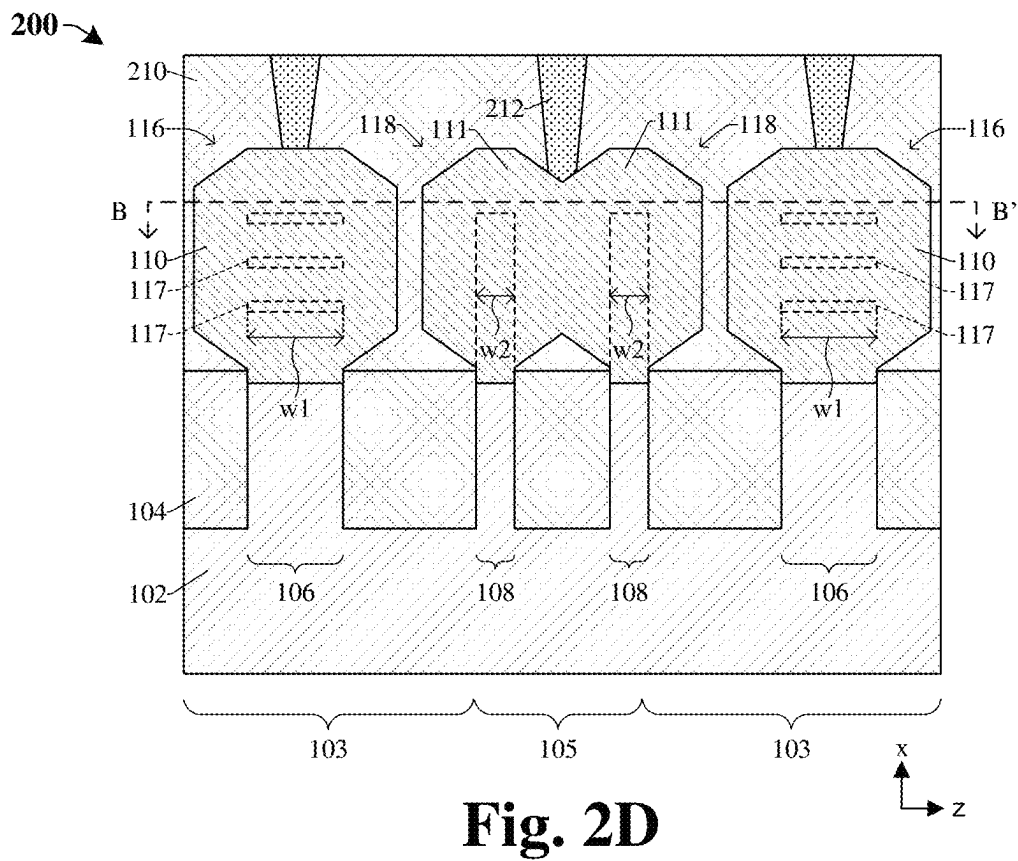
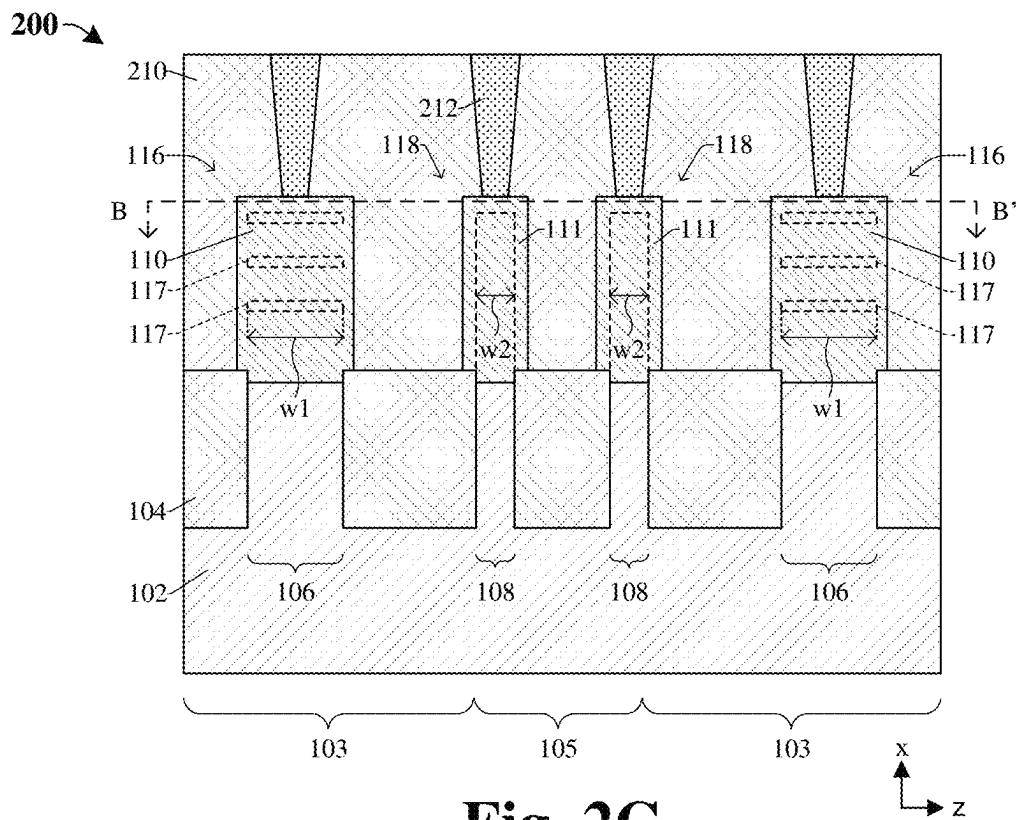


Fig. 1





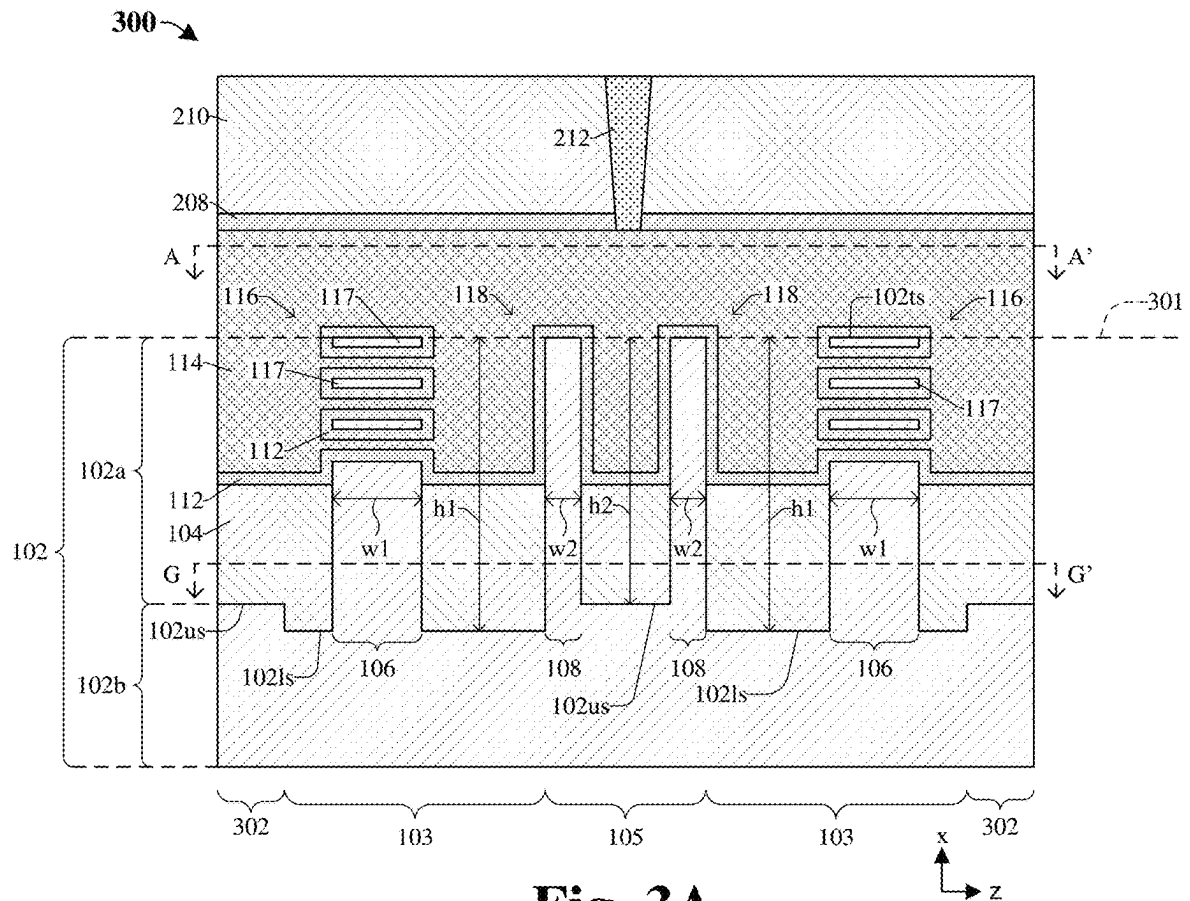


Fig. 3A

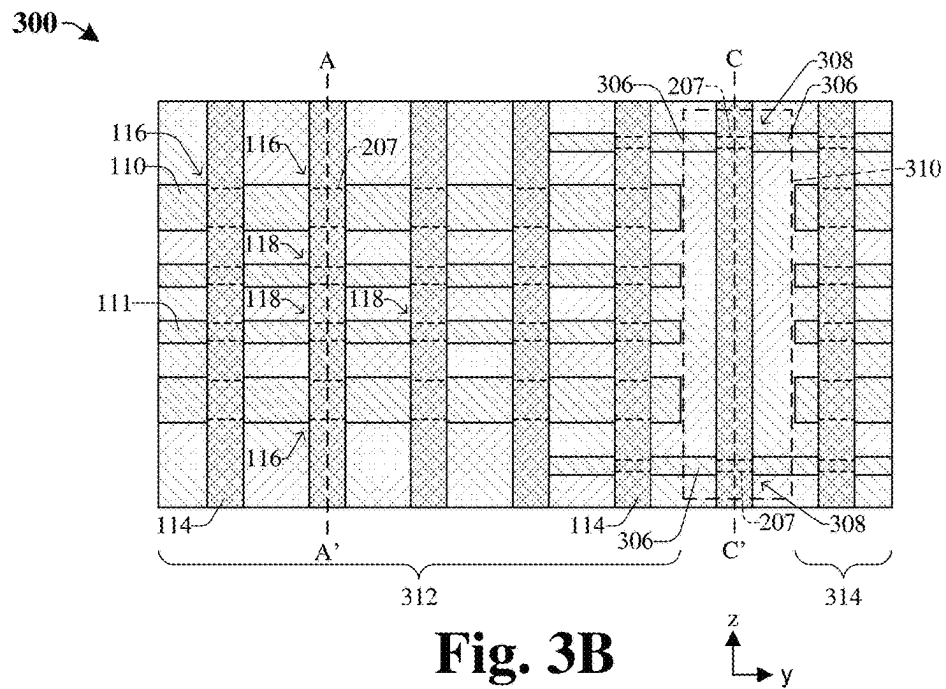
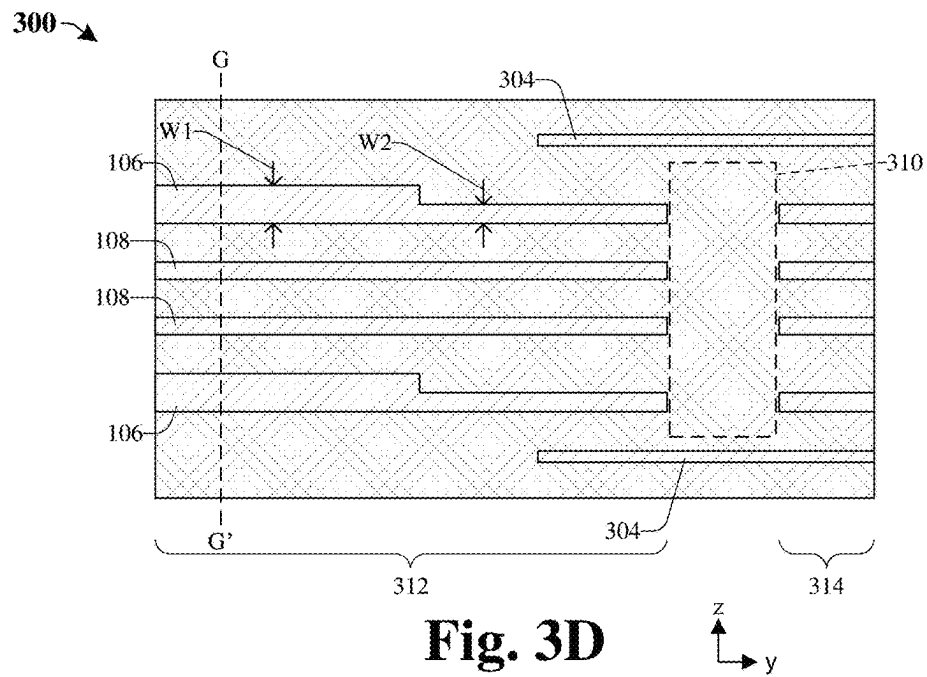
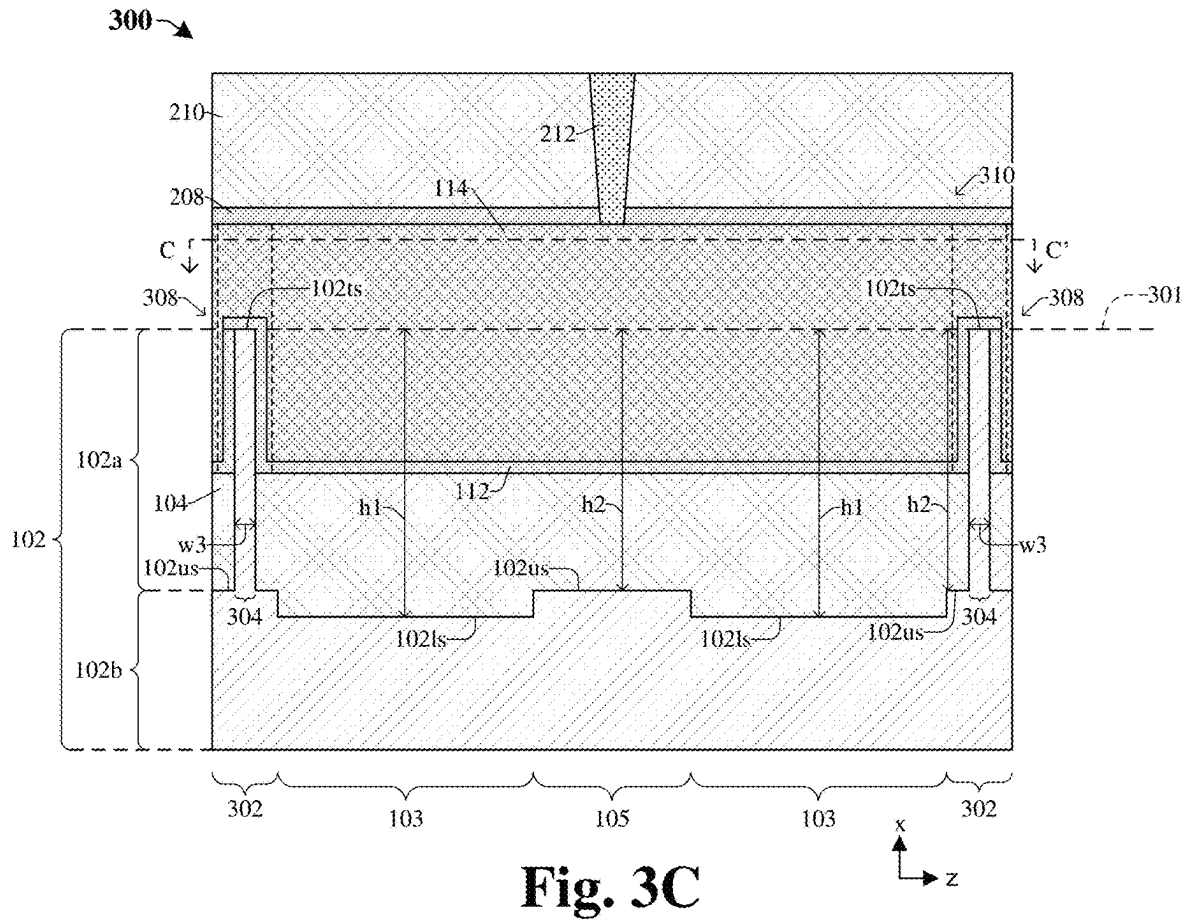
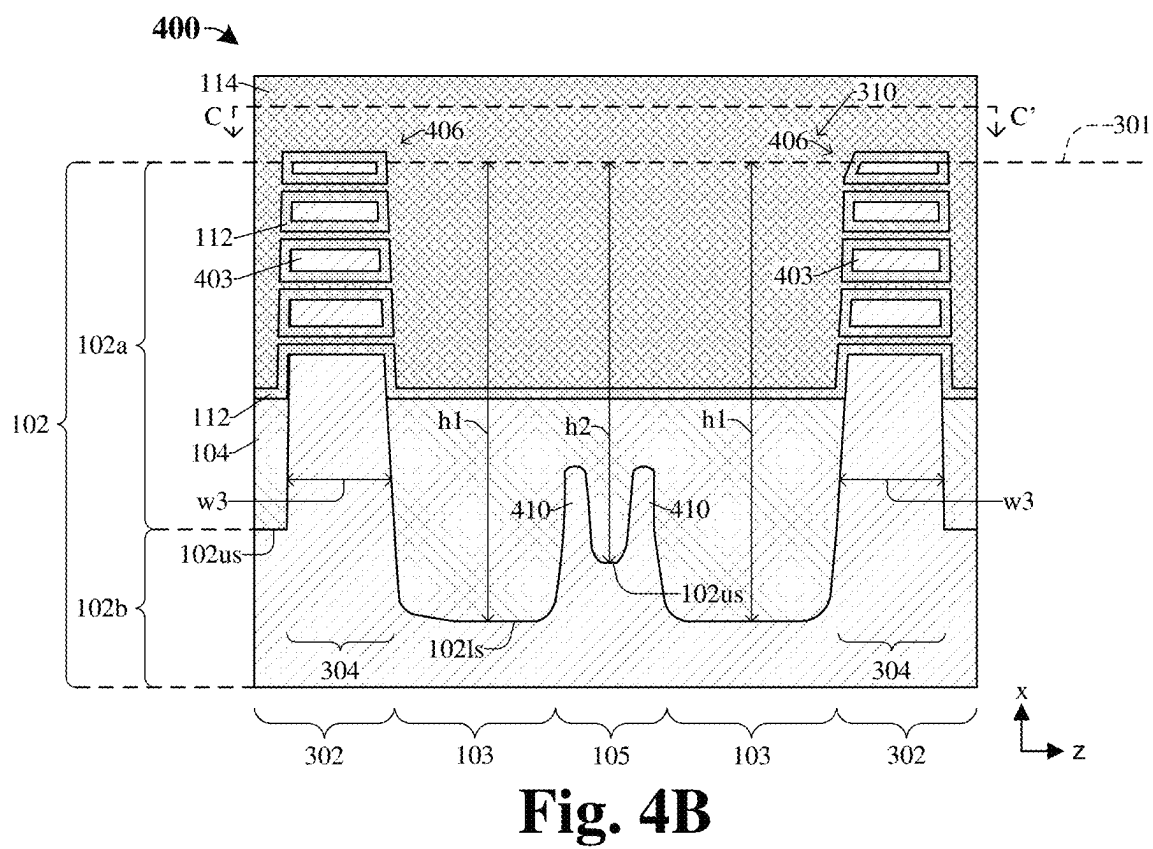
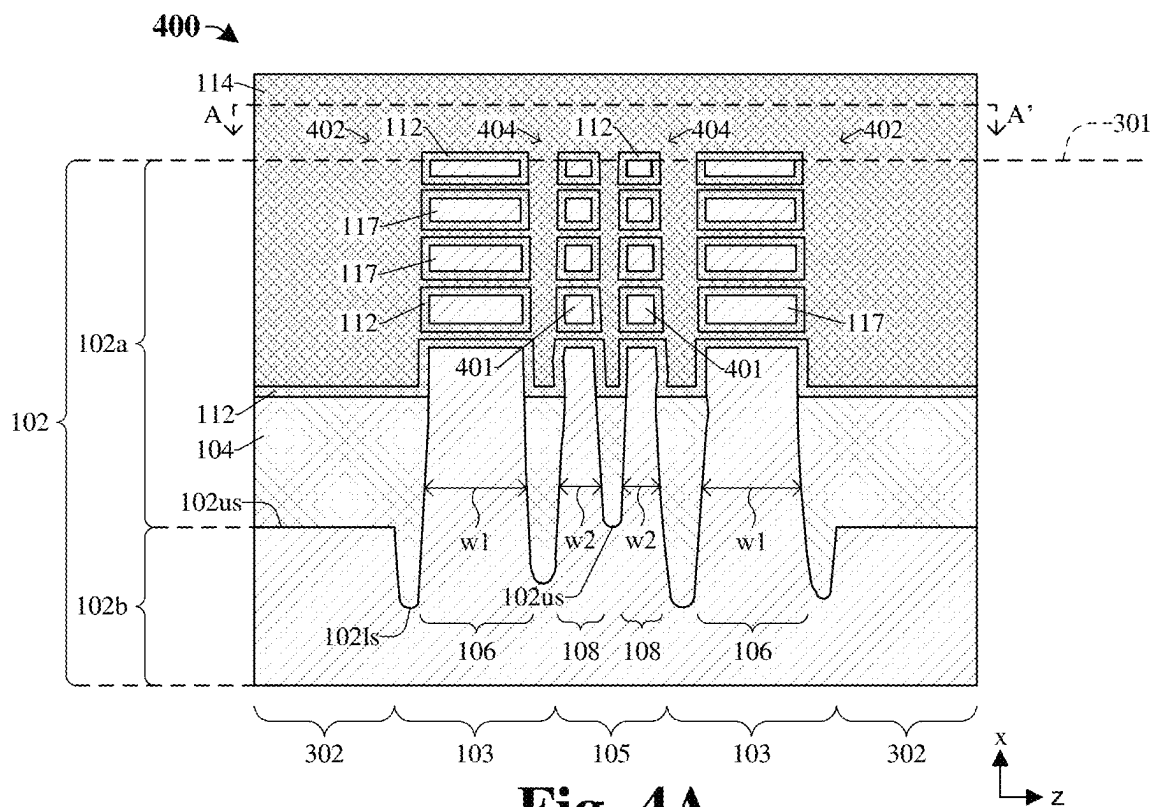


Fig. 3B





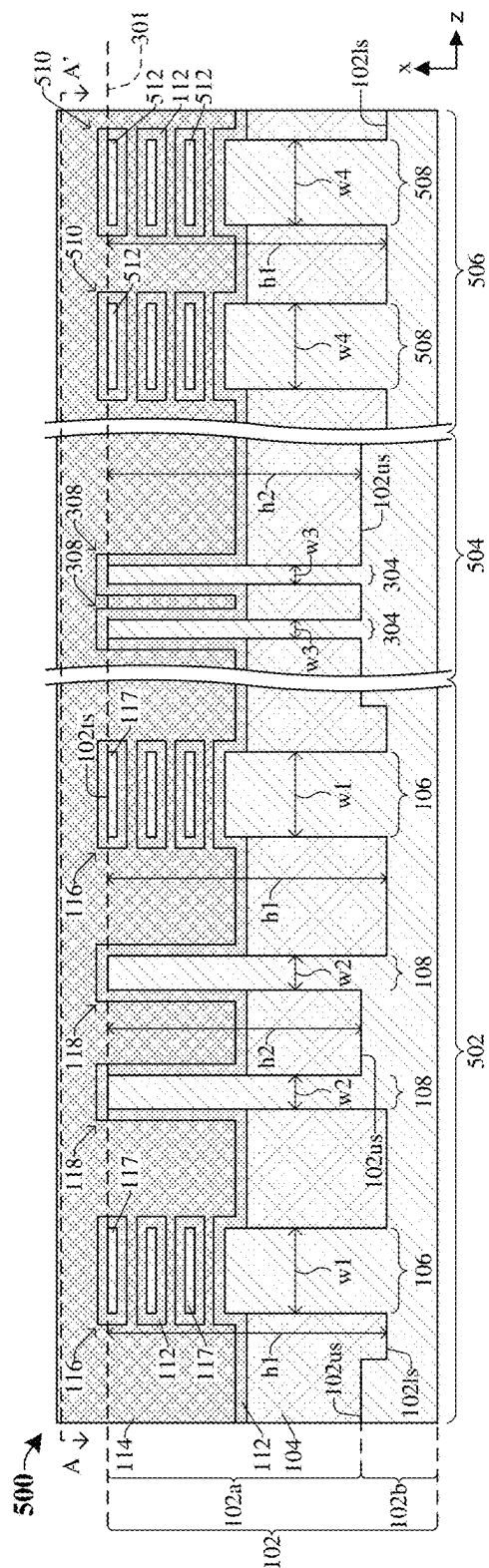


Fig. 5A

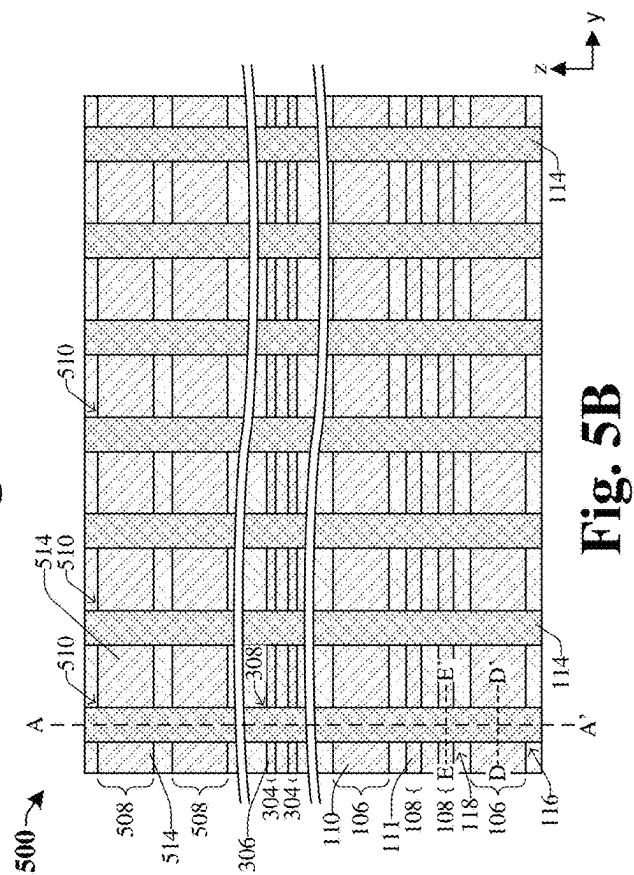


Fig. 5B

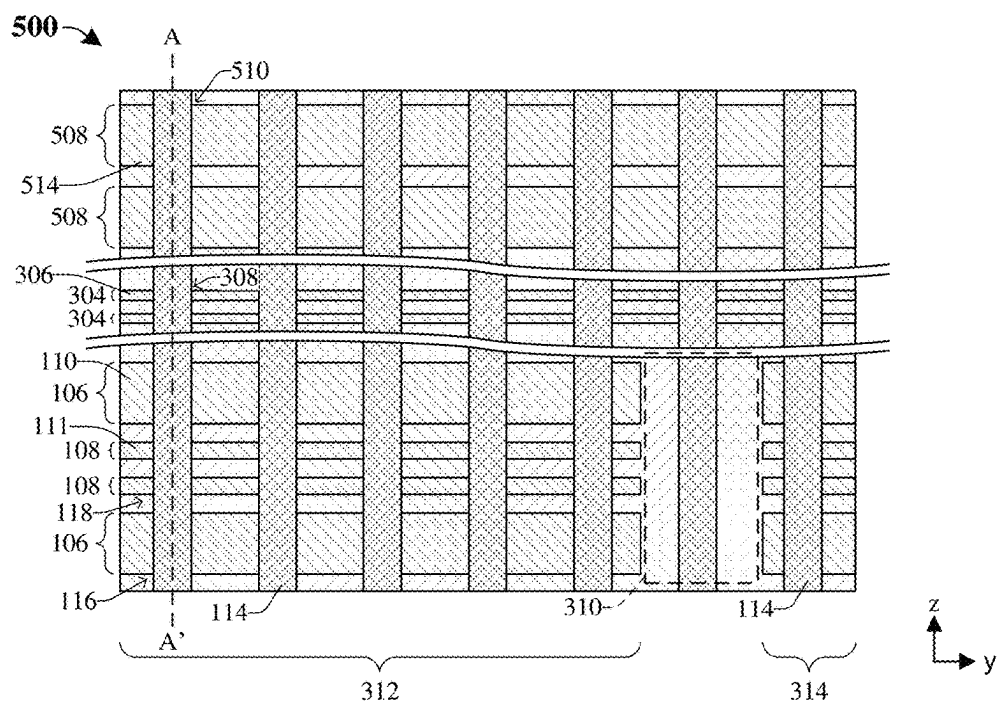


Fig. 5C

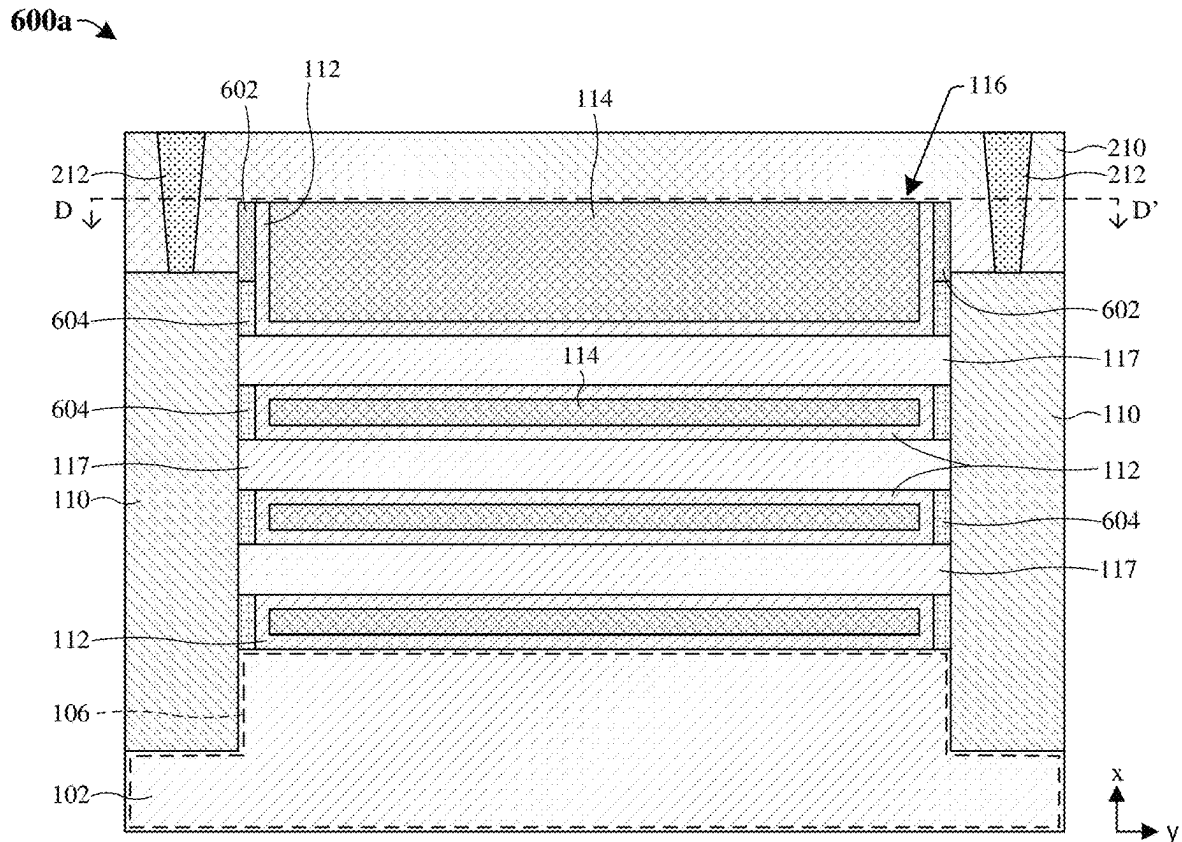


Fig. 6A

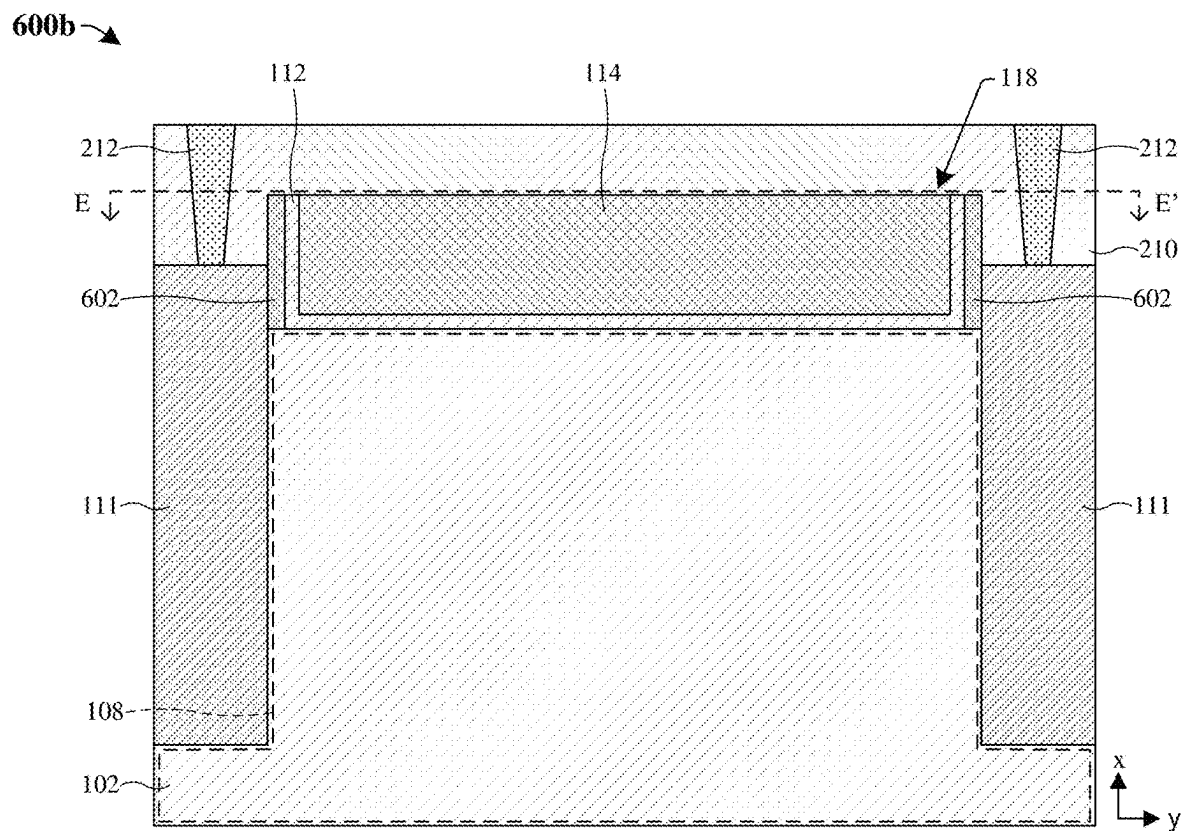


Fig. 6B

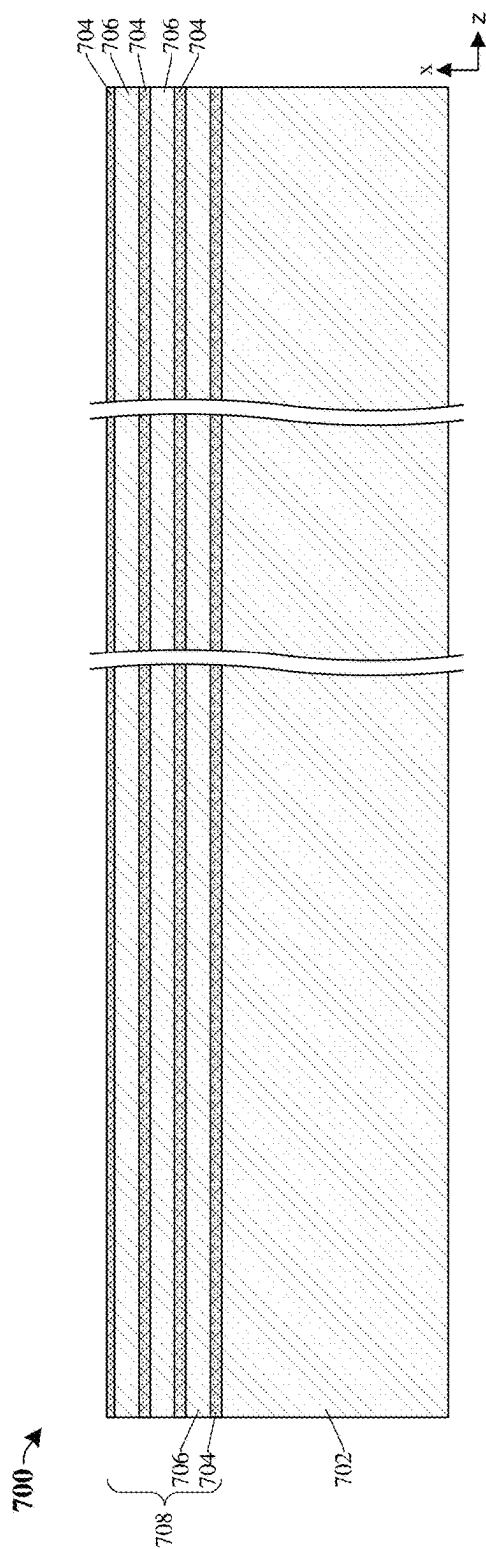


Fig. 7

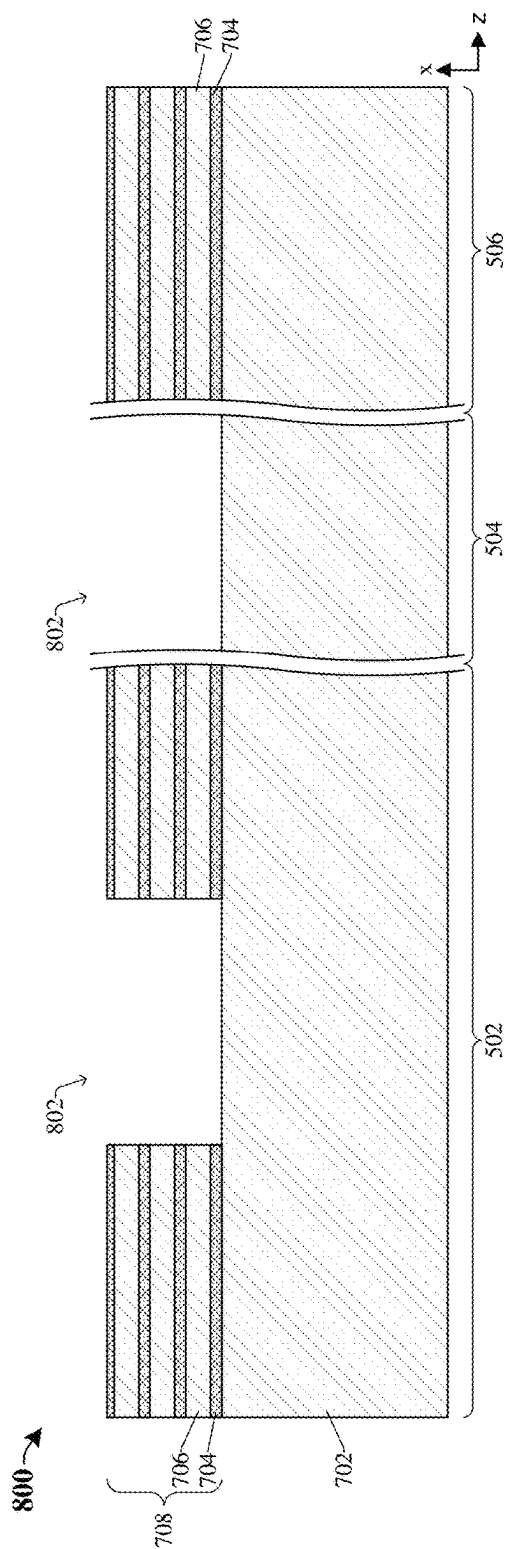


Fig. 8

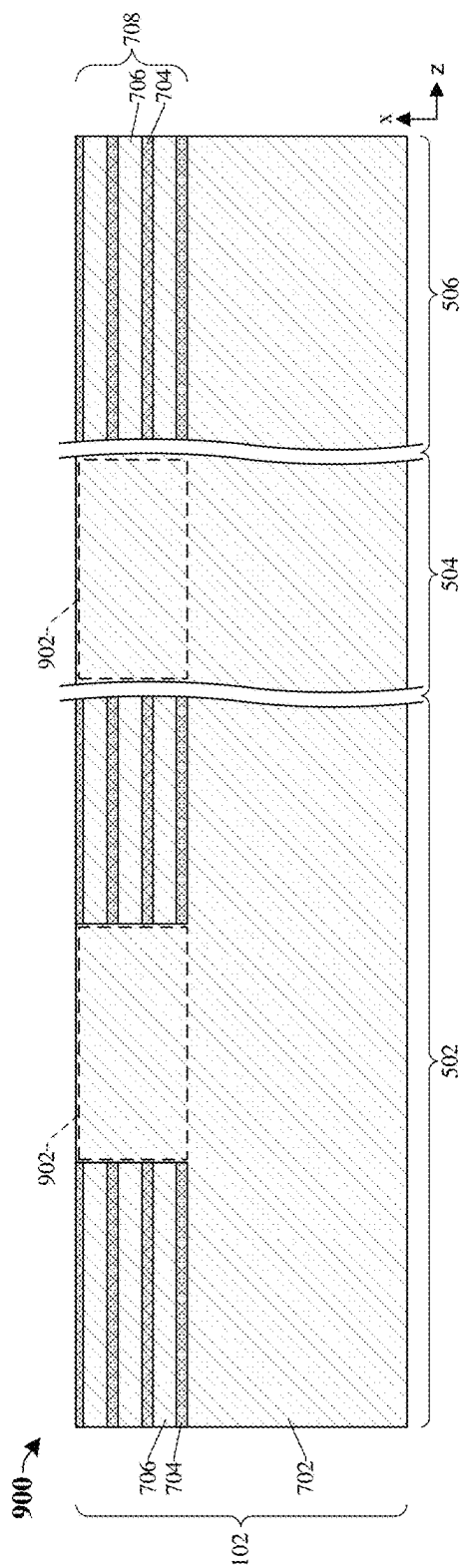


Fig. 9

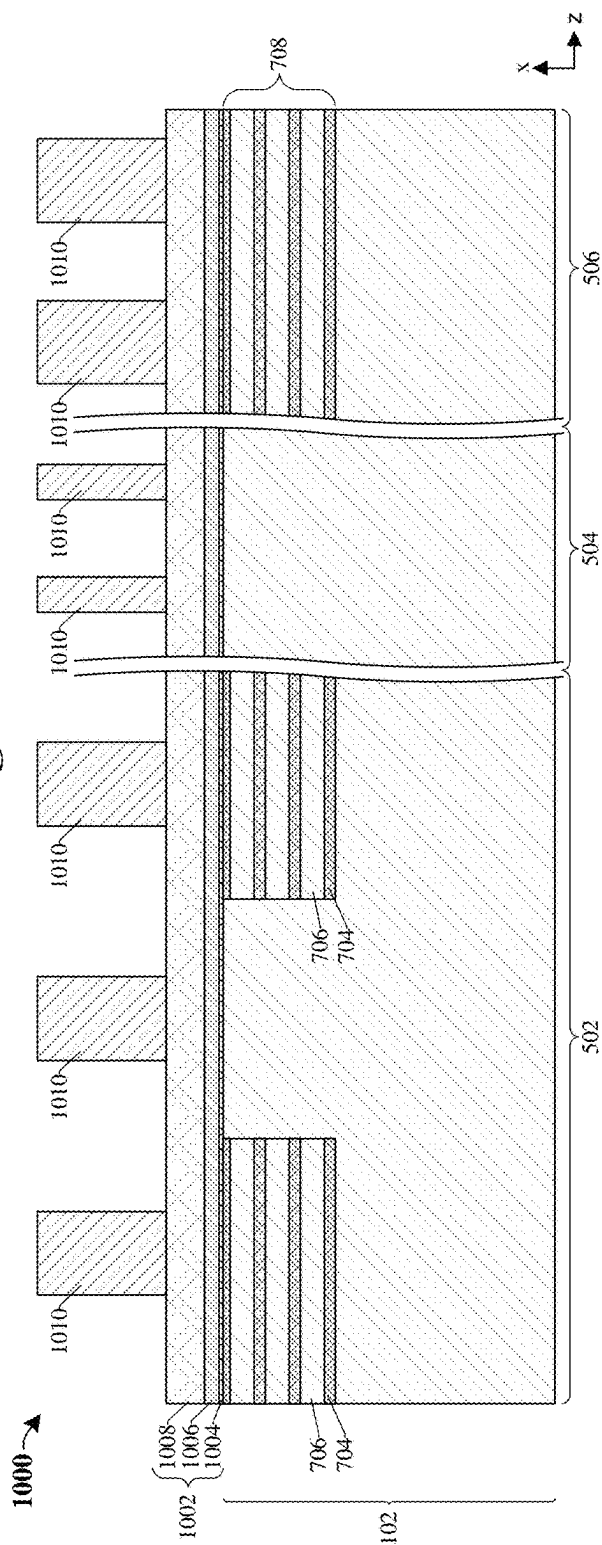


Fig. 10

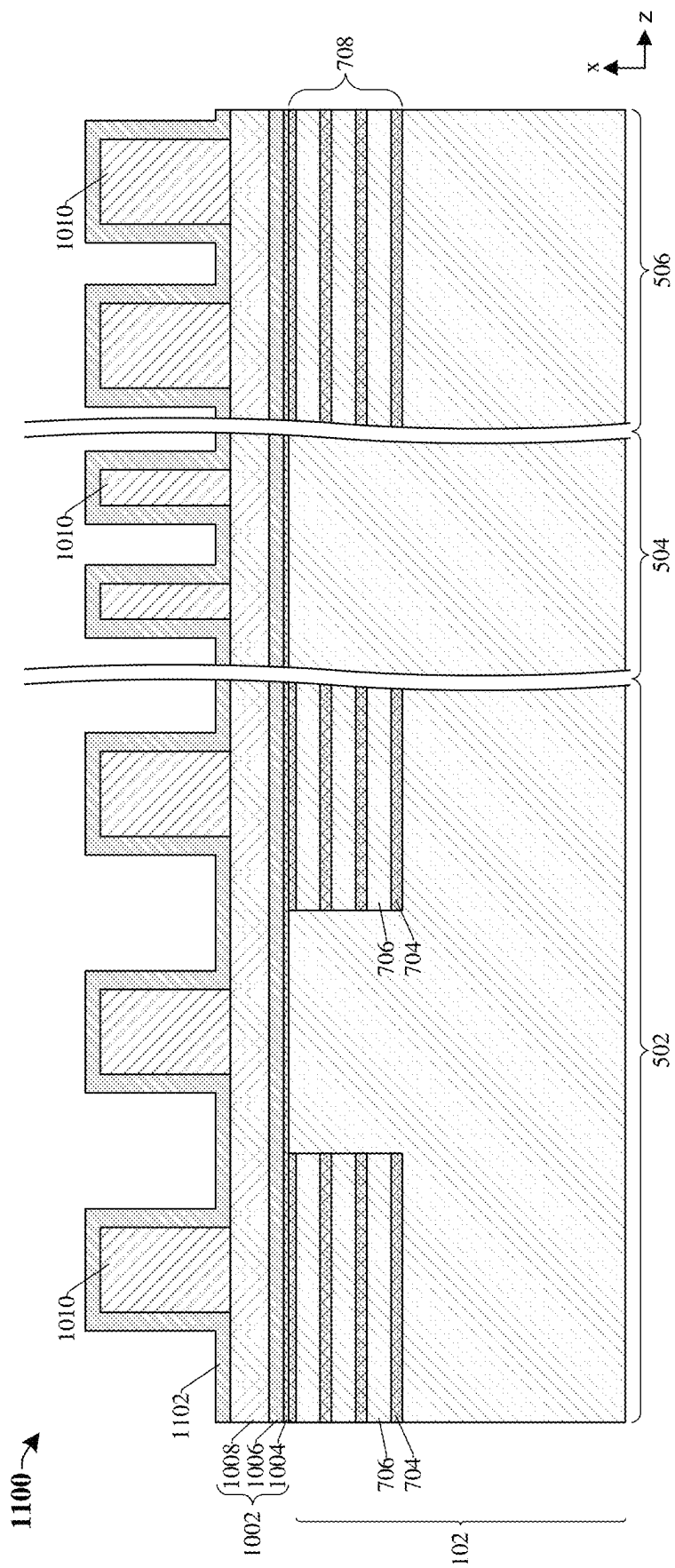


Fig. 11

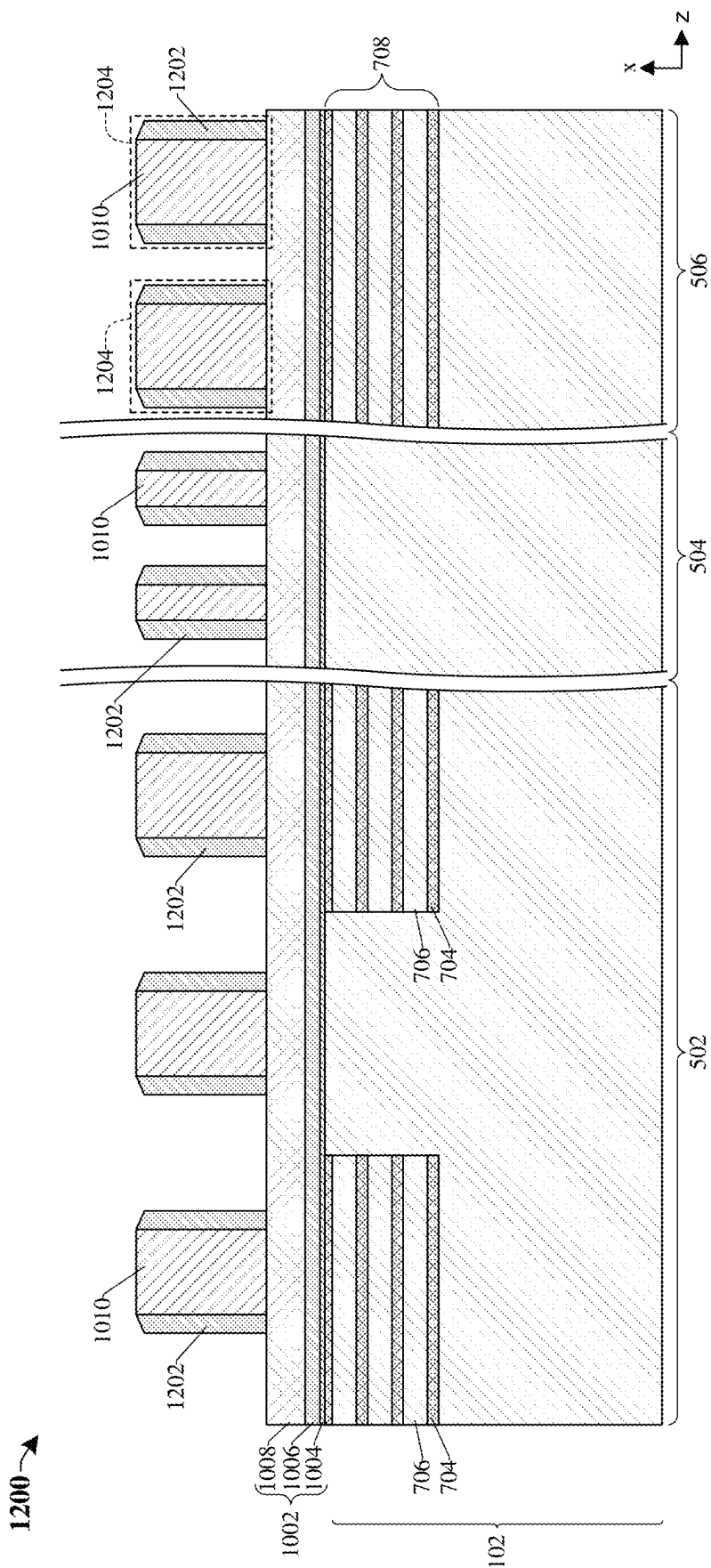


Fig. 12

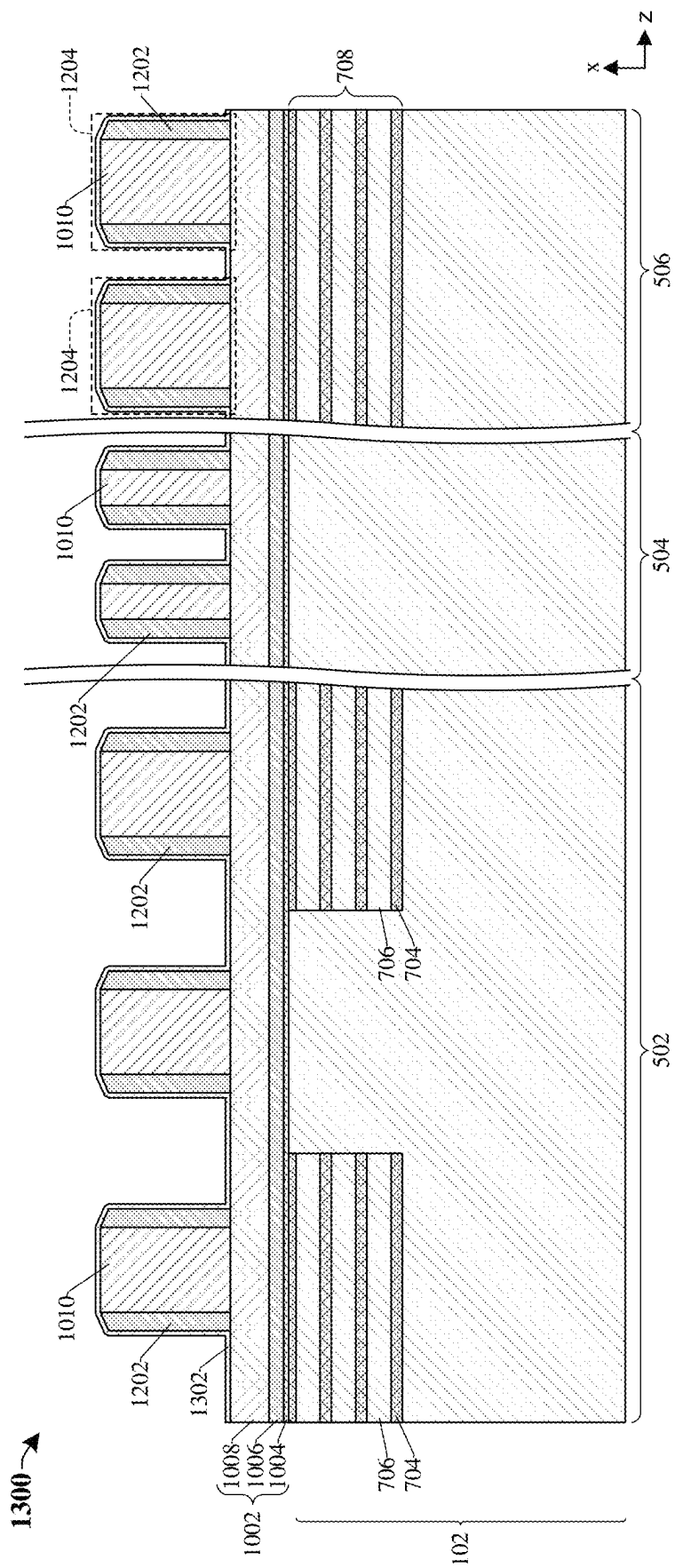


Fig. 13

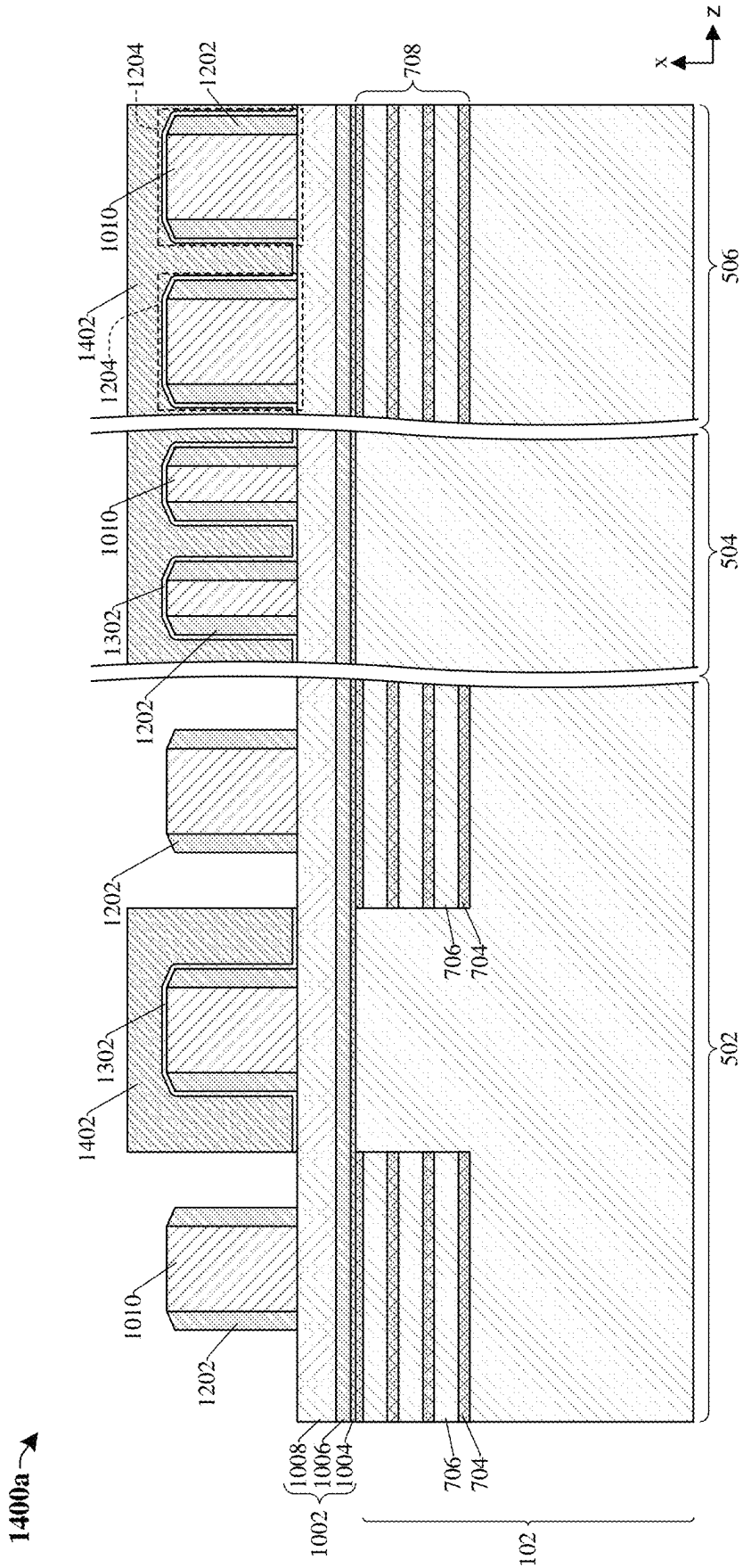


Fig. 14A

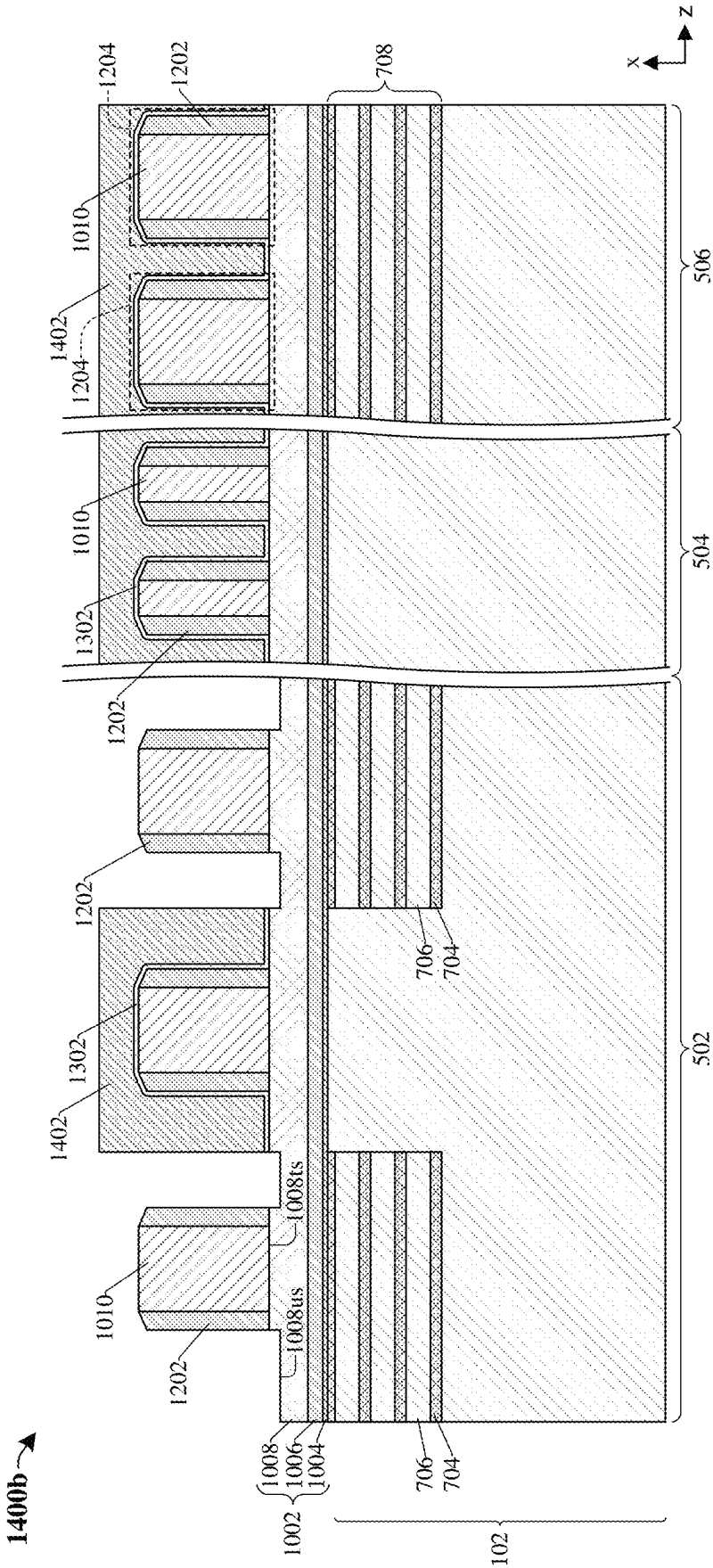


Fig. 14B

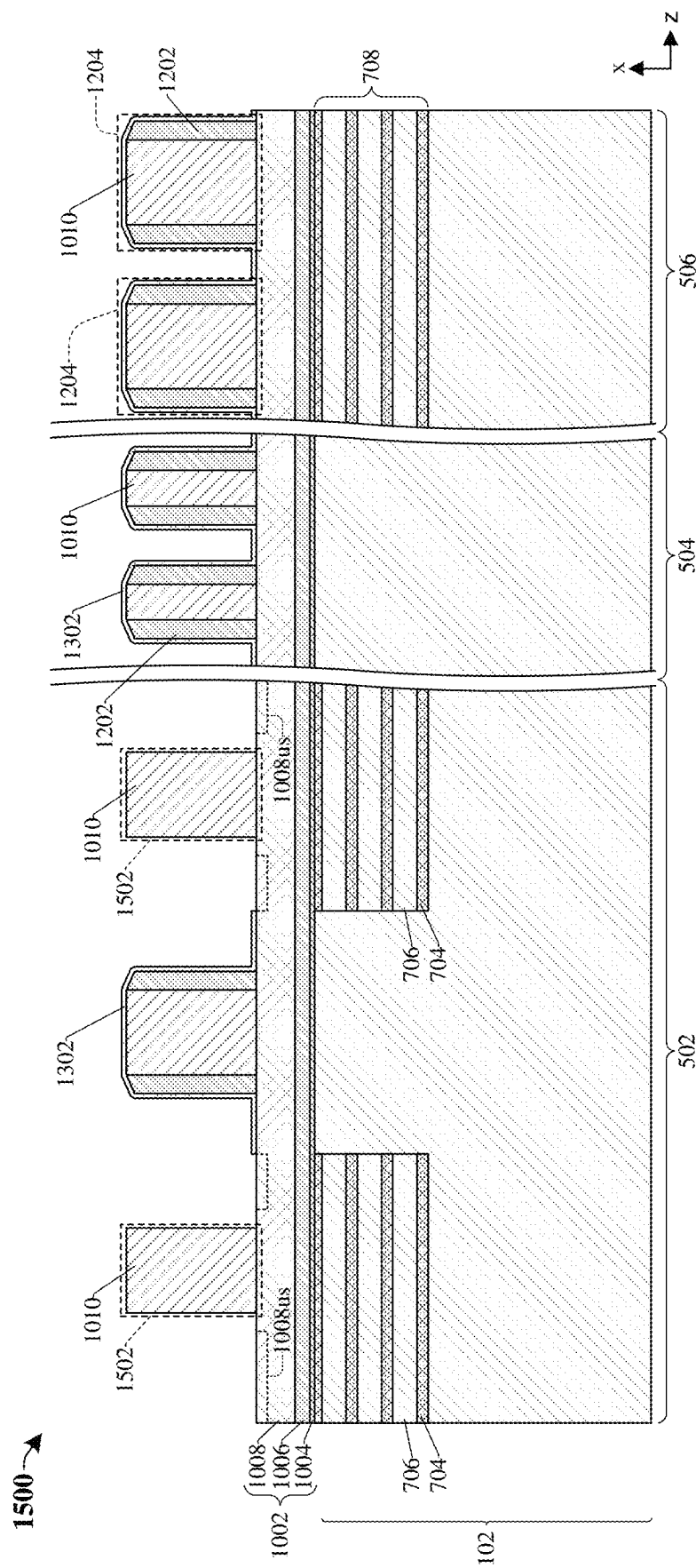


Fig. 15

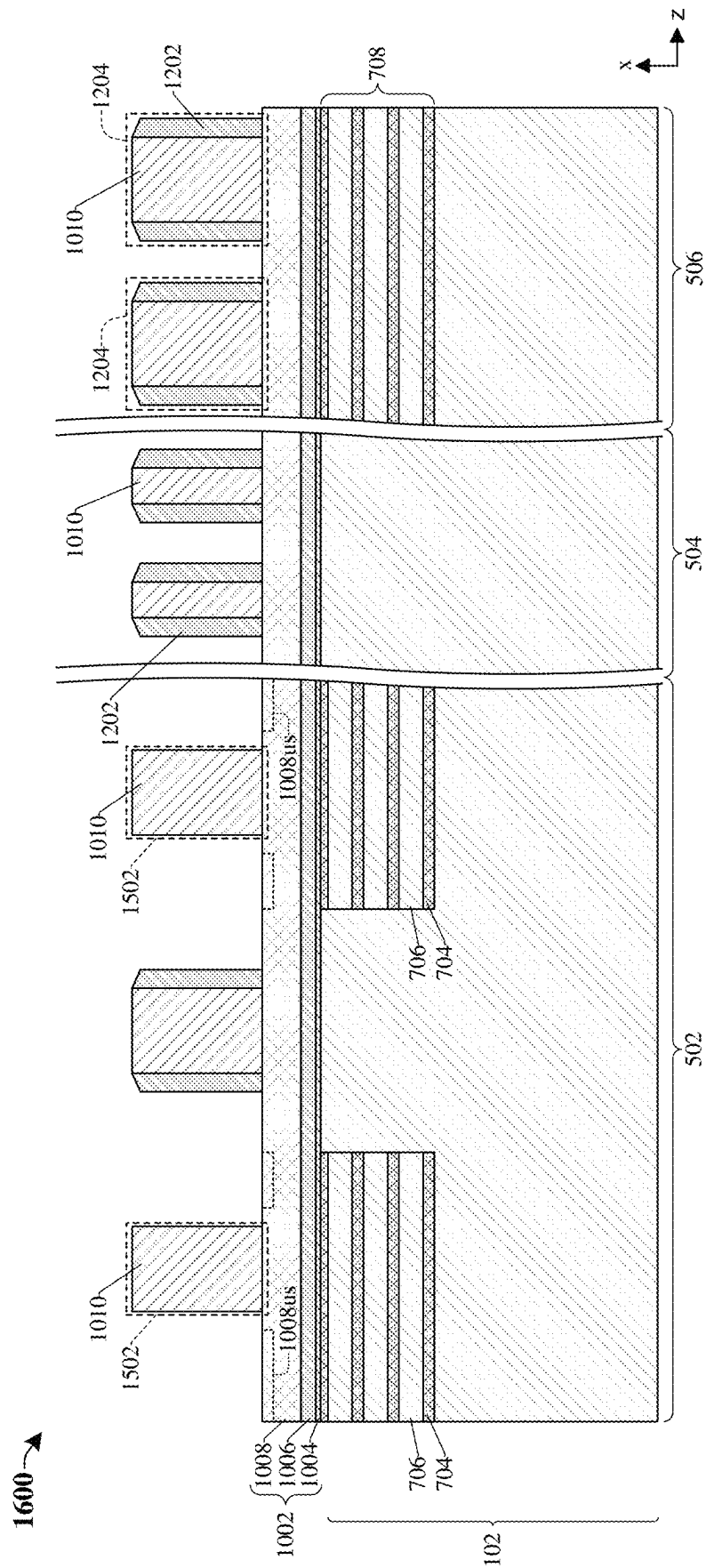


Fig. 16

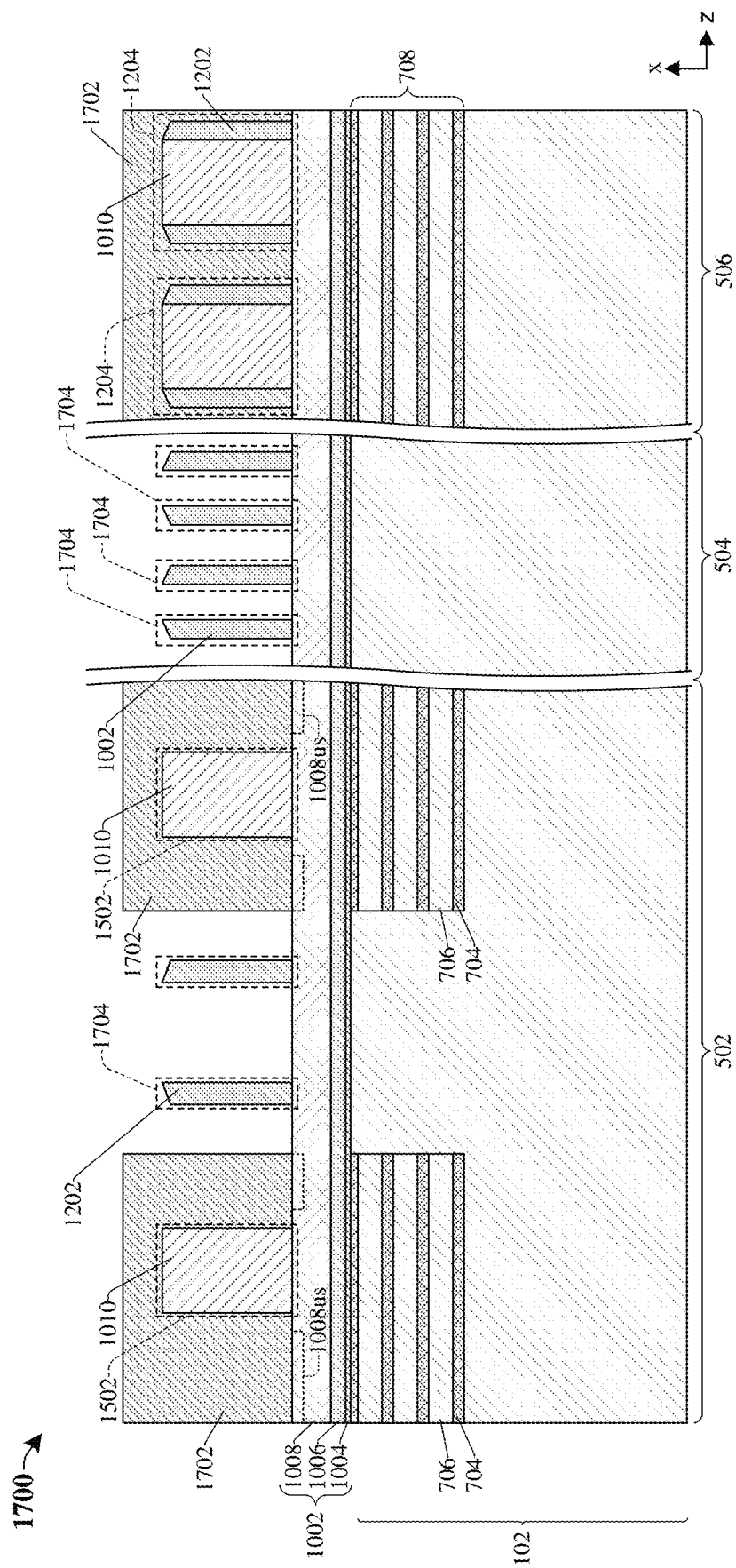


Fig. 17

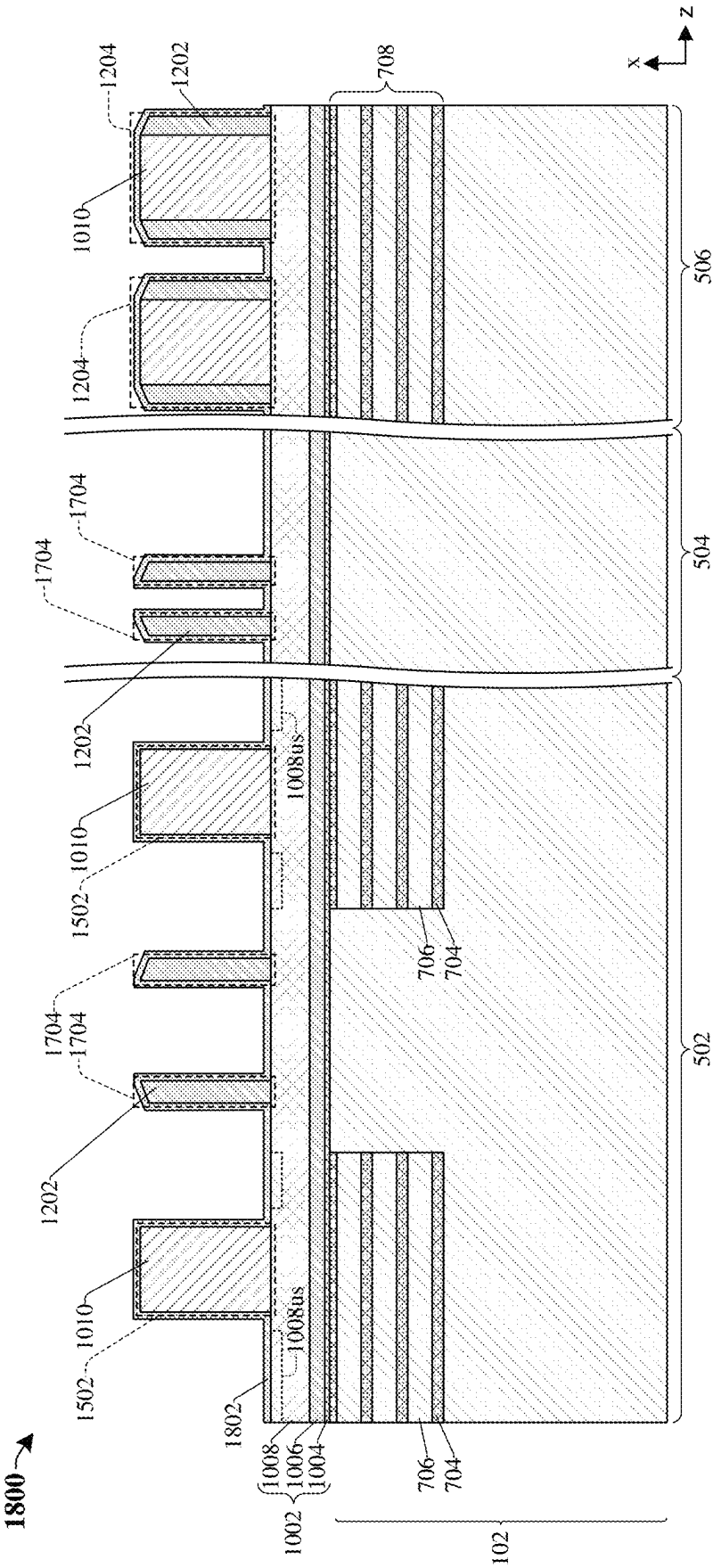


Fig. 18

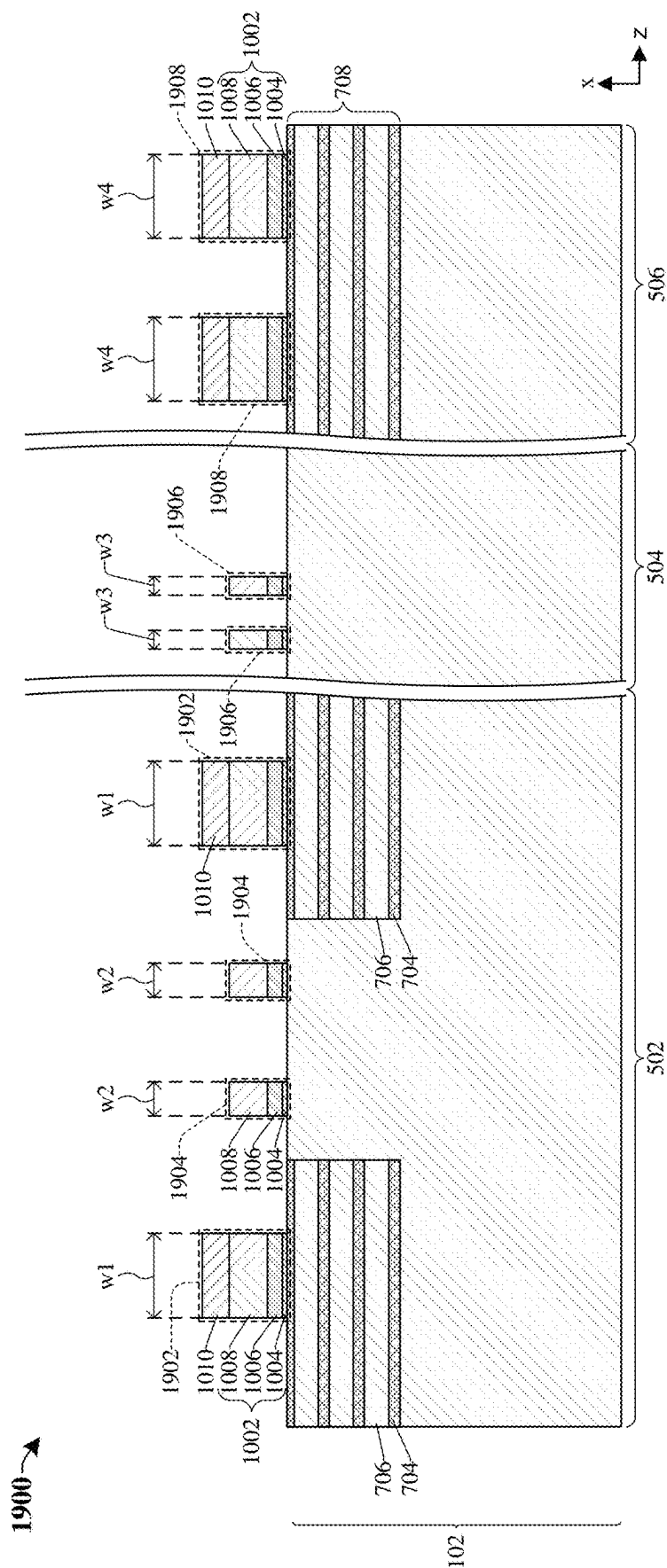


Fig. 19

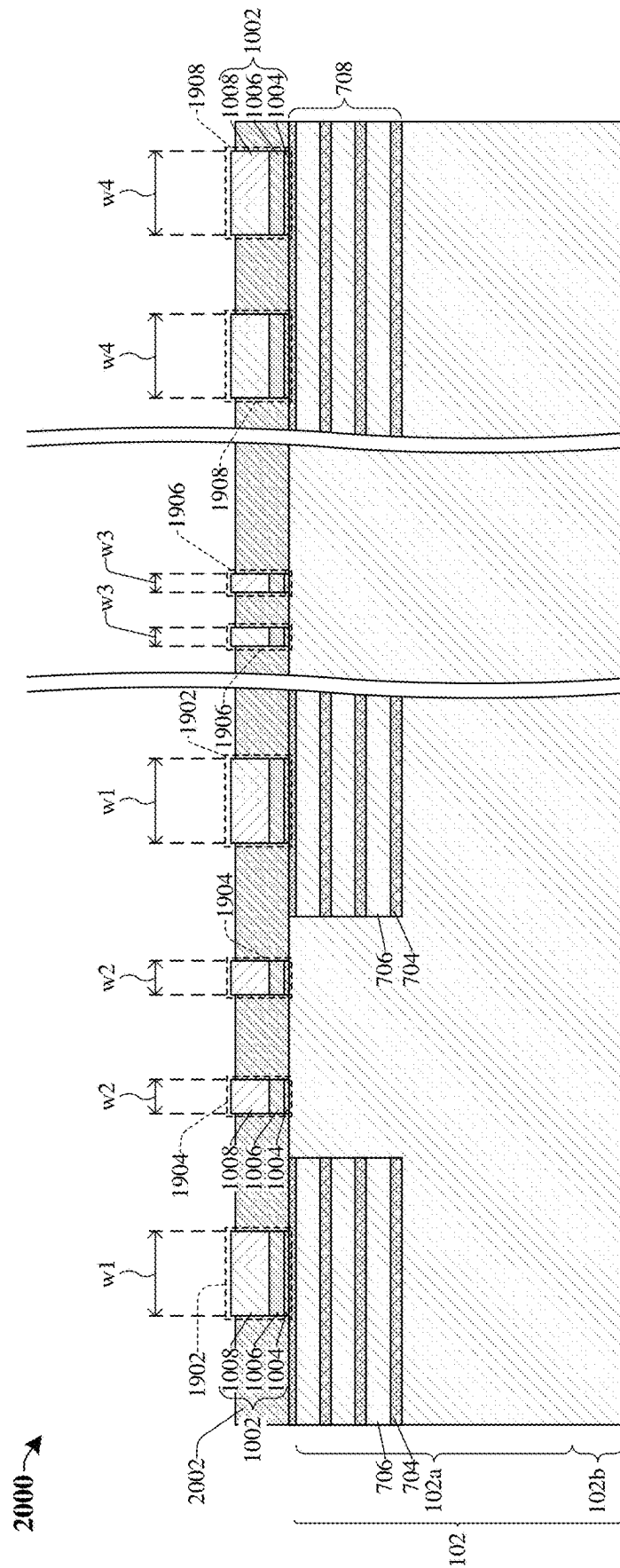


Fig. 20

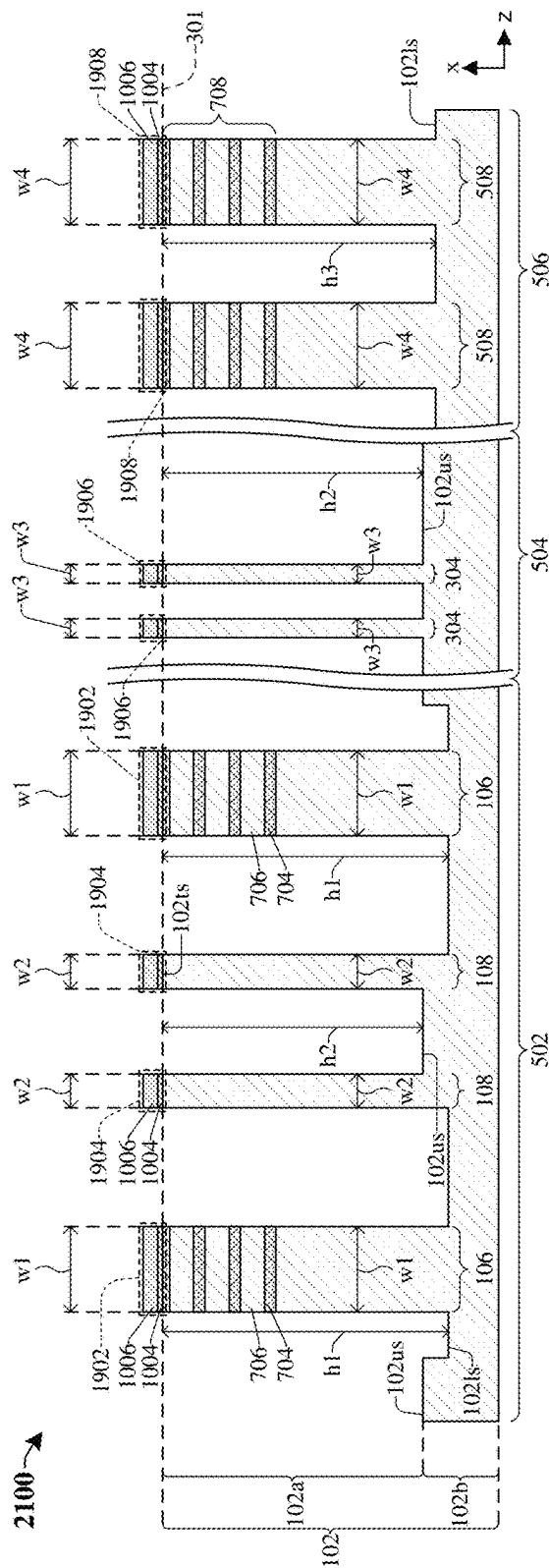


Fig. 21

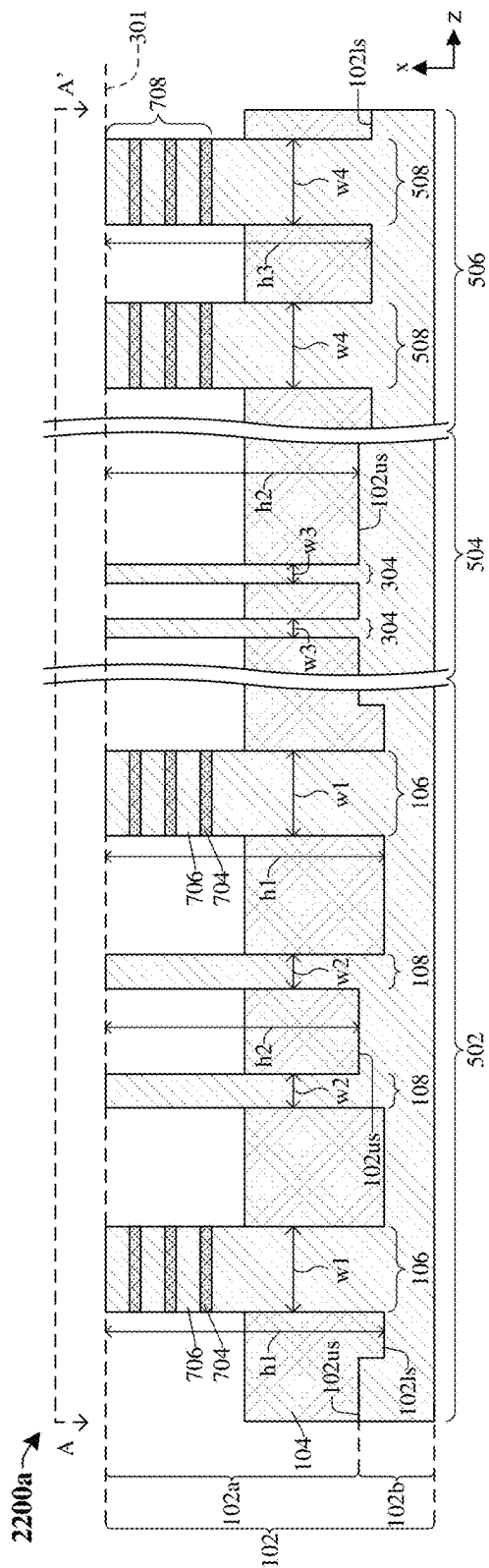
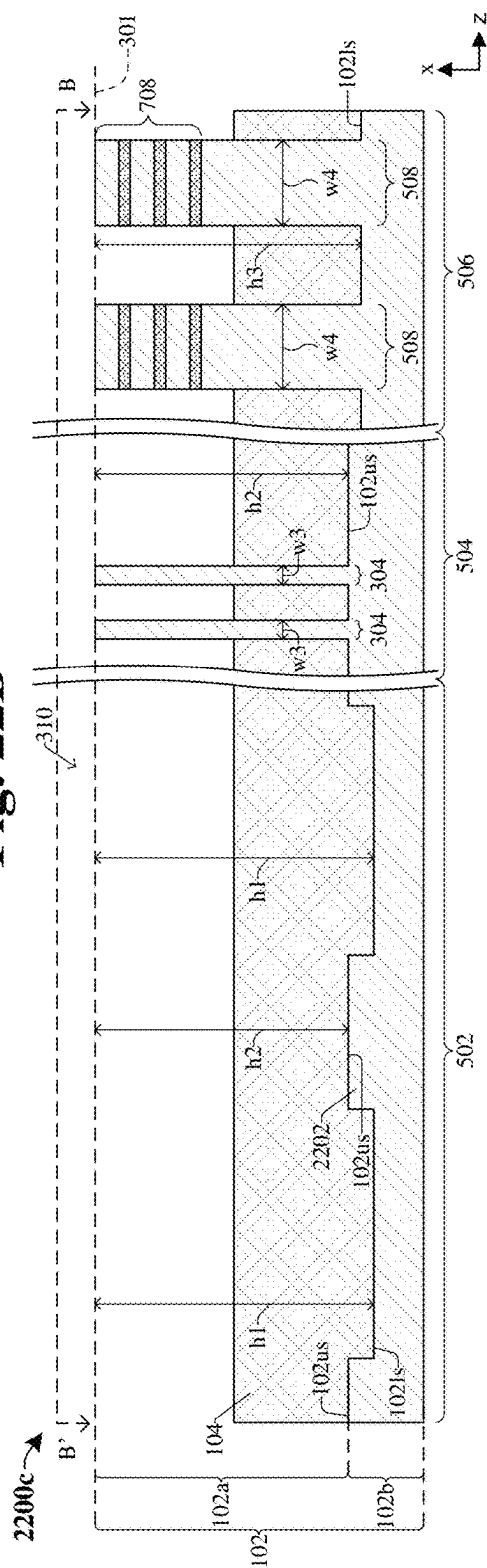
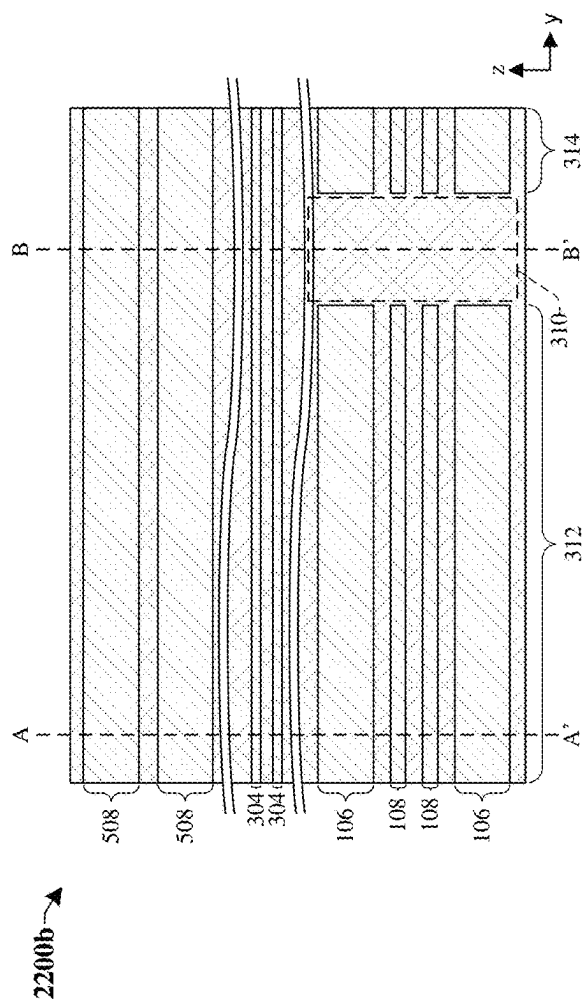


Fig. 22A



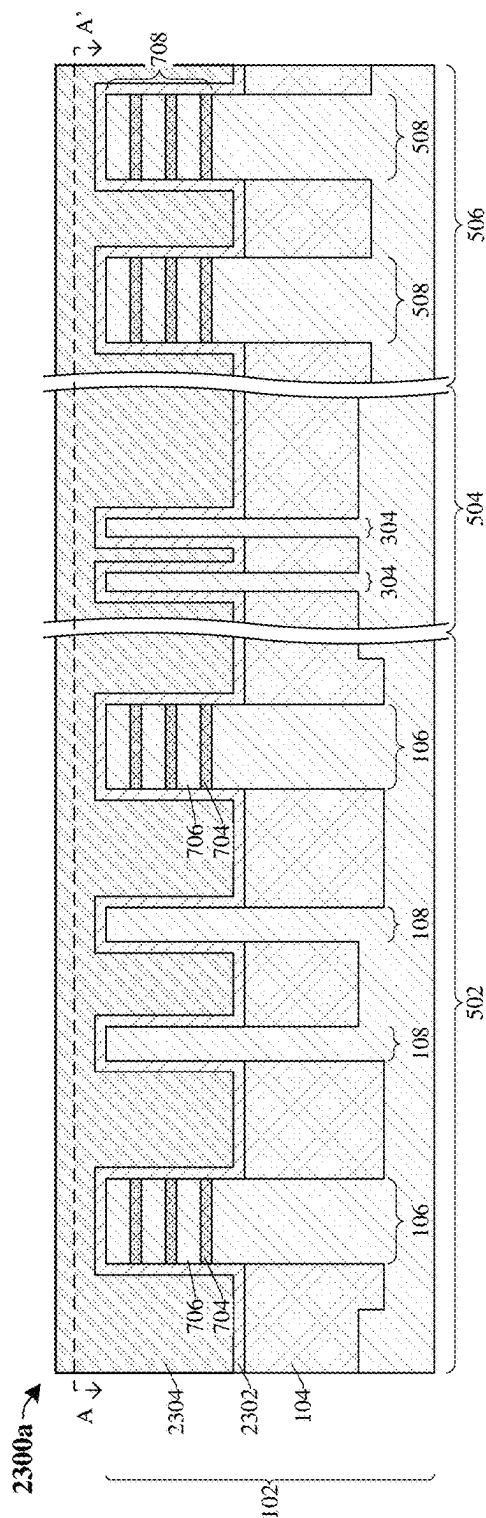


Fig. 23A

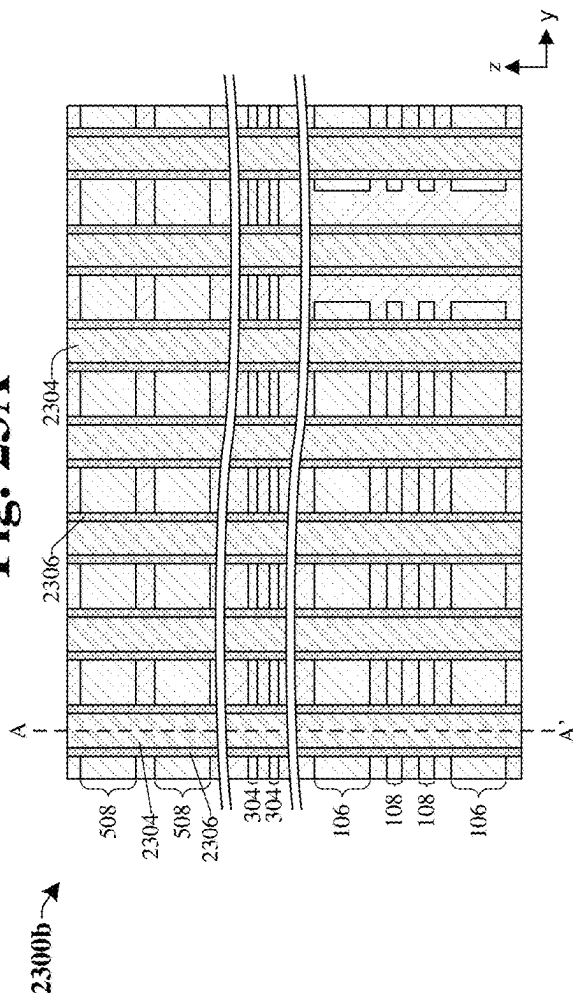


Fig. 23B

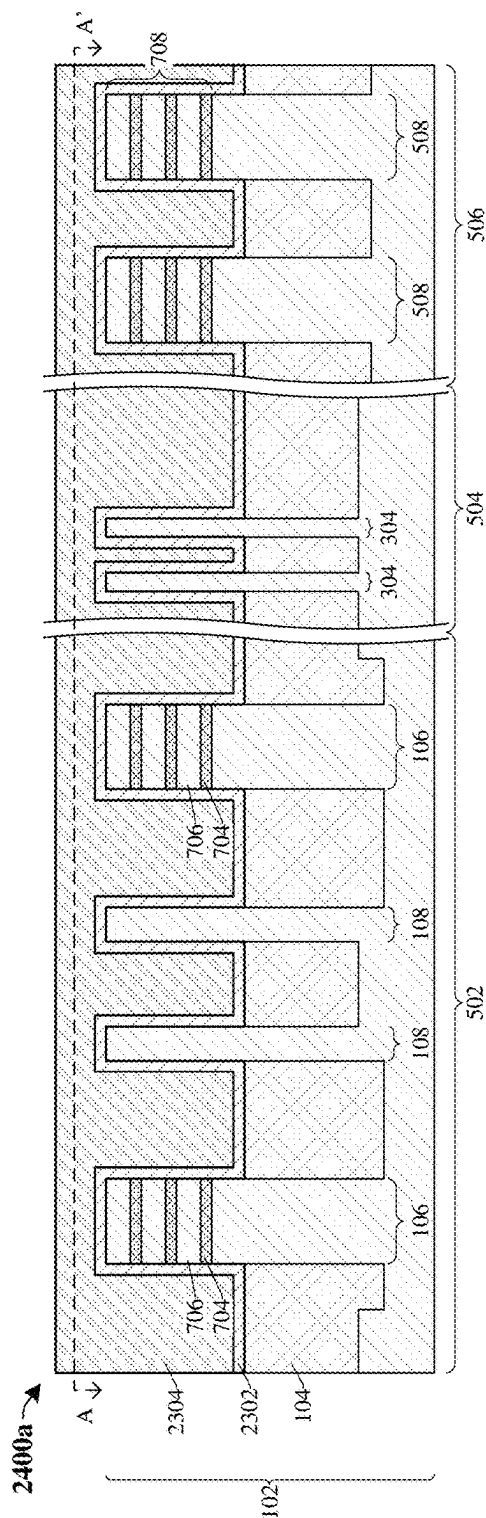


Fig. 24A

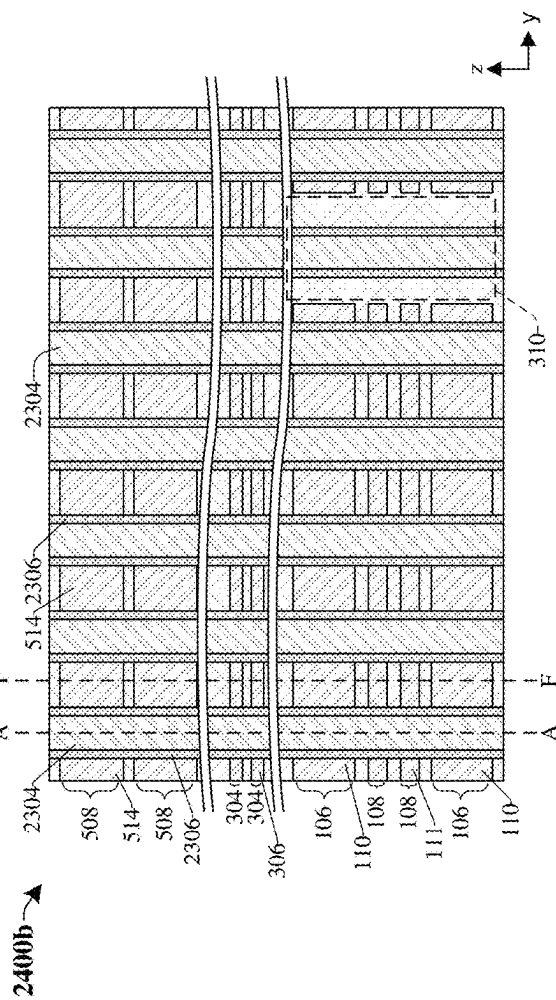
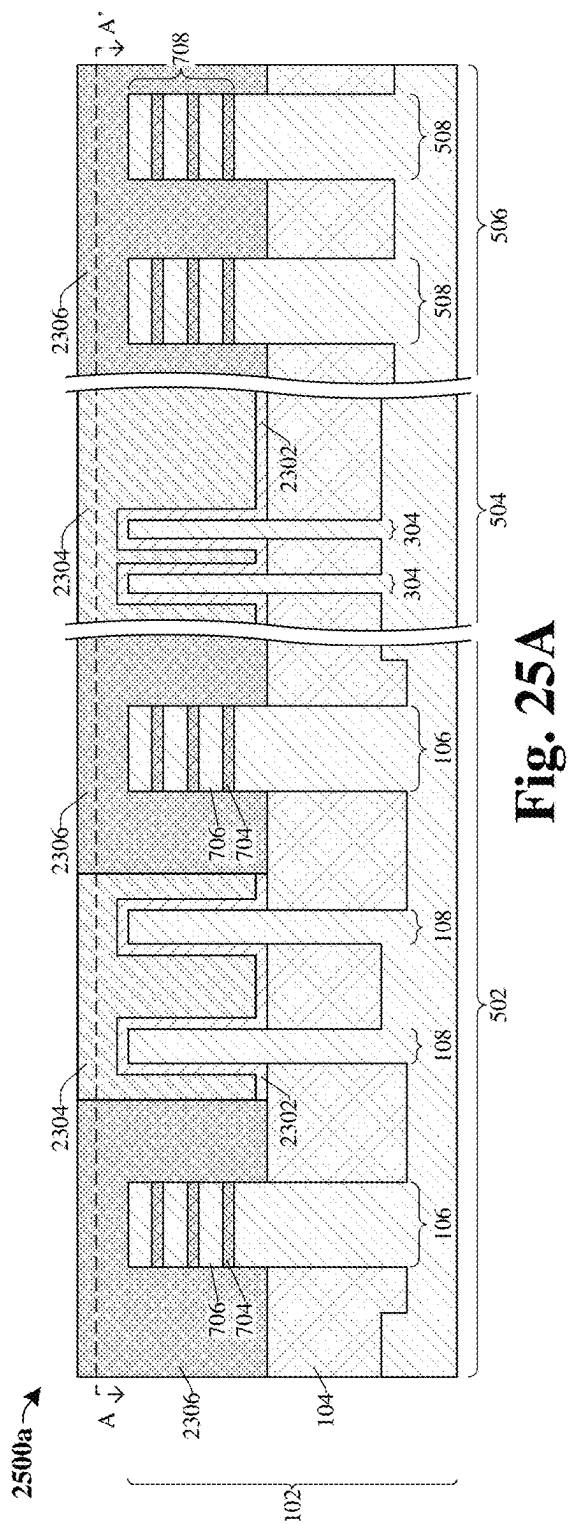
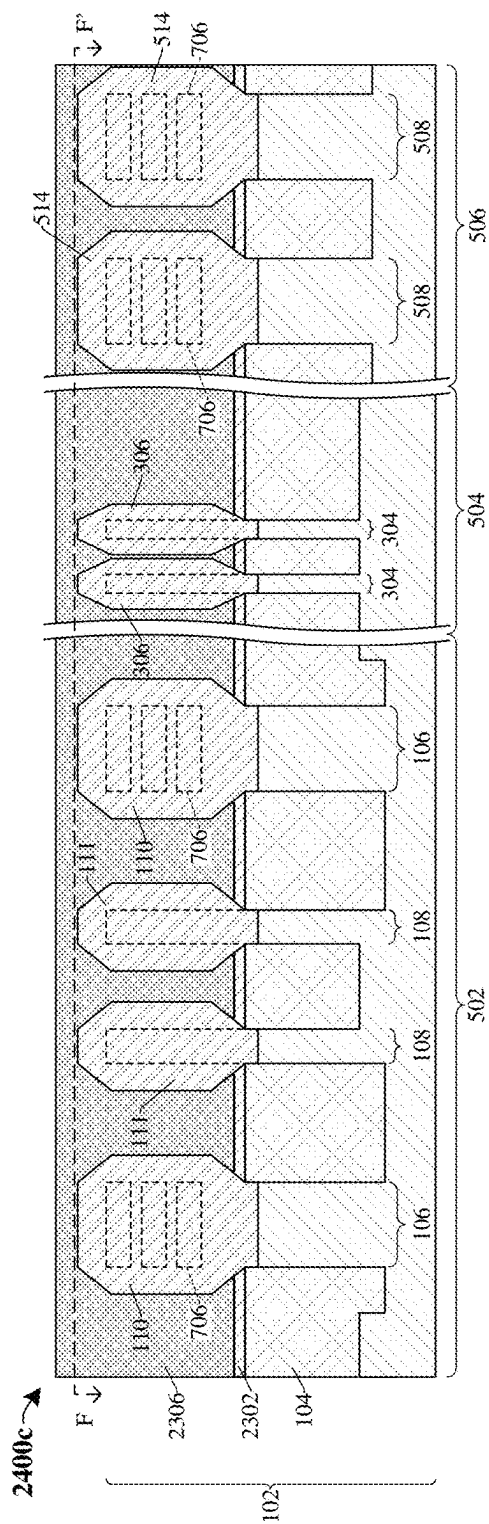


Fig. 24B



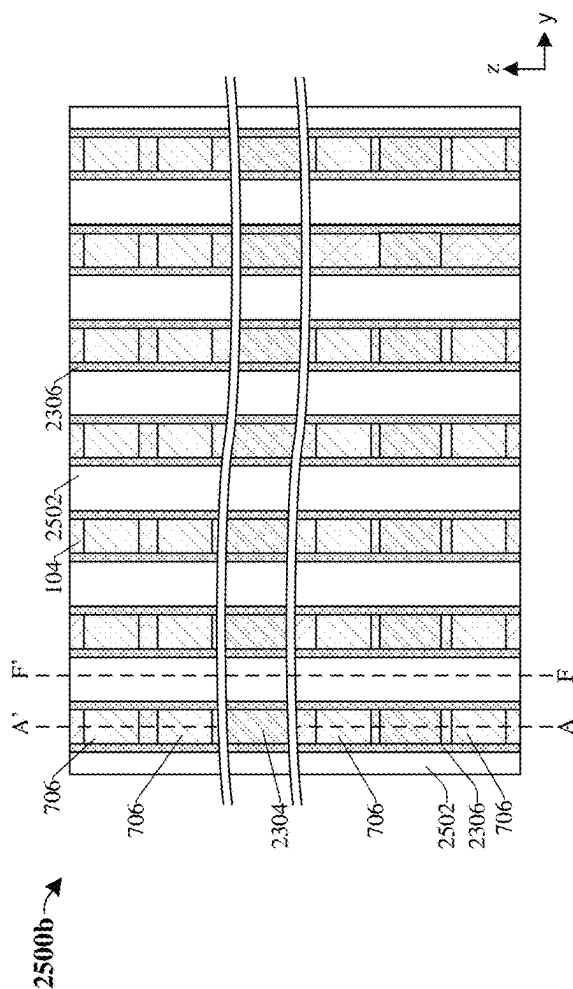


Fig. 25B

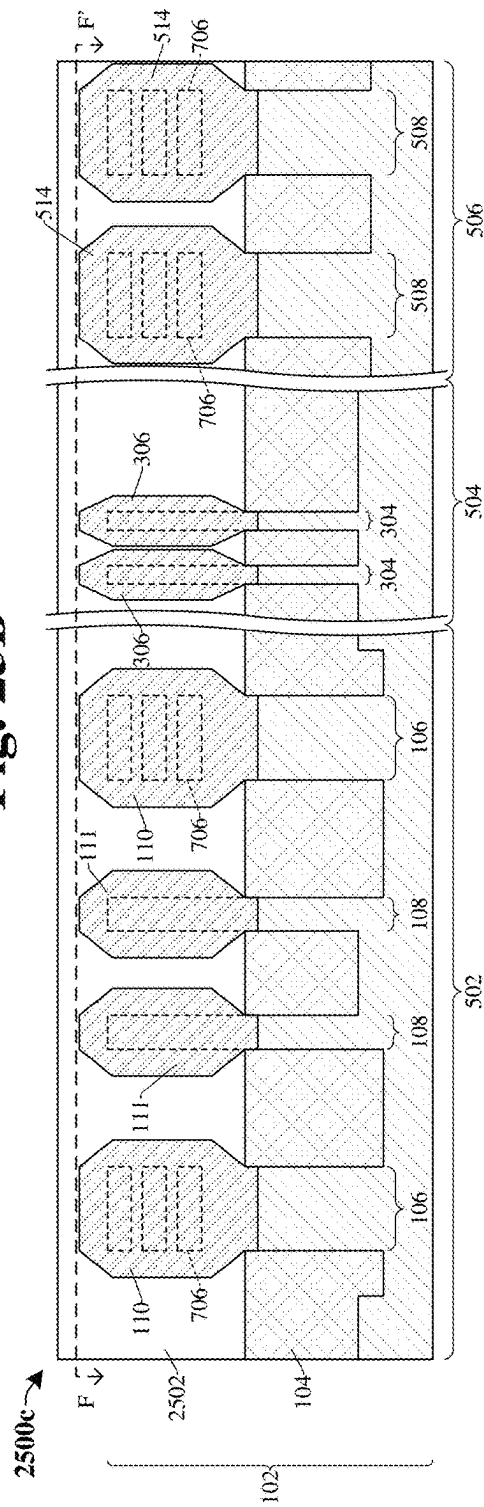
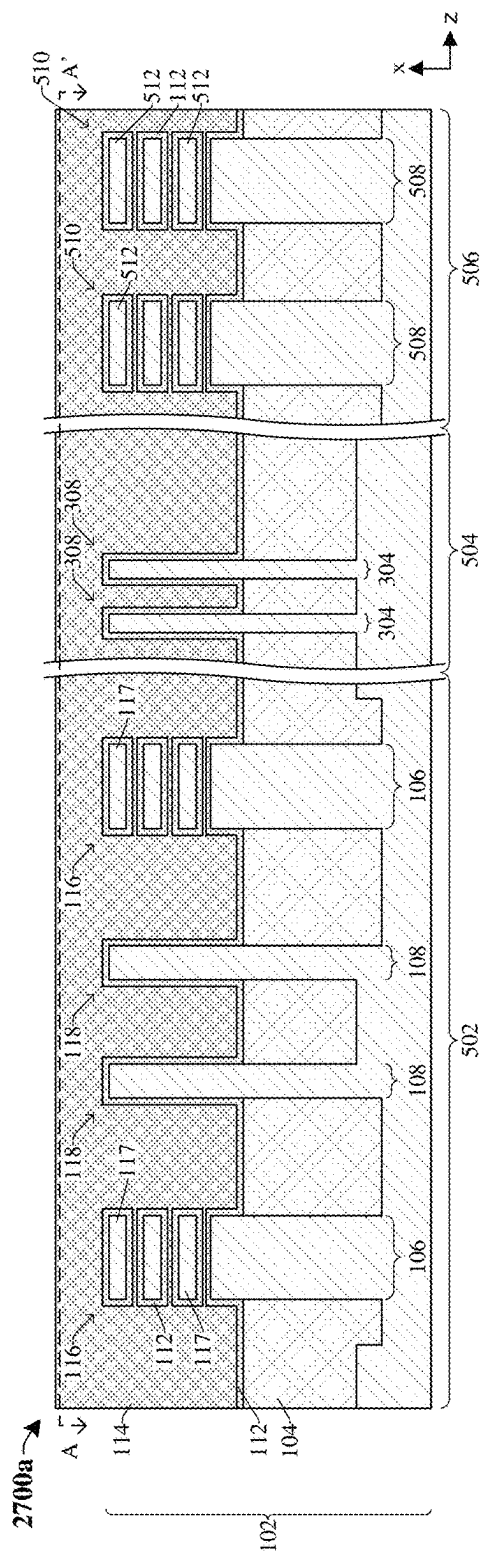
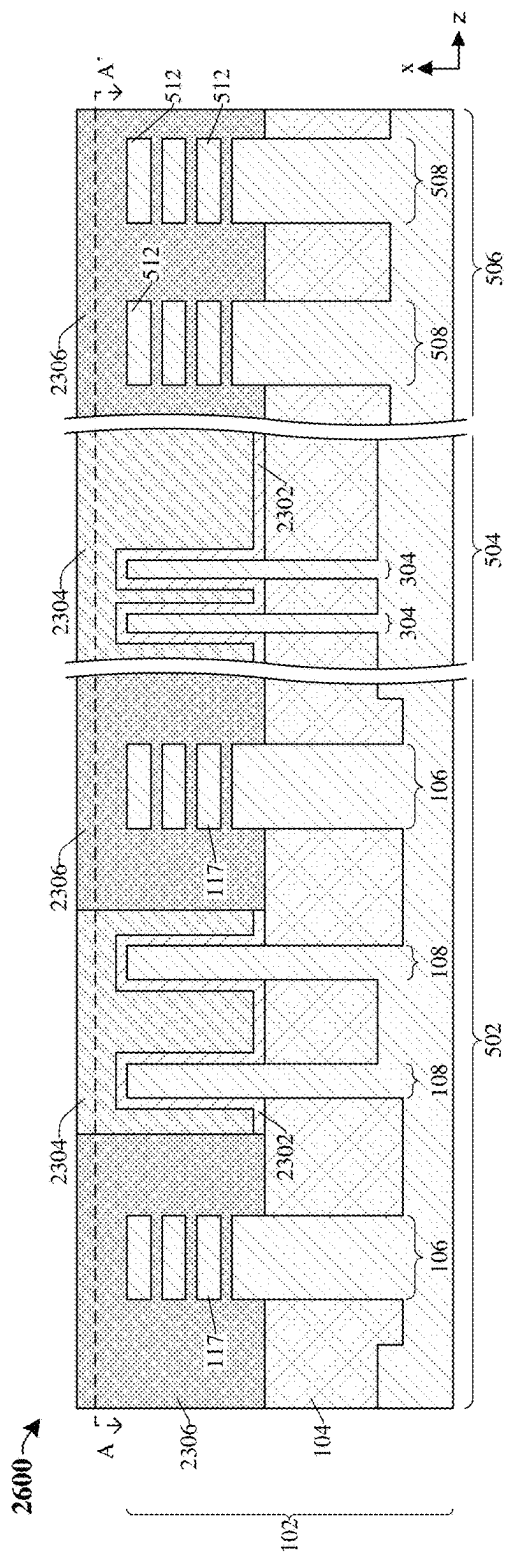


Fig. 25C



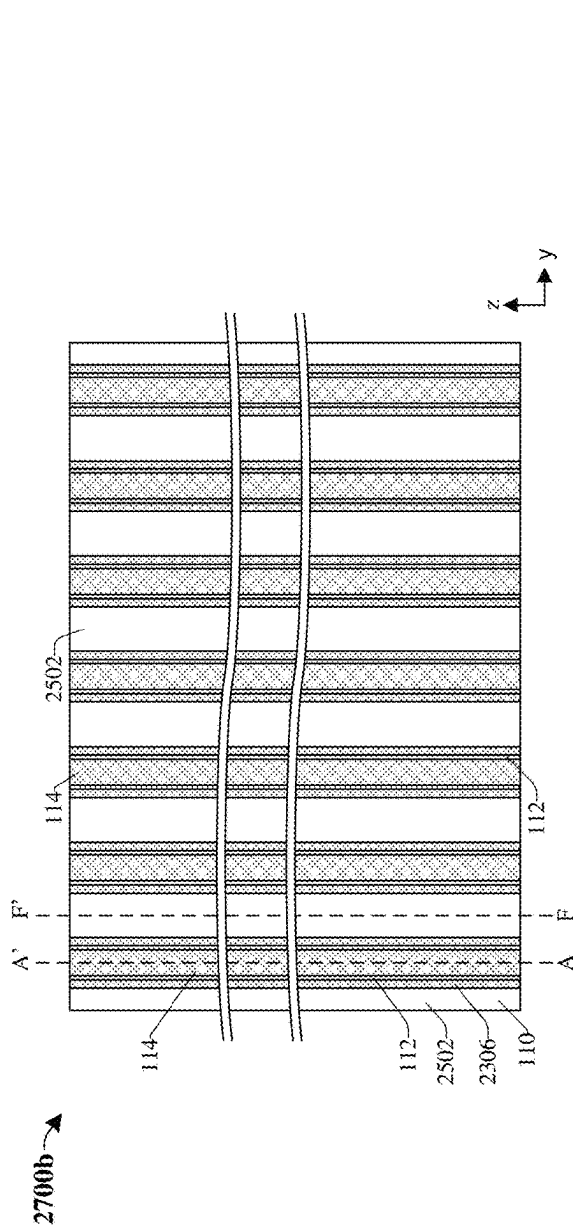


Fig. 27B

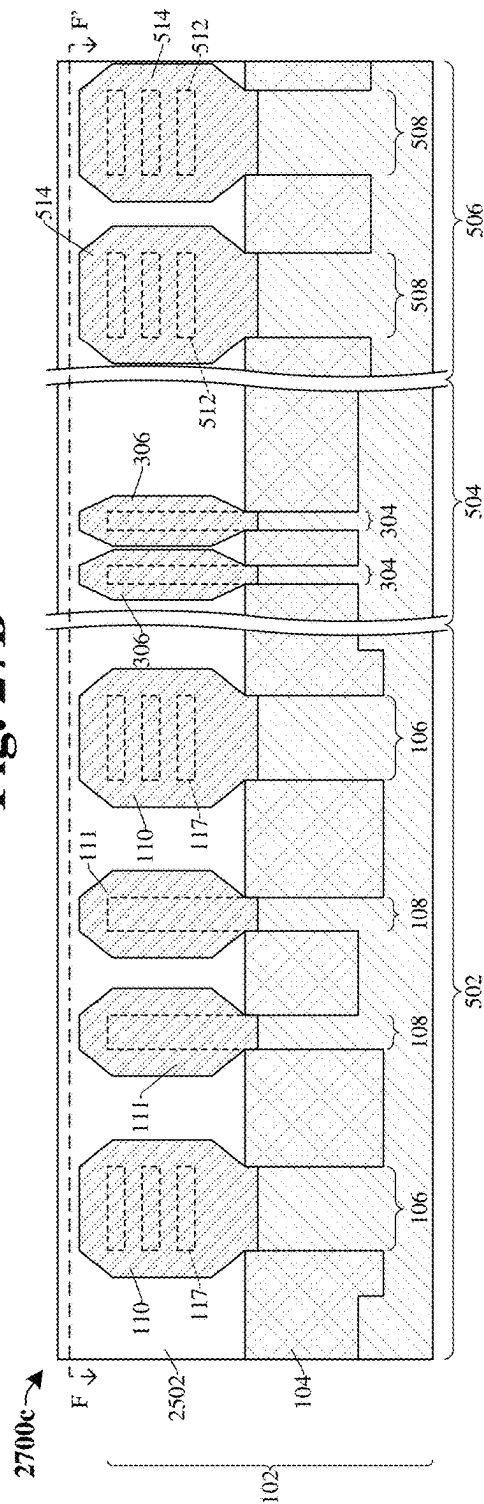
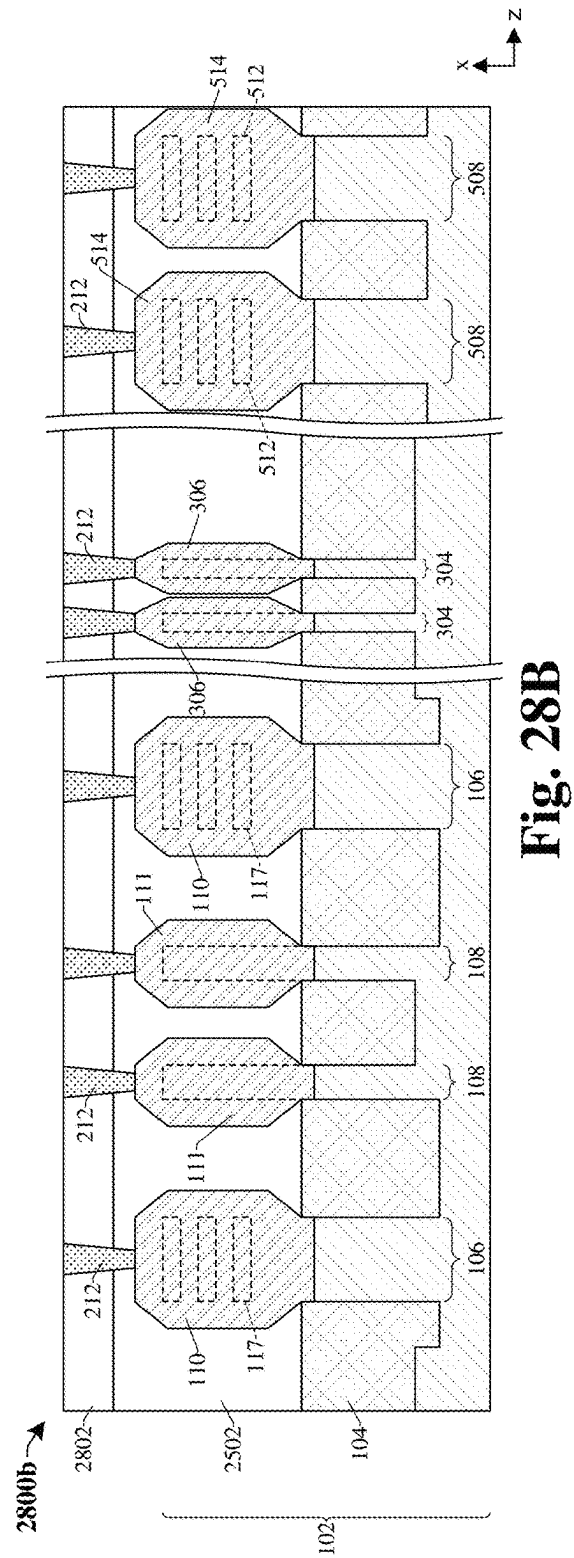
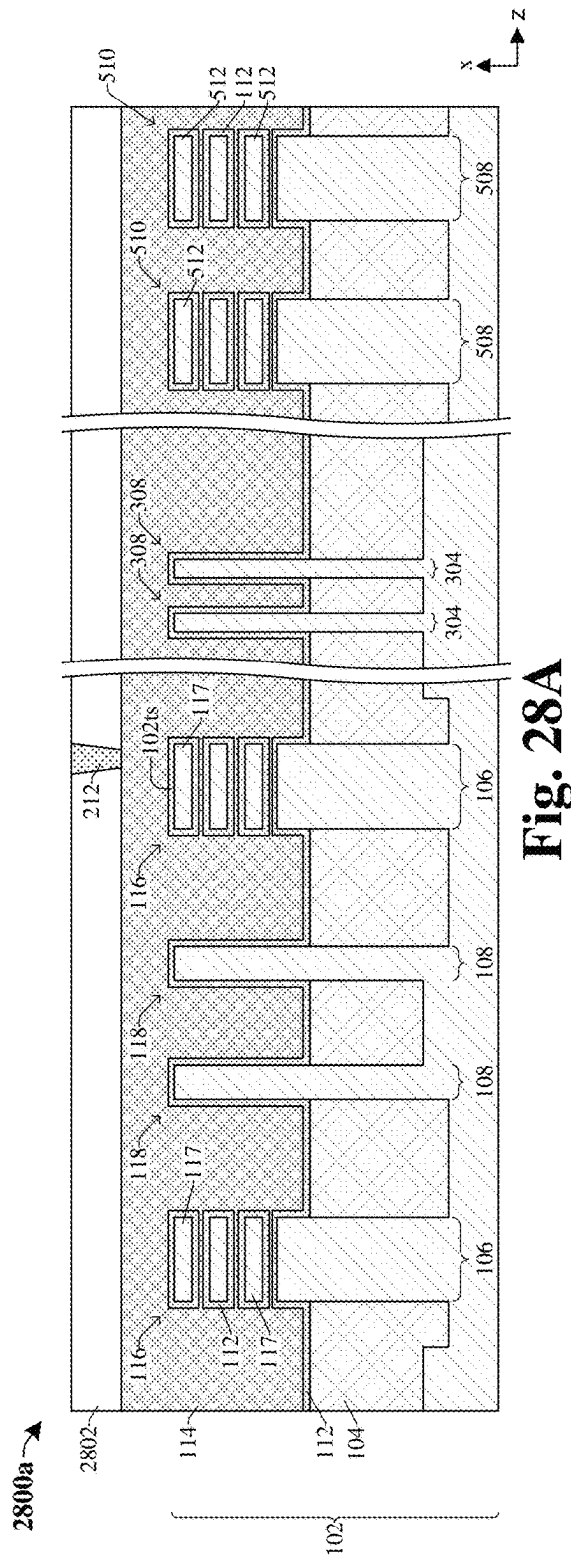
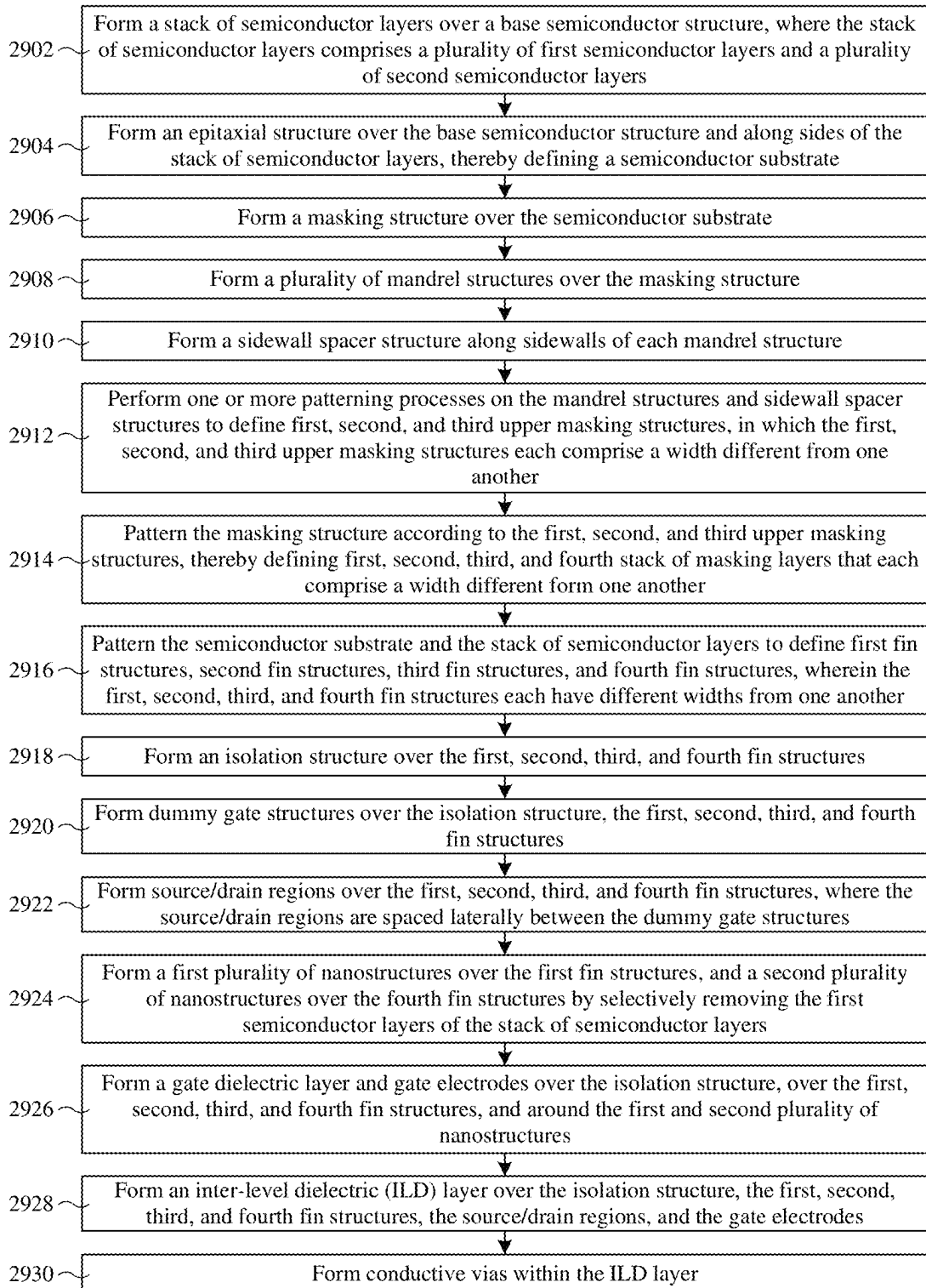


Fig. 27C



2900 →

**Fig. 29**

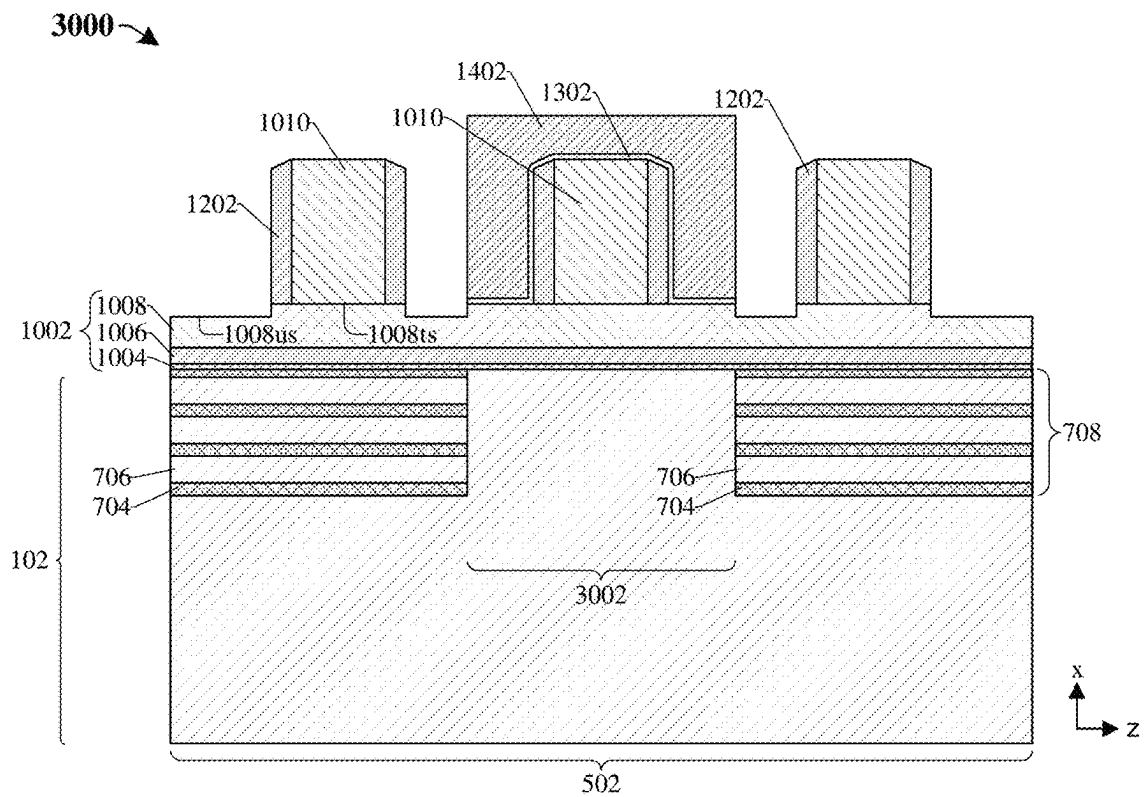


Fig. 30

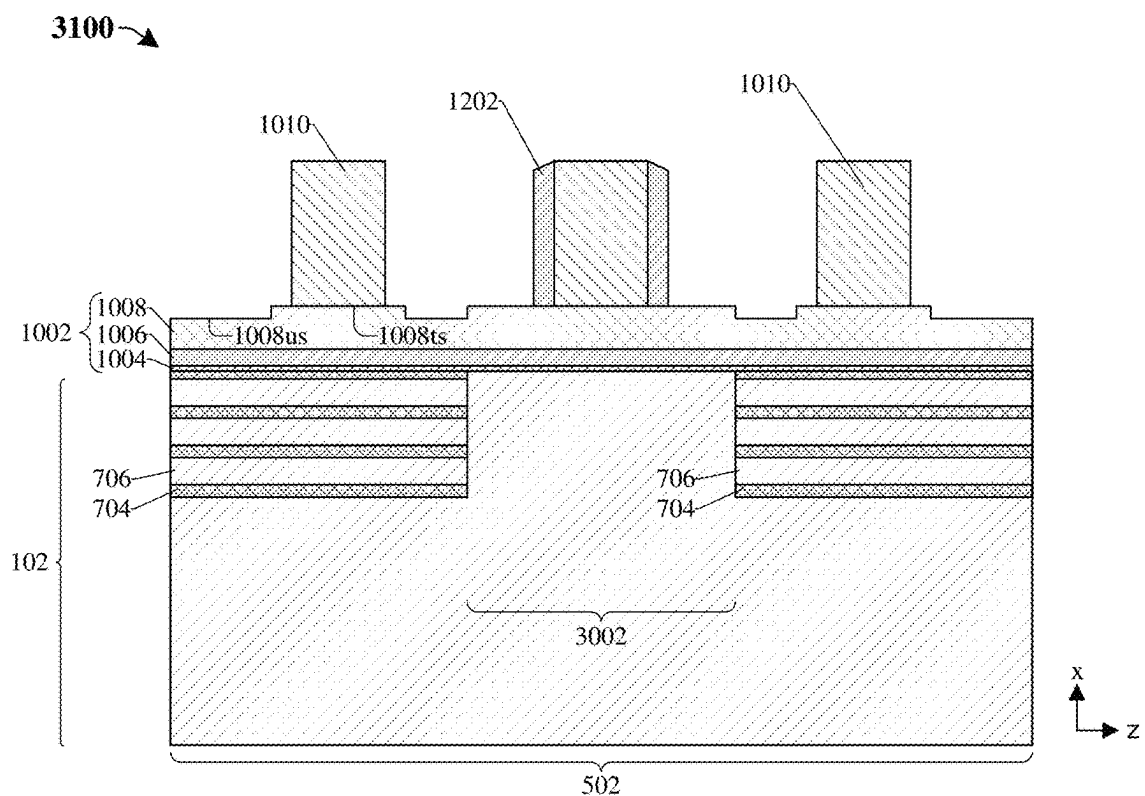


Fig. 31

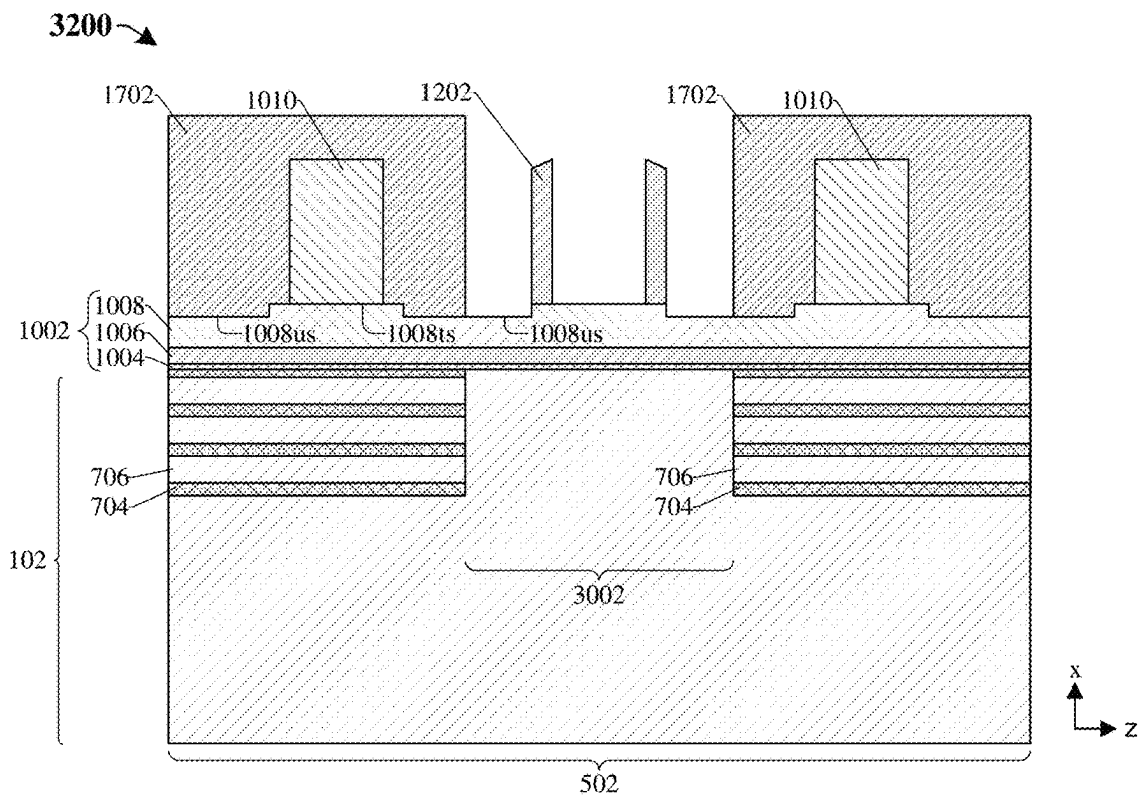


Fig. 32

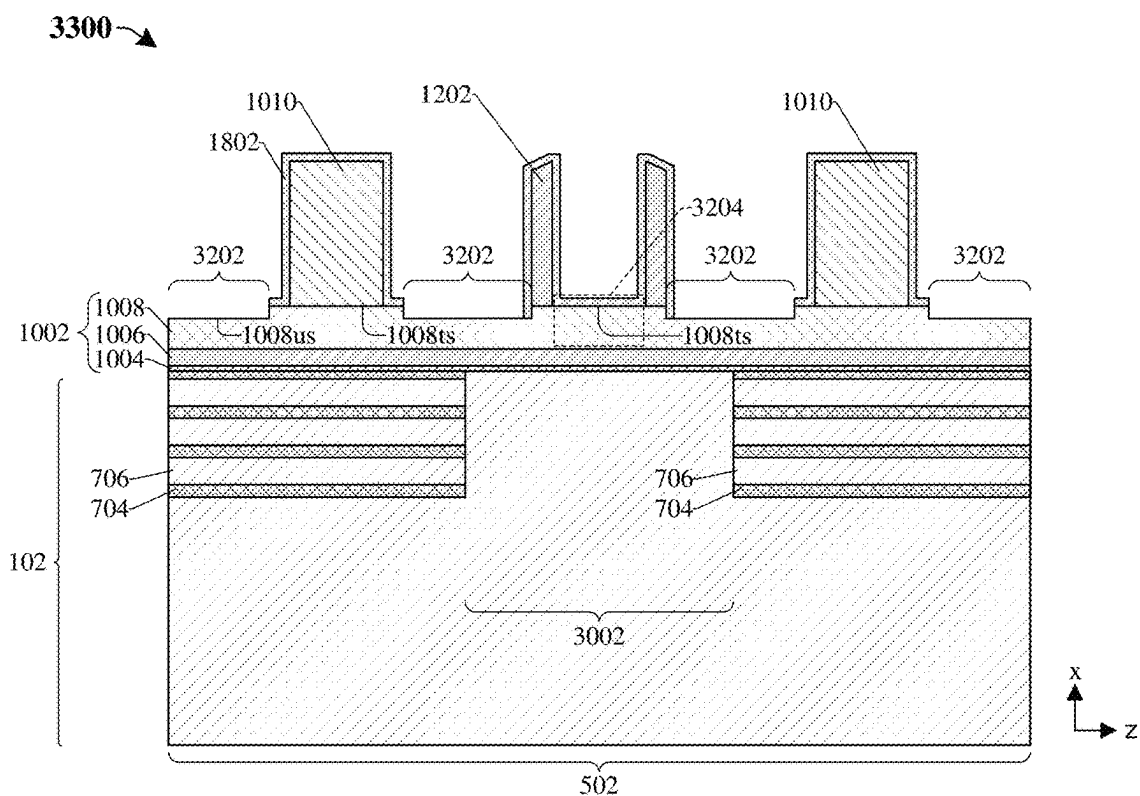


Fig. 33

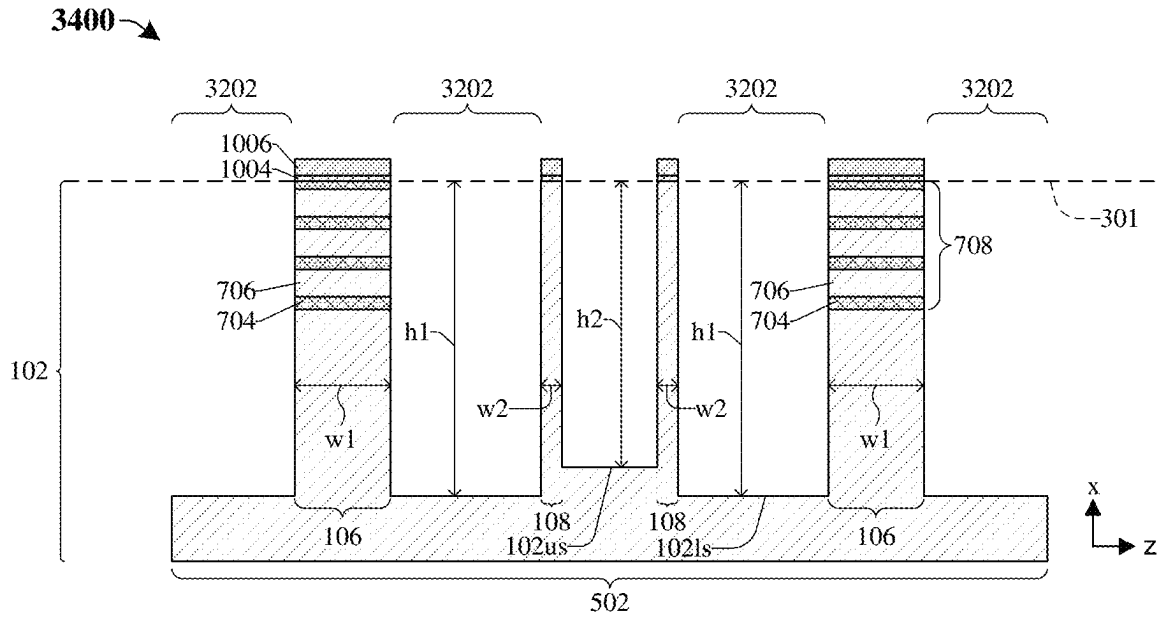


Fig. 34

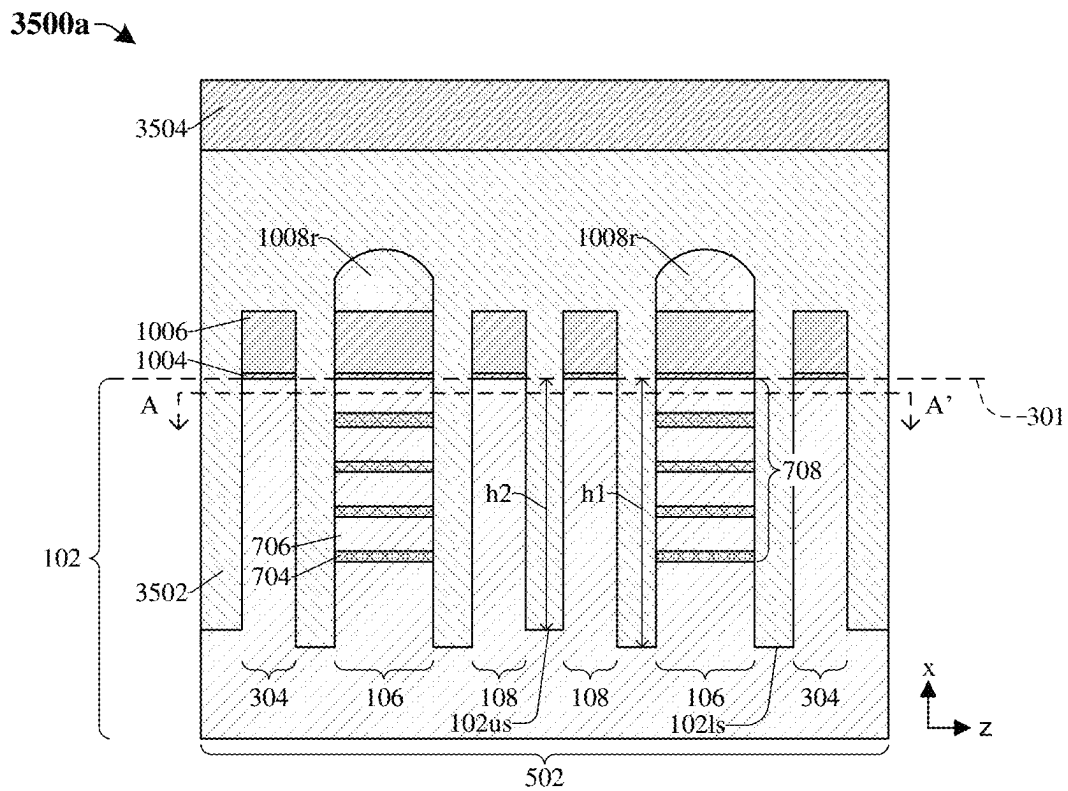


Fig. 35A

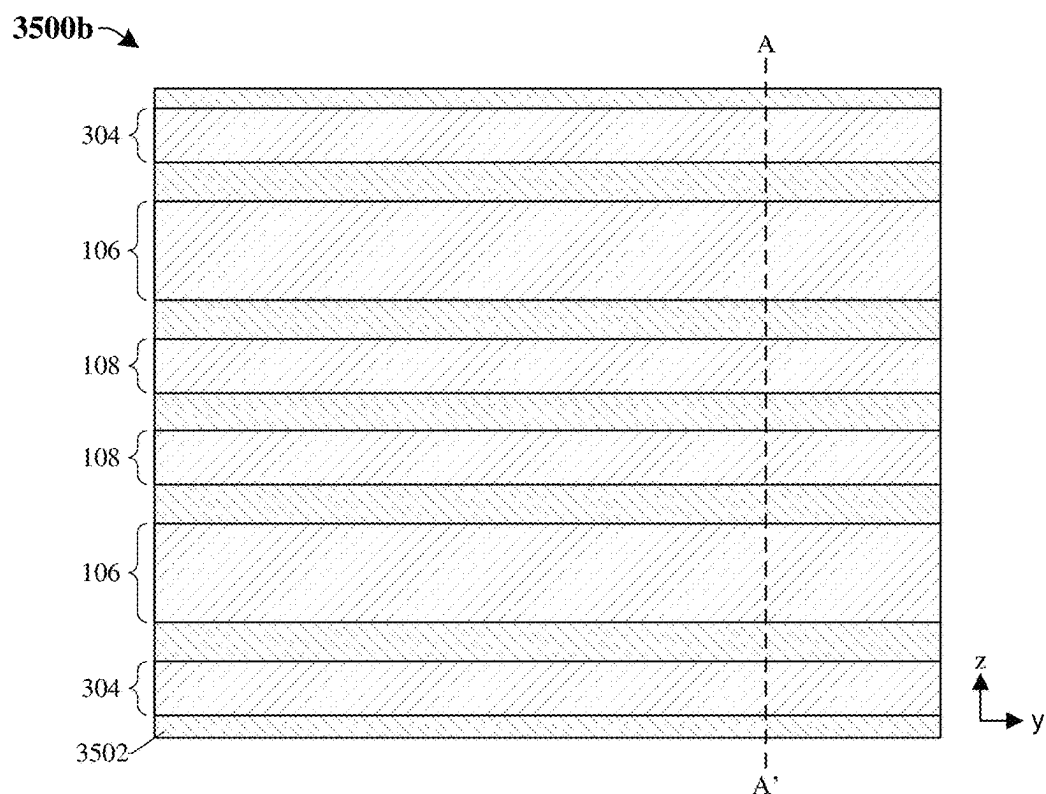


Fig. 35B

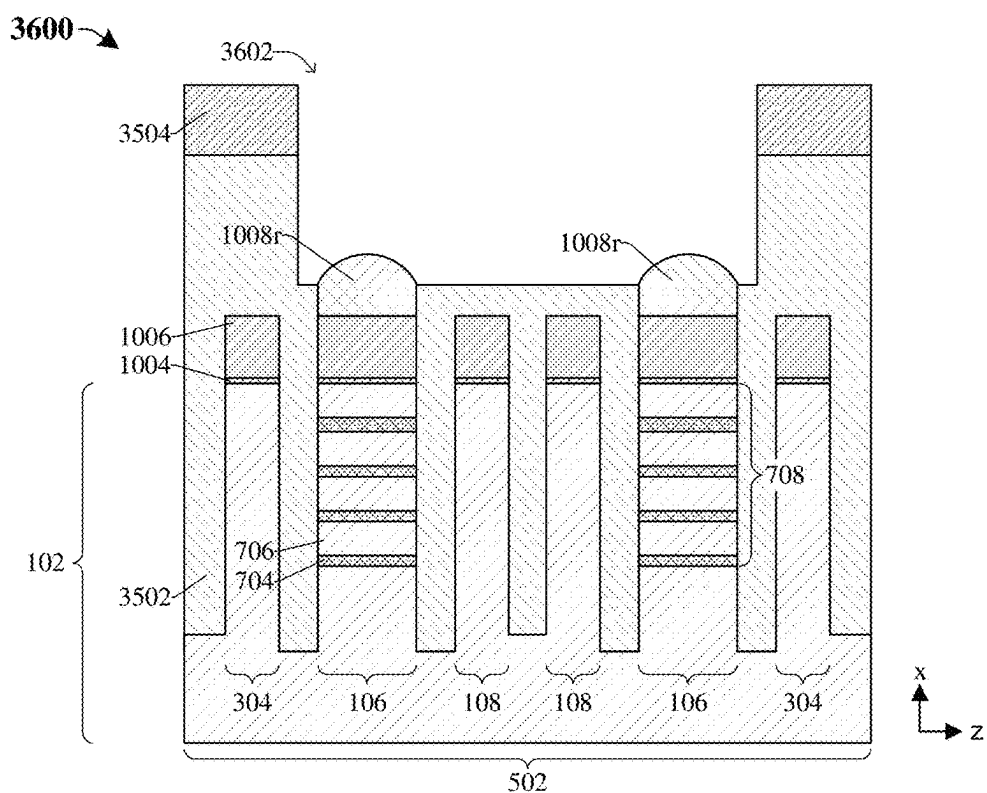


Fig. 36

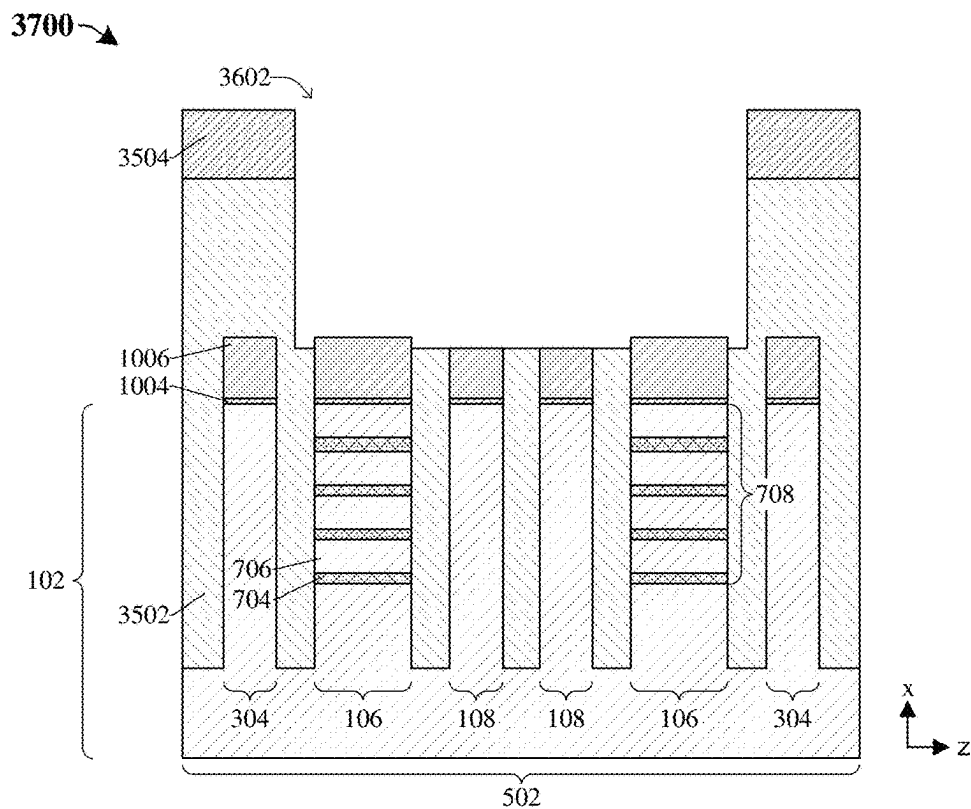


Fig. 37

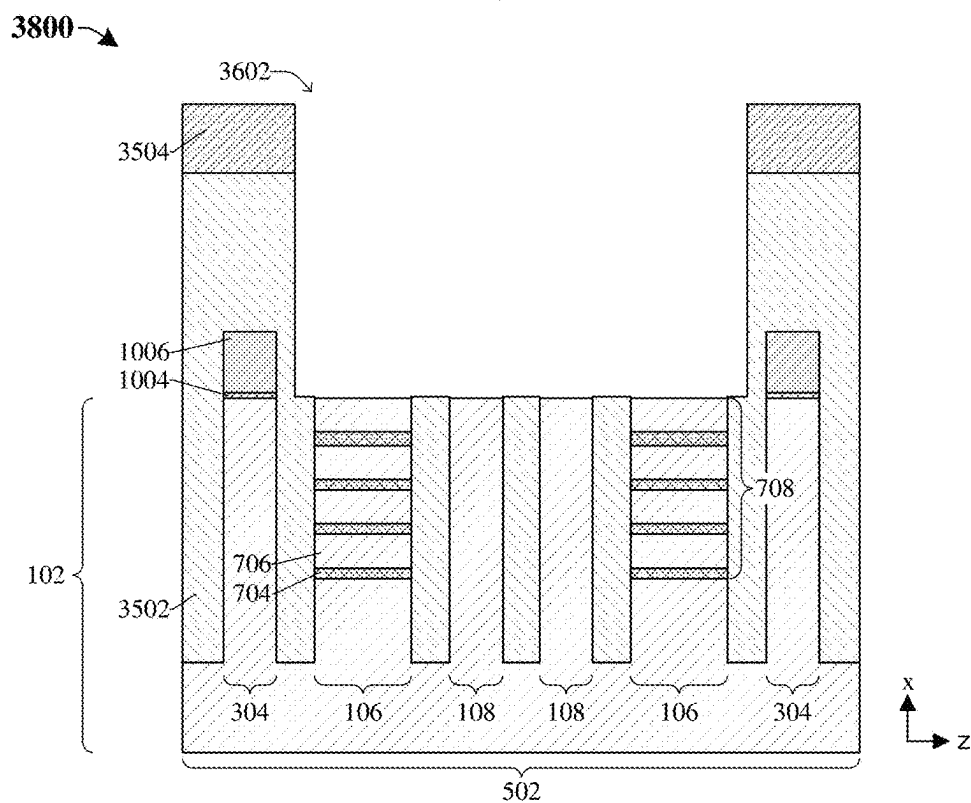


Fig. 38

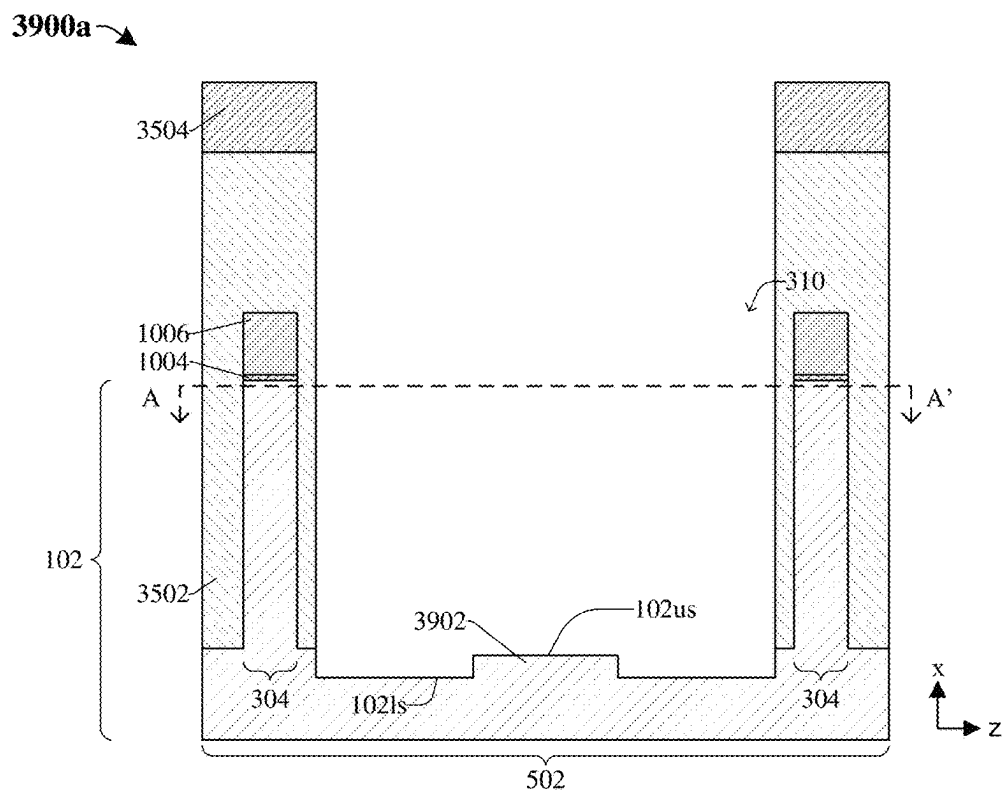


Fig. 39A

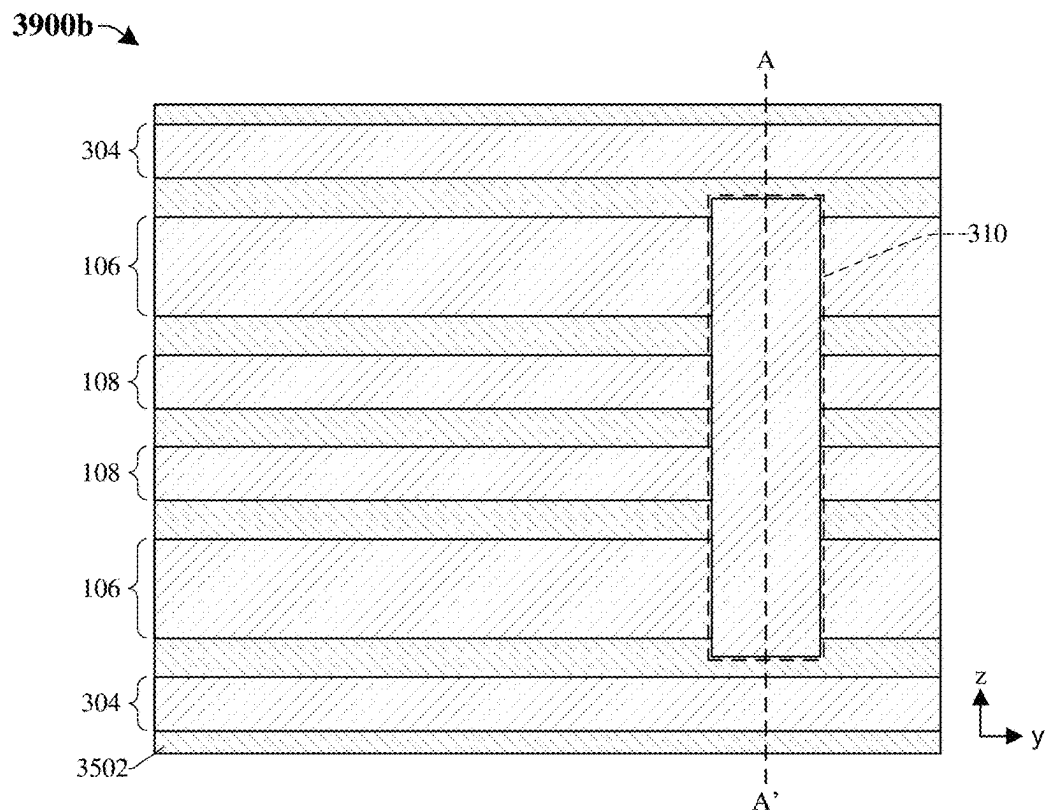


Fig. 39B

1

INTEGRATION OF MULTIPLE FIN STRUCTURES ON A SINGLE SUBSTRATE

REFERENCE TO RELATED APPLICATIONS

This Application is a Continuation of U.S. application Ser. No. 17/748,648, filed on May 19, 2022, which is a Divisional of U.S. application Ser. No. 16/823,581, filed on Mar. 19, 2020 (now U.S. Pat. No. 11,342,325, issued on May 24, 2022). The contents of the above-referenced Patent Applications are hereby incorporated by reference in their entirety.

BACKGROUND

The integrated circuit (IC) manufacturing industry has experienced exponential growth over the last few decades. As ICs have evolved, sizes of semiconductor devices (e.g., an area of a complementary metal-oxide-semiconductor (CMOS) inverter) have been scaled down by, for example, reducing minimum feature sizes and/or reducing spacing between components of the semiconductor devices, which has increased device density (e.g., a number of semiconductor devices integrated in a given area). However, as the sizes of semiconductor devices continue to be scaled down, it is becoming increase difficult to improve device performance of the semiconductors devices (e.g., increase switching speed, reduce current imbalance, reduce read/write times, etc.) without negatively affecting the device density. Thus, advancements in the IC manufacturing industry that improve the device performance of the semiconductors devices without negatively impacting device density are needed.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

FIG. 1 illustrates an isometric view of some embodiments of an integrated chip (IC) having a first fin structure laterally adjacent to a second fin structure on a same substrate, where the first fin structure has a greater width than the second fin structure.

FIGS. 2A-2D illustrate various views of some embodiments of an IC according to alternative embodiments of the IC of FIG. 1.

FIGS. 3A-3D illustrate various views of some embodiments of an IC according to alternative embodiments of the IC of FIGS. 2A-2D.

FIGS. 4A-4B illustrate cross-sectional views of some embodiments of an IC according alternative embodiments of the IC of FIGS. 3A-3D.

FIGS. 5A-5C illustrate various views of some embodiments of an IC comprising a plurality of fin structures having different widths and/or layers disposed laterally adjacent to one another on a same substrate.

FIGS. 6A-6B illustrate cross-sectional views of some embodiments of the IC of FIGS. 5A-5C.

FIGS. 7 through 28A-28B illustrate a series of various views of some embodiments of a first method for forming an IC comprising a plurality of fin structures having different widths, where the fin structures are disposed laterally adjacent to one another on a same substrate.

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FIG. 29 illustrates a flowchart of some embodiments of a method for forming an IC comprising a plurality of fin structures having different widths, where the fin structures are disposed laterally adjacent to one another on a same substrate.

FIGS. 30-34 illustrate a series of cross-sectional views of some alternative embodiments of the method of FIGS. 7 through 28A-28B.

FIGS. 35A-35B through 39A-39B illustrate a series of various views of some alternative embodiments of the method of FIGS. 7 through 28A-28B.

DETAILED DESCRIPTION

The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

A fin field effect transistor (finFET) device may comprise a fin over a substrate, wherein a gate electrode overlies the fin. The gate electrode may directly overlie a selectively conductive channel of the fin, and the selectively conductive channel may be laterally between a source region and a drain region on the fin. A threshold voltage is a voltage that is applied to the gate electrode that turns the finFET device “ON,” in some embodiments, such that mobile charge carriers flow through the selectively conductive channel of the fin from the source region to the drain region.

In some embodiments, a plurality of fins may be defined within a substrate extending along a first direction. The plurality of fins are parallel to one another. Further, a plurality of gate electrodes overlie the plurality of fins and extend along a second direction orthogonal to the first direction. The plurality of gate electrodes are parallel to one another. A finFET device may be defined at the intersection of each fin and gate electrode, such that there are a plurality of finFET devices disposed on the substrate. Each finFET device has source/drain regions defined within and/or on a corresponding fin along opposing sides of an overlying gate electrode, and a selectively conductive channel is defined within the corresponding fin laterally between the source/drain regions. Due to fabrication limitations, each fin will have substantially similar design parameters (e.g., similar

widths) such that the finFET devices disposed along the single substrate may be designed for a single application and/or function. This may decrease a design complexity of the finFETs overlying the single substrate. However, this results in design limitations, performance tradeoffs, and/or an inability to integrate two or more different field effect transistor devices optimized for different applications on a single substrate.

Accordingly, various embodiments of the present disclosure provide an integrated chip (IC) having a plurality of finFET devices disposed laterally adjacent to a plurality of nanosheet field effect transistor (NSFET) devices. The finFET devices comprise fins defined within a substrate and the NSFET devices comprise nanostructures defined over the substrate, where the nanostructures each have a width different from the fins. The fins may extend along a first direction in parallel with one another. Additionally, a plurality of gate electrodes overlie the plurality of fins and overlie/wrap around each of the nanostructures. Further, the gate electrodes extend along a second direction that is orthogonal to the first direction. Thus, the finFET devices are defined at intersections between the gate electrodes and the fins, and the NSFET devices are defined at intersections between the gate electrodes and the nanostructures.

By virtue of the different widths of the fins and the nanostructures, the finFET devices are designed for optimal performance of a first application while the NSFET devices are designed for optimal performance of a second application. For example, the first finFET devices may be designed as a pull-up transistor for a static random access memory (SRAM) device, while the NSFET devices may be designed as an access transistor for an SRAM device. Because the width of the nanostructures are larger than the widths of the fins, the NSFET devices may be configured to operate with higher currents than the finFET devices. Thus, the fins and the nanostructures may be formed in such a manner to maximize performance of finFET and NSFET devices for the first and second applications, respectively while minimizing an area of the substrate the fins and the nanostructures occupy. This, in part, increases a number of different semiconductor devices that may be formed over a single substrate, and increases design flexibility and performance of the semiconductor devices disposed on the substrate.

FIG. 1 illustrates an isometric view of some embodiments of an integrated chip (IC) **100** having a first fin structure laterally adjacent to a second fin structure on a same substrate, where the first fin structure has a greater width than the second fin structure.

As shown in FIG. 1, the IC **100** includes a semiconductor substrate **102**. In some embodiments, the semiconductor substrate **102** may, for example, be or comprise any type of semiconductor body, such as monocrystalline silicon, CMOS bulk, silicon-germanium (SiGe), silicon carbide, or another suitable semiconductor material. In some embodiments, the semiconductor substrate **102** comprises a first doping type (e.g., p-type). The semiconductor substrate **102** comprises a center device region **105** and a peripheral device region **103**, where the peripheral device region **103** is disposed on opposing sides of the center device region **105**. Further, the semiconductor substrate **102** comprises a plurality of first fin structures **106** disposed within the peripheral device region **103**, and a plurality of second fin structures **108** disposed within the center device region **105**.

Each of the first and second fin structures **106**, **108** extend in parallel with one another in a first direction (e.g., along the “y” direction). In some embodiments, the first and second fin structures **106**, **108** are referred to as fins of the semicon-

ductor substrate **102**, respectively. The first and second fin structures **106**, **108** are laterally spaced from one another along a second direction (e.g., along the “z” direction). In some embodiments, the first direction is orthogonal to the second direction. Each of the first and second fin structures **106**, **108** comprise at least a portion of an upper region of the semiconductor substrate **102**, respectively. The upper region of the semiconductor substrate **102** extends vertically from a lower region of the semiconductor substrate **102** along a third direction (e.g., along the “x” direction). For example, a first fin structure **106** comprises a first portion of the upper region of the semiconductor substrate **102** that extends vertically from the lower region of the semiconductor substrate **102**, and a second fin structure **108** comprises a second portion of the upper region of the semiconductor substrate **102** that extends vertically from the lower region of the semiconductor substrate **102**. Further, the upper region of the semiconductor substrate **102** extends continuously through an isolation structures **104**. The isolation structure **104** is configured to electrically isolate the first and second fin structures **106**, **108** from one another. In some embodiments, the isolation structure **104** may, for example, be or comprise silicon dioxide, silicon nitride, an oxy-nitride, some other dielectric material, or any combination of the foregoing. The first and/or second fin structures **106**, **108** may, for example, respectively be or comprise silicon, germanium, silicon-germanium, some other semiconductor material, or any combination of the foregoing. In some embodiments, the first and/or second fin structures **106**, **108** may be referred to as semiconductor fins.

A plurality of nanostructures **117** are respectively disposed over the first fin structures **106**. The nanostructures **117** are vertically stacked over one another (in the “x” direction). Further, in some embodiments, the nanostructures **117** may be vertically spaced from a corresponding underlying first fin structure **106** by a non-zero distance. In some embodiments, the plurality of nanostructures **117** comprise between two and twenty nanostructures. For example, the plurality of nanostructures **117** overlying a corresponding first fin structure **106** comprises three nanostructures. The plurality of nanostructures **117** may, for example, respectively be or comprise silicon, germanium, silicon-germanium, some other semiconductor material, or a combination of the foregoing. In further embodiments, the nanostructures **117** may be referred to as semiconductor nanostructures. In yet further embodiments, the nanostructures **117** may each comprise a same material as the semiconductor substrate **102**.

Pairs of first source/drain regions **110** are disposed on/over the plurality of first fin structures **106**. The first source/drain regions **110** are laterally spaced (in the “y” direction). For example, a pair of the first source/drain regions **110** may be disposed on opposite sides of a corresponding plurality of nanostructures **117**, such that the corresponding plurality of nanostructures **117** continuously laterally extend between the pair of first source/drain regions **110**. The first source/drain regions **110** may, for example, be or comprise silicon germanium, silicon-germanium, silicon carbide, some other semiconductor material, or any combination of the foregoing and/or may comprise a second doping type (e.g., n-type) opposite the first doping type (e.g., p-type). In further embodiments, the first source/drain regions **110** may, for example, be or comprise an epitaxial semiconductor material (e.g., a semiconductor material formed by an epitaxy, such as epitaxial silicon, epitaxial germanium, epitaxial silicon-germanium, epitaxial silicon carbide, etc.).

A first plurality of selectively-conductive channels (not shown) are disposed within the plurality of nanostructures 117. The first plurality of selectively-conductive channels each extend (in the “y” direction) between each pair of first source/drain regions 110. In some embodiments, each nano-structure 117 comprises a selectively-conductive channel extending between a corresponding pair of first source/drain regions 110.

Pairs of second source/drain regions 111 are disposed on/over the plurality of second fin structures 108. The second source/drain regions 111 are laterally spaced (in the “y” direction). The second source/drain regions 111 may, for example, be or comprise silicon germanium, silicon-germanium, silicon carbide, some other semiconductor material, or any combination of the foregoing and/or may comprise the second doping type (e.g., n-type) opposite the first doping type (e.g., p-type). In further embodiments, the second source/drain regions 111 may, for example, be or comprise an epitaxial semiconductor material (e.g., a semiconductor material formed by an epitaxy, such as epitaxial silicon, epitaxial germanium, epitaxial silicon-germanium, epitaxial silicon carbide, etc.).

A second plurality of selectively-conductive channels (not shown) are disposed within the second fin structures 108. The second plurality of selectively-conductive channels each extend (in the “y” direction) between each pair of second source/drain regions 111.

A gate electrode 114 overlies the first and second fin structures 106, 108 along the second direction (e.g., along the “z” direction). A gate dielectric layer 112 extends along the second direction (e.g., along the “z” direction) and is disposed between the gate electrode 114 and the first fin structures 106, the second fin structures 108, and the nanostructures 117. In further embodiments, the gate dielectric layer 112 continuously wraps around each of the nanostructures 117, such that the gate dielectric layer 112 continuously wraps around an outer perimeter of each nanostructure 117.

A plurality of nanosheet field effect transistors (NSFETs) 116 are defined at intersections between the gate electrode 114 and the first fin structures 106. In some embodiments, the NSFETs 116 respectively comprise a corresponding pair of first source/drain regions 110, segment(s) of the gate dielectric layer 112, a segment of the gate electrode 114, and a stack of nanostructures 117 overlying a corresponding first fin structure 106. In yet further embodiments, the NSFETs 116 are disposed within the peripheral device region 103 of the semiconductor substrate 102. In some embodiments, application of a suitable threshold voltage to the gate electrode 114 may result in the formation of a selectively-conductive channel (not shown) within each nanostructure 117 laterally between a corresponding pair of first source/drain regions 110. Upon application of the suitable threshold voltage, charge carrier (e.g., electrons) may travel through the selectively-conductive channels between the pair of first source/drain regions 110.

Further, it will be appreciated that in some instances, each NSFETs 116 may be also known as and/or referred to as, for example, a gate-all-around FET (GAAFET), a gate-surrounding FET, a multi-bridge channel FET (MBCFET), a nanowire FET, a nanoring FET, or the like.

A plurality of fin field effect transistors (finFETs) 118 are defined at intersections between the gate electrode 114 and the second fin structures 108. In some embodiments, the finFETs 118 comprise a pair of second source/drain regions 111, a segment of a second fin structure 108, a segment of the gate dielectric layer 112, and a segment of the gate electrode 114, respectively. In yet further embodiments, the

finFETs 118 are disposed within the center device region 105 of the semiconductor substrate 102. Thus, in some embodiments, the finFETs 118 are spaced laterally between the NSFETs 116. In further embodiments, application of a suitable threshold voltage to the gate electrode 114 may result in the formation of a selectively-conductive channel (not shown) within the second fin structure 108 laterally between the second source/drain regions 111. Upon application of the suitable threshold voltage, charge carrier (e.g., electrons) may travel through the selectively-conductive channel between the second source/drain regions 111.

The first fin structures 106 each comprise a first width w1, and the second fin structures 108 each comprise a second width w2 that is different from the first width w1. In further embodiments, each nanostructure 117 may comprise the first width w1. In some embodiments, the first width w1 is greater than the second width w2. In other embodiments, a width of each nanostructure 117 may be less than the first width w1 and greater than the second width w2. Because the first width w1 is greater than the second width w2, the NSFETs 116 may be configured to operate at higher currents than the finFETs 118. Thus, the first fin structures 106 may be configured for optimal performance of a first application and the second fin structures 108 may be configured for optimal performance of a second application, while minimizing an area of the semiconductor substrate 102 that the NSFETs 116 and the finFETs 118 occupy. In some embodiments, the first application may, for example, include acting as an access transistor for a static random access memory (SRAM) device, and the second application may, for example, include acting as a pull-up transistor for an SRAM device. In such embodiments, the first application may require operating at higher currents than the second application. This, in part, increases a performance of the NSFETs 116 and the finFETs 118 while increasing a number of semiconductor devices (e.g., NSFETs, finFETs, etc.) that may be disposed within and/or over the semiconductor substrate 102.

In addition, during a method for forming the IC 100, the NSFETs 116 and the finFETs 118 may be formed concurrently with one another, where a first masking layer is utilized to define the first fin structures 106 (and/or the nanostructures 117) and a second masking layer is utilized to define the second fin structures 108. The first masking layer is configured such that the first fin structures 106 each comprise the first width w1, and the second masking layer is configured such that the second fin structures 108 each comprise the second width w2 different from the first width w1. By forming the NSFETs 116 and the finFETs 118 concurrently with one another, time and cost associated with forming the IC 100 may be reduced. Additionally, by utilizing the first and second masking layers, the NSFETs 116 and the finFETs 118 may be configured to meet different design parameters. This, in part, increases a design flexibility and performance of the IC 100.

FIGS. 2A-2D illustrate various views of some embodiments of an IC 200 having a first fin structure laterally adjacent to a second fin structure on a same substrate, where the first fin structure has a greater width than the second fin structure. FIGS. 2A, 2C, and 2D illustrate cross-sectional views of some embodiments of the IC 200. FIG. 2B illustrates a top view of some embodiments of the IC 200 of FIG. 2A taken along the line A-A' of FIG. 2A. FIG. 2A illustrates a cross-sectional view of some embodiments of the IC 200 taken along the line A-A' of FIG. 2B. FIGS. 2C and 2D illustrate cross-sectional views of various embodiments of the IC 200 taken along the line B-B' of FIG. 2B.

An isolation structure **104** is disposed over the semiconductor substrate **102** and is spaced laterally between the first and second fin structures **106**, **108**. The isolation structure **104** is configured to electrically isolate the first fin structures **106** and the second fin structures **108** from one another. In some embodiments, the isolation structure **104** may be configured as a shallow trench isolation (STI) structure, a deep trench isolation (DTI) structure, or another suitable isolation structure. In further embodiments, the isolation structure **104** may, for example, be or comprise silicon dioxide, silicon nitride, silicon carbide, silicon oxy-nitride, silicon oxy-carbide, another suitable dielectric material, or any combination of the foregoing. In further embodiments, the isolation structure **104** may be a multi-layer structure, for example, comprising one or more liner layers. The first and second fin structures **106**, **108** continuously laterally extend in parallel with one another along a first direction (e.g., along the “y” direction, see FIG. 2B). In further embodiments, a plurality of nanostructures **117** vertically overlies each of the second fin structures **108**.

A gate electrode **114** continuously laterally extends along the first and second fin structures **106**, **108**, the nanostructures **117**, and the isolation structure **104**. In some embodiments, the gate electrode **114** may be a part of a plurality of gate electrodes **114** that each extend continuously over the semiconductor substrate **102** (e.g., see the top view of FIG. 2B). A plurality of NSFETs **116** are defined at intersections between the gate electrodes **114** and the first fin structures **106**, and a plurality of finFETs **118** are defined at intersections between the gate electrodes **114** and the second fin structures **108**. The plurality of gate electrodes **114** are each a continuous structure that are disposed between the first source/drain regions **110** and the second source/drain regions **111** of the NSFETs **116** and the finFETs **118**. Further, the plurality of gate electrodes **114** extend along a second direction (e.g., along the “z” direction) that is substantially orthogonal to the first direction. Gate dielectric layers **112** are disposed between the gate electrodes **114** and the first fin structures **106**, the second fin structures **108**, and the nanostructures **117**. Furthermore, the gate dielectric layers **112** are disposed between the gate electrodes **114** and the isolation structure **104**.

In further embodiments, the gate dielectric layer **112** may, for example, be or comprise an oxide, such as silicon dioxide, a high-k dielectric material (e.g., a dielectric material with a dielectric constant greater than 3.9), some other suitable dielectric material, or any combination of the foregoing. The gate dielectric layer **112** may, for example, be a multi-layered structure comprising one or more interfacial layers. In yet further embodiments, the gate electrodes **114** may, for example, respectively be or comprise polysilicon, a metal (e.g., tungsten), some other suitable conductive material, or any combination of the foregoing. For example, the gate electrodes may comprise a work function metal layer and a metal fill layer.

The gate electrodes **114** and the gate dielectric layers **112** comprise a plurality of gate structures **207** that overlie a corresponding selectively-conductive channel (not shown) of each of the NSFETs **116** and the finFETs **118**. For example, the selectively-conductive channel of each NSFET **116** is disposed laterally between a pair of the first source/drain regions **110**, and the selectively-conductive channel of each finFETs **118** is disposed laterally between a pair of the second source/drain regions **111**. In some embodiments, the selectively-conductive channel of the NSFETs **116** and the finFETs **118** comprise undoped regions of the semiconductor substrate **102** and/or the nanostructures **117**. Further, as

illustrated in the cross-sectional view of FIG. 2A, each of the nanostructures **117** may have a rectangular-like shaped profile. However, other shapes are amendable, for example, the nanostructures **117** may each have an ellipse-like shaped profile, a square-like shaped profile, a stadium-like shaped (e.g., geometric stadium shape) profile, a hexagonal-like shaped profile, a circle-like shaped profile, or another suitable shape.

Each of the gate structures **207** comprise a portion of the gate dielectric layer **112**, which may be referred to as a gate dielectric structure, and a portion of the gate electrode **114**, which may be referred to as a gate electrode structure. Each of the gate structures **207** disposed along a gate electrode **114** may be electrically coupled together by portions of the gate electrode **114** disposed between each of the gate structures **207**. The gate structures **207** are configured to control a conductivity of the selectively-conductive channel (e.g., switch between one or more conducting states and a non-conducting state) of a fin structure and/or nanostructure disposed between each pair of source/drain regions. In some embodiments, the gate structures **207** include a first gate structure **207a** and a second gate structure **207b**. For example, a first gate structure **207a** is configured to control the conductivity of a first selectively-conductive channel(s) of one of the NSFETs **116**. In another example, a second gate structure **207b** is configured to control the conductivity of a second selectively-conductive channel of one of the finFETs **118**.

An etch stop layer **208** is disposed along an upper surface of the gate electrodes **114**. In some embodiments, the etch stop layer **208** may, for example, be or comprise silicon nitride, silicon carbide, or another suitable dielectric material. Further, an inter-level dielectric (ILD) layer **210** overlies the etch stop layer **208**. In further embodiments, the ILD layer **210** may, for example, be or comprise a low-k dielectric material (e.g., a dielectric material having a dielectric constant less than about 3.9), an extreme low-k dielectric material, an oxide, such as silicon dioxide, another suitable dielectric material, or any combination of the foregoing. A conductive via **212** is disposed within the ILD layer **210** and the etch stop layer **208**. The conductive via **212** is electrically coupled to the gate electrode **114** and may be configured to facilitate application of a suitable bias voltage to the gate structures **207** of the NSFETs **116** and the finFETs **118**. In further embodiments, the conductive via **212** may, for example, be or comprise copper, aluminum, titanium nitride, tantalum nitride, tungsten, another suitable conductive material, or any combination of the foregoing.

In some embodiments, because the first fin structures **106** and/or the nanostructures **117** respectively comprise the first width **w1**, and the second fin structures **108** comprise the second width **w2**, the NSFETs **116** may be optimized for a first application while the finFETs **118** are optimized for a second application. A width of the fin structure (and/or nanostructure) may correspond to a maximum current that may travel through the selectively-conductive channel within the fin structure. As the width of the fin structure increases, the maximum current that may travel through the selectively-conductive channel of the fin structure, without damage to the fin structure, increases. In some embodiments, because the first width **w1** is greater than the second width **w2**, the NSFETs **116** are configured to conduct a higher current than the finFETs **118**. In such embodiments, the first application of the NSFETs **116** may operate at higher currents than the second application of the finFETs **118**. Further, the location and/or dimensions of the first fin structures **106**, the second fin structures **108**, and/or the

nanostructures **117** may be configured to minimize an area of the semiconductor substrate **102** they occupy, respectively. This, in part, increases a number of different fin structures that may be formed over and/or on the semiconductor substrate **102**, and increases design flexibility and performance of the different fin structures disposed over and/or on the semiconductor substrate **102**.

As illustrated in the top view of FIG. 2B, the first source/drain regions **110** are over and/or on the first fin structures **106** and are spaced laterally between the plurality of gate electrodes **114**. Further, the second source/drain regions **111** are disposed over and/or on the second fin structures **108** and are spaced laterally between the plurality of gate electrodes **114**.

FIG. 2C illustrates a cross-sectional view of some embodiments of the IC **200** taken along the line B-B' of FIG. 2B. As illustrated in the cross-sectional view of FIG. 2C, the first and second source/drain regions **110**, **111** may each have a rectangular-shaped profile. Further, an upper surface of the first and second source/drain regions **110**, **111** may be disposed above a top surface of the nanostructures **117** and/or above a top surface of the second fin structures **108**. In further embodiments, a width of the first source/drain regions **110** may be greater than the first width w_1 of the first fin structures **106** and/or the nanostructures **117**, and a width of the second source/drain regions **111** may be greater than the second width w_2 of the second fin structures **108**.

FIG. 2D illustrates a cross-sectional view of some alternative embodiments of the IC **200** taken along the line B-B' of FIG. 2B. As illustrated in the cross-sectional view of FIG. 2D, the first source/drain regions **110** may each have a hexagon-like shaped profile. In other embodiments, the first source/drain regions **110** each have a diamond-like shaped profile. Further, the second source/drain regions **111** may each have a hexagon-like shaped profile. In other embodiments, the second source/drain regions **111** may each have a diamond-like shaped profile. Further, adjacent second source/drain regions **111** may directly contact one another, such that adjacent finFETs **118** may share one or more common source/drain regions.

FIGS. 3A-3D illustrate various views of some embodiments of an IC **300** corresponding to some alternative embodiments of the IC **200** of FIGS. 2A-2D. FIGS. 3A and 3C illustrate cross-sectional views of the IC **300**, and FIG. 3B illustrates a top view of the IC **300**. FIG. 3A illustrates a cross-sectional view of some embodiments of the IC **300** taken along the line A-A' of FIG. 3B. FIG. 3C illustrates a cross-sectional view of some embodiments of the IC **300** taken along the line C-C' of FIG. 3B. In some embodiments, FIG. 3A illustrates a cross-sectional view of some alternative embodiments of the IC **200** of FIG. 2A. In further embodiments, FIG. 3B illustrates a top view of some alternative embodiments of the IC **200** of FIG. 2B.

The semiconductor substrate **102** comprises a top region **102a** that vertically extends from a bottom region **102b**. Further, the semiconductor substrate **102** has an upper surface **102us** that is disposed above a lower surface **102ls**. In some embodiments, the upper surface **102us** defines a topmost surface of the bottom region **102b** of the semiconductor substrate **102**. The first fin structures **106** extend continuously from the lower surface **102ls** of the semiconductor substrate **102** to a first point disposed above the upper surface **102us** of the semiconductor substrate **102**. In some embodiments, the lower surface **102ls** of the semiconductor substrate **102** contacts outer sidewalls of the first fin structures **106**, respectively. In some embodiments, a top surface **102ts** of the semiconductor substrate **102** defines a topmost

surface of the top region **102a** of the semiconductor substrate **102**. In some embodiments, the top surface **102ts** of the semiconductor substrate **102** is aligned with a substantially straight line **301**. Further, top surfaces of the second fin structures **106** are aligned with the substantially straight line **301**, respectively. In addition, in some embodiments, a top surface of each topmost nanostructure **117** overlying a corresponding first fin structure **106** is aligned with the substantially straight line **301**.

In some embodiments, the upper surface **102us** of the semiconductor substrate **102** within the center device region **105** contacts at least a sidewall of each of the second fin structures **108**. Further, the center device region **105** and the peripheral device regions **103** are spaced laterally between outermost device regions **302**. In further embodiments, the upper surface **102us** of the semiconductor substrate **102** disposed laterally within the outermost device regions **302** is vertically aligned with the upper surface **102us** of the semiconductor substrate **102** disposed laterally within the center device region **105**.

A first height h_1 of each of the NSFETs **116** is defined between the substantially straight line **301** and the lower surface **102ls** of the semiconductor substrate **102**. A second height h_2 of each of the finFETs **118** is defined between the substantially straight line **301** and the upper surface **102us** of the semiconductor substrate **102**. In some embodiments, the first height h_1 is greater than the second height h_2 . In some embodiments, a difference between the first height h_1 and the second height h_2 (e.g., h_1-h_2) may be within a range of about 5 to 30 nanometers. In some embodiments, if the difference between the first height h_1 and the second height h_2 is relatively high (e.g., greater than about 30 nanometers), then a height of the first fin structures **106** and/or the nanostructures **117** may be too small, thereby impeding formation of a selectively-conductive channel in the first fin structures **106** and/or the nanostructures **117**. In further embodiments, during fabrication of the IC **300**, one or more masking layers may be disposed over the semiconductor substrate **102** while utilizing one or more etch processes to define the first fin structures **106** (and/or nanostructures **117**) and the second fin structures **108**. For example, the first fin structures **106** (and/or nanostructures **117**) and the second fin structures **108** may be defined by a single etch process according to a first masking layer and a second masking layer, such that the first fin structures **106** (and/or nanostructures **117**) and the second fin structures **108** are formed concurrently. The first masking layer overlies the first fin structures **106**, and the second masking layer overlies the second fin structures **108**. In some embodiments, the first masking layer comprises a first material (e.g., amorphous silicon), the second masking layer comprises a second material (e.g., silicon nitride) different from the first material, and a width of the first masking layer is greater than a width of the second masking layer. Due to the difference in widths of the first and second masking layers and/or the difference in materials of the first and second masking layers the peripheral device regions **103** of the semiconductor substrate **102** will be etched more quickly than the center device region **105** and/or the outermost device regions **302** of the semiconductor substrate **102**. This, in part, causes the lower surface **102ls** of the semiconductor substrate **102** to be disposed below the upper surface **102us** of the semiconductor substrate **102** such that the first height h_1 is greater than the second height h_2 .

As illustrated in the top view of FIG. 3B and the cross-sectional view of FIG. 3C, the first and second fin structures **106**, **108** are spaced laterally between a plurality of third fin

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structures 304. A plurality of third source/drain regions 306 are disposed within and/or on each of the third fin structures 304 on opposing sides of an overlying gate electrode 114. In some embodiments, the third source/drain regions 306 comprise the second doping type (e.g., n-type). Further, a selectively-conductive channel is defined within each of the third fin structures 304 and is spaced laterally between the third source/drain regions 306. The third fin structures 304 respectively have a third width w_3 and comprise a single segment of the top region 102a of the semiconductor substrate 102 that comprises the first semiconductor material (e.g., silicon). A plurality of second finFETs 308 are defined at intersections between the third fin structures 304 and the gate electrodes 114. In further embodiments, the third width w_3 is less than the second width w_2 of the second fin structures 108, such that the finFETs 118 are configured for a higher current during operation than the second finFETs 308. In yet further embodiments, the second finFETs 308 are configured for a same application as the finFETs 118 (e.g., as a pull-up transistor for an SRAM device). In other embodiments, the second width w_2 is equal to the third width w_3 .

Further, in some embodiments, the first and second fin structures 106, 108 are each laterally offset from an isolation region 310 by a non-zero distance. Thus, the first and second fin structures 106, 108 may extend continuously in the first direction (e.g., along the "y" direction) across a first region 312 and may extend continuously in the first direction across a second region 314. In such embodiments, the first region 312 is on a first side of the isolation region 310 and the second region 314 is disposed on a second side of the isolation region 310, such that the first and second fin structures 106, 108 are discontinuous across the isolation region 310. This improves isolation between the NSFETs 116 and the finFETs 118 disposed laterally within the first region 312 and the NSFETs 116 and the finFETs 118 disposed laterally within the second region 314, thereby increasing a performance of the IC 300. In further embodiments, the third fin structures 304 extend continuously in the first direction from the first region 312 to the second region 314, such that the third fin structures 304 are not discontinuous across the isolation region 310. At least one of the gate electrodes 114 extends continuously in the second direction (e.g., along the "z" direction) across the isolation region 310.

As illustrated in the cross-sectional view of FIG. 3C, an upper surface of the 102_{us} of the semiconductor substrate 102 within the isolation region 310 is vertically disposed above a lower surface 102_{ls} of the semiconductor substrate 102 within the isolation region 310. Further, within the isolation region 310, the first height h_1 is defined between the substantially straight line 301 and the lower surface 102_{ls} of the semiconductor substrate 102, and the second height h_2 is defined between the substantially straight line 301 and the upper surface 102_{us} of the semiconductor substrate 102. In some embodiments, the first height h_1 is greater than the second height h_2 . In further embodiments, within the isolation region 310, a difference between the first height h_1 and the second height h_2 (e.g., h_1-h_2) may be within a range of about 5 to 30 nanometers.

FIG. 3D illustrates a top view of some alternative embodiments of the IC 300 taken along the line G-G' of FIG. 3A. As illustrated in FIG. 3D, a width of each of the first fin structures 106 may discretely decrease. For example, the first fin structures 106 may each have a first width W_1 and a second width W_2 that is less than the first width W_1 . Further, nanostructures (117 of FIG. 3A) may have a same

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layout as the first fin structures 106, such that a width of each nanostructure (117 of FIG. 3A) may discretely decrease (not shown). Further, a width of each of the second and/or third fin structures 108, 304 may discretely decrease (not shown).

FIGS. 4A-4B illustrate various cross-sectional views of some embodiments of an IC 400 corresponding to some alternative embodiments of the IC 300 of FIGS. 3A-3C. FIG. 4A illustrates a cross-sectional view of some other embodiments of the IC of FIG. 3B taken along the line A-A'. FIG. 4B illustrates a cross-sectional view of some other embodiments of the IC of FIG. 3B taken along the line C-C'.

As illustrated in the cross-sectional views of FIGS. 4A and 4B, a plurality of first nanostructures 117 directly overlie each of the first fin structures 106, a plurality of second nanostructures 401 directly overlie each of the second fin structures 108, and a plurality of third nanostructures 403 directly overlie each of the third fin structures 304. Thus, a plurality of first NSFETs 402 are defined between intersections of the gate electrode 114 and the first fin structures 106, a plurality of second NSFETs 404 are defined between intersections of the gate electrode 114 and the second fin structures 108, and a plurality of third NSFETs 406 are defined between intersections of the gate electrode 114 and the third fin structures 304. The first NSFETs 402 are disposed laterally within the peripheral device region 103, the second NSFETs 404 are disposed laterally within the center device region 105, and the third NSFETs 406 are disposed laterally within the outermost device region 302.

In further embodiments, a first width w_1 of the first fin structures 106 and the first nanostructures 117 is greater than a second width w_2 of the second fin structures 108 and the second nanostructures 401. In some embodiments, the first width w_1 of the first fin structures 106 and the first nanostructures 117 continuously increases from the substantially straight line 301 to the lower surface 102_{ls} of the semiconductor substrate 102. In yet further embodiments, the second width w_2 of the second fin structures 108 and the second nanostructures 401 continuously increases from the substantially straight line 301 to the upper surface 102_{us} of the semiconductor substrate 102. In various embodiments, the upper surface 102_{us} and the lower surface 102_{ls} of the semiconductor substrate 102 may respectively be curved, concave, and/or U-shaped.

In yet further embodiments, a third width w_3 of the third fin structures 304 and the third nanostructures 403 may continuously increase from the substantially straight line 301 to the upper surface 102_{us} of the semiconductor substrate 102. In other embodiments, the third width w_3 of the third fin structures 304 may be greater than the first width w_1 and/or the second width w_2 . A plurality of protrusions 410 may be disposed within the isolation region 310 and may be spaced laterally between the third fin structures 304. The protrusions 410 are segments of the semiconductor substrate 102 that directly underlie the gate electrode 114. In further embodiments, during fabrication of the IC 400, the protrusions 410 may be remnants of the second fin structures 108 that remain in the center device region 105 after performing an etch process to define the isolation region 310.

FIGS. 5A-5C illustrated various views of some embodiments of an IC 500 corresponding to some alternative embodiments of the IC 300 of FIGS. 3A-3C. FIG. 5A illustrates a cross-sectional view of some embodiments of the IC 500, and FIGS. 5B-5C illustrate top views of some embodiments of the IC 500. FIG. 5A illustrates a cross-sectional view of some embodiments of the IC 500 taken along the line A-A' of FIG. 5B or FIG. 5C. In some embodiments, the IC 500 includes a first plurality of

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NSFETs **116**, a first plurality of finFETs **118**, a second plurality of finFETs **308**, and a second plurality of NSFETs **510** disposed laterally adjacent to one another.

The semiconductor substrate **102** comprises a first device region **502** laterally adjacent to a second device region **504**, and a third device region **506** laterally adjacent to the second device region **504**. In some embodiments, the first and second fin structures **106**, **108** are disposed laterally within the first device region **502** and/or may be configured as the first and second fin structures **106**, **108** of FIGS. **1**, **2A-2D**, **3A-3D**, and/or **4A-4B**. Thus, the first plurality of NSFETs **116** and the first plurality of finFETs **118** are disposed laterally within the first device region **502**. In further embodiments, the third fin structures **304** are disposed laterally within the second device region **504** and/or may be configured as the third fin structures **304** of FIGS. **3A-3D** and/or **4A-4B**. Thus, the second plurality of finFETs **308** are disposed laterally within the second device region **504**.

In yet further embodiments, a plurality of fourth fin structures **508** is disposed laterally within the third device region **506**. The fourth fin structures **508** may be configured as the first fin structures **106**, such that a second plurality of nanostructures **512** is disposed over each of the fourth fin structures **508**. In some embodiments, the fourth fin structures **508** and the second plurality of nanostructures **512** each comprise a fourth width w_4 , where the fourth width w_4 may be greater than the first width w_1 and/or the first width w_1 may be equal to the fourth width w_4 . In further embodiments, a second plurality of NSFETs **510** are defined between intersections of the fourth fin structures **508** and the gate electrodes **114**. In some embodiments, the second plurality of NSFETs **510** may be configured as the first plurality of NSFETs **116** of FIGS. **1**, **2A-2D**, **3A-3D**, and/or **4A-4B**.

In some embodiments, the first and second fin structures **106**, **108** continuously laterally extend along the first direction (e.g., along the “y” direction) in an unbroken path. In contrast, and as illustrated in the top view of FIG. **5C**, in further embodiments, the first and second fin structures **106**, **108** are laterally offset from the isolation region **310** as illustrated and described in FIGS. **3B** and **3D**.

FIG. **6A** illustrates a cross-sectional view **600a** of some embodiments of a NSFET **116** of FIGS. **5A-5C**. In some embodiments, the cross-sectional view **600a** of FIG. **6A** is taken along the line D-D' of the top view of FIG. **5B**.

As illustrated in FIG. **6A**, the nanostructures **117** are vertically stacked over the first fin structure **106**. Further, the nanostructures **117** each laterally extend between a pair of the first source/drain regions **110**. Further, the gate electrode **114** is disposed vertically between each of the nanostructures **117**. The gate dielectric layer **112** surrounds segments of the gate electrode **114** that are disposed vertically between each of the nanostructures **117**. Thus, the gate dielectric layer **112** is configured to separate the gate electrode **114** from the nanostructures **117**. A first sidewall spacer structure **602** is disposed on opposing sidewalls of the gate electrode **114** and is disposed between the gate electrode **114** and the first source/drain regions **110**. In addition, a plurality of second sidewall spacers **604** are disposed along sidewalls of the gate dielectric layer **112** and are configured to laterally separate the gate electrode **114** from the first source/drain regions **110**. In some embodiments, the first sidewall spacer structure **602** may, for example, be or comprise an oxide (e.g., silicon dioxide), silicon nitride, silicon oxynitride, silicon carbide, some other suitable dielectric material, or any combination of the foregoing. In further embodiments, the plurality of second sidewall spacers **604** may, for

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example, respectively be or comprise an oxide (e.g., silicon dioxide), silicon nitride, silicon carbide, silicon oxynitride, silicon oxycarbide, another suitable dielectric material, or any combination of the foregoing.

Although the cross-sectional view **600a** of FIG. **6A** is described in relation to one of the NSFETs **116** of FIGS. **5A-5C**, it will be appreciated that any NSFET (e.g., each of the first and/or second plurality of NSFETs **116**, **510**) of FIGS. **1**, **2A-2D**, **3A-3D**, **4A-4B**, and/or **5A-5C** may be illustrated and/or described as the NSFET **116** of FIG. **6A**.

FIG. **6B** illustrates a cross-sectional view **600b** of some embodiments of a finFET **118** of FIGS. **5A-5C**. In some embodiments, the cross-sectional view **600b** of FIG. **6B** is taken along the line E-E' of the top view of FIG. **5B**.

As illustrated in FIG. **6B**, the gate electrode **114** overlies the second fin structure **108**. The gate dielectric layer **112** extends along opposing sidewalls and a bottom surface of a segment of the gate electrode **114** that overlies the second fin structure **108**. The first sidewall spacer structure **602** is disposed on opposing sides of the gate electrode **114**.

Although the cross-sectional view **600b** of FIG. **6B** is described in relation to one of the finFETs **118** of FIGS. **5A-5C**, it will be appreciated that any finFET (e.g., each of the first and/or second plurality of finFETs **118**, **308**) of FIGS. **1**, **2A-2D**, **3A-3D**, **4A-4B**, and/or **5A-5C** may be illustrated and/or described as the finFET **118** of FIG. **6B**.

FIGS. **7** through **28A-28B** illustrate various views of some embodiments of a method for forming an IC comprising a plurality of fin structures having different widths that are laterally adjacent to one another on a same substrate according to the present disclosure. Although the various views shown in FIGS. **7** through **28A-28B** are described with reference to a method, it will be appreciated that the structures shown in FIGS. **7** through **28A-28B** are not limited to the method but rather may stand alone separate of the method. Although FIGS. **7** through **28A-28B** are described as a series of acts, it will be appreciated that these acts are not limited in that the order of the acts can be altered in other embodiments, and the methods disclosed are also applicable to other structures. In other embodiments, some acts that are illustrated and/or described may be omitted in whole or in part.

As shown in the cross-sectional view **700** of FIG. **7**, a stack of semiconductor layers **708** is formed over a base semiconductor structure **702** (e.g., a semiconductor wafer, semiconductor substrate, etc.). The base semiconductor structure **702** is or comprises silicon, or some other suitable material. The stack of semiconductor layers **708** comprises alternating layers of a first semiconductor layer **704** and a second semiconductor layer **706**. In some embodiments, the first semiconductor layer **704** comprises a different material than the second semiconductor layer **706**. In further embodiments, the first semiconductor layer **704** comprises silicon-germanium and/or the second semiconductor layer **706** comprises silicon. Thus, in some embodiments, the second semiconductor layers **706** may comprise a same material as the base semiconductor structure **702**.

In some embodiments, a process for forming the stack of semiconductor layers **708** comprises epitaxially forming the first semiconductor layers **704** and the second semiconductor layers **706**. For example, a first one of the first semiconductor layers **704** is grown on the base semiconductor structure **702** by a first epitaxial process, such as, vapor-phase epitaxy (VPE), liquid-phase epitaxy (LPE), molecular-beam epitaxy (MBE), some other epitaxial process, or a combination of the foregoing. Thereafter, a first one of the second semiconductor layers **706** is grown on the first one of

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the first semiconductor layers **704** by a second epitaxial process, such as, VPE, LPE, MBE, some other epitaxial process, or a combination of the foregoing. The first epitaxial process and the second epitaxial process are repeated in an alternative manner until the stack of semiconductor layers **708** is formed.

As shown in the cross-sectional view **800** of FIG. **8**, a patterning process is performed on the stack of semiconductor layers **708**, thereby defining a plurality of openings **802**. In some embodiments, the patterning process includes: forming a first patterned masking layer (not shown) (e.g., positive/negative photoresist, hardmask, etc.) over the stack of semiconductor layers **708**; performing a first etching process on the stack of semiconductor layers **708** according to the first patterned masking layer, thereby defining the openings **802**; and performing a removal process to remove the first patterned masking layer. In further embodiments, the first patterned masking layer may be formed by forming a masking layer (not shown) over the stack of semiconductor layers **708**; exposing the masking layer to a pattern (e.g., via a lithography process, such as photolithography, extreme ultraviolet lithography, or the like), and developing the masking layer to form the first patterned masking layer. In some embodiments, the first etching process may be a dry etching process, a wet etching process, a reactive ion etching (RIE) process, some other etching process, or any combination of the foregoing. In yet further embodiments, the first etching process may remove at least a portion of the base semiconductor structure **702**, such that an upper surface of the base semiconductor structure **702** is disposed vertically below the stack of semiconductor layers **708** (not shown).

As shown in the cross-sectional view **900** of FIG. **9**, epitaxial semiconductor structures **902** are formed in over the base semiconductor structure **702** and on sides of the stack of semiconductor layers **708**, thereby defining a semiconductor substrate **102**. In some embodiments, the epitaxial semiconductor structures **902** comprise an epitaxial semiconductor material (e.g., a semiconductor material formed by an epitaxial process, such as epitaxial silicon, epitaxial germanium, epitaxial silicon-germanium, etc.). In some embodiments, the epitaxial semiconductor structures **902** each comprise a same material as the base semiconductor structure **702**. In further embodiments, the base semiconductor structure **702** has a first lattice orientation (e.g., a first crystal plane of the semiconductor material of the base semiconductor structure **702**). In some embodiments, the first lattice orientation may be, for example, (100), (110), or the like. In yet further embodiments, the epitaxial semiconductor structures **902** may have a second lattice orientation (e.g., (100), (110), or the like) that may, for example, be different from the first lattice orientation. In yet further embodiments, the first lattice orientation may be the same as the second lattice orientation. Further, in various embodiments, a bottom surface of the epitaxial semiconductor structures **902** may be disposed below a top surface of the base semiconductor structure **702**.

In some embodiments, a process for forming the epitaxial semiconductor structures **902** may, for example, include performing VPE, LPE, MBE, some other epitaxial process, or a combination of the foregoing to define the epitaxial semiconductor structures **902** over the base semiconductor structure **702** and/or on sides of the stack of semiconductor layers **708**. Subsequently, in some embodiments, a planarization process (e.g., chemical-mechanical polishing (CMP)) is then performed on the epitaxial semiconductor structures **902** to co-planarize an uppermost surface of the epitaxial semiconductor structures **902** and an uppermost

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surface of the stack of semiconductor layers **708**. In some embodiments, the semiconductor substrate **102** comprises the stack of semiconductor layers **708**, the base semiconductor structure **702**, and the epitaxial semiconductor structures **902**. In further embodiments, a process for forming the semiconductor substrate **102** includes performing the processing steps illustrated and/or described in FIGS. **7-9**. Further, the semiconductor substrate **102** comprises a first device region **502** laterally adjacent to a second device region **504**, and a third device region **506** laterally adjacent to the second device region **504**.

As shown in the cross-sectional view **1000** of FIG. **10**, a masking structure **1002** is formed over the semiconductor substrate **102**. In some embodiments, the masking structure **1002** continuously laterally extends over the first, second, and third device regions **502**, **504**, **506** along an unbroken path. In further embodiments, the masking structure **1002** comprises a single layer or a stack of layers, for example, the masking structure **1002** may comprise a lower masking layer **1004**, a middle masking layer **1006**, and/or an upper masking layer **1008**. In yet further embodiments, the masking structure **1002** may be configured as a hard mask. Further, multiple mandrel structures **1010** are formed over the masking structure **1002**. In some embodiments, a process for forming the mandrel structures **1010** may include: depositing (e.g., by physical vapor deposition (PVD), chemical vapor deposition (CVD), or another suitable growth or deposition process) a mandrel layer (not shown) over the masking structure **1002**; patterning the mandrel layer according to a masking layer (not shown), thereby defining the multiple mandrel structures **1010**; and performing a removal process to remove the masking layer. The patterning process may include performing a wet etch process and/or a dry etch process. In some embodiments, the mandrel structures **1010** may, for example, respectively be or comprise amorphous silicon, or another suitable material.

In some embodiments, the lower masking layer **1004** may, for example, be or comprise an oxide, such as silicon dioxide, or another suitable dielectric material. In further embodiments, the middle masking layer **1006** may, for example, be or comprise silicon nitride, silicon carbide, or another suitable dielectric material. In yet further embodiments, the upper masking layer **1008** may, for example, be or comprise an oxide, such as silicon dioxide, or another suitable dielectric material. In various embodiments, the lower masking layer **1004**, the middle masking layer **1006**, and/or the upper masking layer **1008** may, for example, respectively be formed by PVD, CVD, atomic layer deposition (ALD), an oxidation process, or another suitable deposition or growth process. In further embodiments, a thickness of the lower masking layer **1004** is less than a thickness of the middle masking layer **1006**, and a thickness of the upper masking layer **1008** is greater than the thickness of the middle masking layer **1006**.

As shown in the cross-sectional view **1100** of FIG. **11**, a sidewall spacer layer **1102** is deposited over the masking structure **1002** and the mandrel structures **1010**. In some embodiments, the sidewall spacer layer **1102** is formed by, for example, a CVD process, a PVD process, an ALD process, and/or another suitable growth or deposition process. In further embodiments, the sidewall spacer layer **1102** may, for example, be or comprise silicon nitride, silicon carbide, or another suitable dielectric material.

As shown in the cross-sectional view **1200** of FIG. **12**, the sidewall spacer layer (**1102** of FIG. **11**) is patterned, thereby defining sidewall spacer structures **1202** that laterally enclose each mandrel structure **1010**. This, in some embodi-

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ments, defines first upper masking structures **1204** disposed laterally within the third device region **506**. The first upper masking structures **1204** may, for example, each include a mandrel structure **1010** and a sidewall spacer structure **1202** that laterally encloses the mandrel structure **1010**. In some embodiments, the patterning process includes performing an anisotropic etch (e.g., a vertical etch) on the sidewall spacer layer (**1102** of FIG. **11**) to remove lateral stretches of the sidewall spacer layer (**1102** of FIG. **11**), thereby resulting in the sidewall spacer structures **1202** disposed along sidewalls of the mandrel structures **1010**.

As shown in the cross-sectional view **1300** of FIG. **13**, a dielectric capping layer **1302** is deposited over the masking structure **1002**, the mandrel structures **1010**, and the sidewall spacer structures **1202**. In some embodiments, the dielectric capping layer **1302** is configured to protect underlying layers and/or may, for example, be or comprise an oxide, such as silicon dioxide, or another suitable dielectric material. In further embodiments, the dielectric capping layer **1302** may, for example, be deposited by a CVD process, a PVD process, an ALD process, or another suitable deposition or growth process. In some embodiments, the dielectric capping layer **1302** is formed solely by a PVD process. In yet further embodiments, the dielectric capping layer **1302** may continuously laterally extend along the first, second, and third device regions **502**, **504**, **506** along an unbroken path.

As shown in the cross-sectional view **1400a** of FIG. **14A**, a first protection layer **1402** is formed to cover a first set of the mandrel structures **1010** and corresponding sidewall spacer structures **1202** while exposing a second set of the mandrel structures **1010** and corresponding sidewall spacer structures **1202**. In some embodiments, a process for forming the first protection layer **1402** includes: depositing (e.g., by CVD, PVD, ALD, spin-on or coating techniques, etc.) a first protection film; and patterning the first protection film according to a photolithographic mask (not shown), thereby defining the first protection layer **1402** and exposing the second set of mandrel structures **1010**. Subsequently, the dielectric capping layer **1302** is patterned according to the first protection layer **1402** such that the dielectric capping layer **1302** is removed from over the second set of the mandrel structures **1010** while the dielectric capping layer **1302** remains over the first set of mandrel structures **1010**. In various embodiments, the dielectric capping layer **1302** is remove from segments of the first device region **502**. In further embodiments, the first protection layer **1402** may comprise a single layer and/or may comprise a multi-layer structure. For example, the multi-layer stack may be or comprise one or more layers such as a first layer, a second layer, and a third layer (not shown). The second layer may be disposed between the first and third layers. In yet further embodiments, the first layer may comprise an allyl monomer, the second layer may comprise diazonaphthoquinone (DNQ), and the third layer may comprise an off-stoichiometry thiol-ene polymer.

Cross-sectional view **1400b** of FIG. **14B** illustrates various alternative embodiments of the processing step of the cross-sectional view **1400a** of FIG. **14A**, in which the patterning process utilized to remove segments of the dielectric capping layer **1302** may over-etch and remove portions of the upper masking layer **1008**. In some embodiments, the portions of the upper masking layer **1008** removed by the patterning process are laterally offset from the second set of mandrel structures **1010**. Thus, an upper surface **1008_{us}** of the upper masking layer **1008** is disposed vertically below a top surface **1008_s** of the upper masking layer **1008**. As

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illustrated in FIG. **15**, the upper surface **1008_{us}** of the upper masking layer **1008** may be illustrated by a dashed line in subsequent processing steps.

As shown in cross-sectional view **1500** of FIG. **15**, a removal process is performed to remove the first protection layer (**1402** of FIG. **14A**). In some embodiments, after removing the first protection layer (**1402** of FIG. **14A**), the sidewall spacer structures **1202** laterally enclosing the second set of mandrel structures **1010** are stripped, thereby exposing outer sidewalls of each of the mandrel structures **1010** in the second set. This, in part, defines second upper masking structures **1502** within the first device region **502**. The second upper masking structures **1502** each comprise a mandrel structure **1010**. In some embodiments, stripping the sidewall spacer structures **1202** that laterally enclose the second set of mandrel structures **1010** comprises performing one or more etch processes that may include a wet etch process and/or a dry etch process. The dielectric capping layer **1302** is configured to protect the sidewall spacer structures **1202** that laterally enclose the first set of mandrel structures **1010** such that they are not removed during the one or more etch processes. In some embodiments, the one or more etch processes includes exposing layers overlying the semiconductor substrate **102** to one or more etchants, such as, for example, phosphoric acid (e.g., H_3PO_4). In yet further embodiments, the one or more etch processes may etch the sidewall spacer structures **1202** more quickly than the dielectric capping layer **1302**, the upper masking layer **1008**, and/or the mandrel structures **1010**.

As shown in cross-sectional view **1600** of FIG. **16**, a removal process is performed on the structure of FIG. **15**, such that the dielectric capping layer (**1302** of FIG. **15**) is removed. In some embodiments, the removal process includes performing a wet etch process and/or a dry etch process.

As shown in cross-sectional view **1700** of FIG. **17**, a second protection layer **1702** is formed to cover the first and second upper masking structures **1204**, **1502**. In some embodiments, a process for forming the second protection layer **1702** includes: depositing (e.g., by CVD, PVD, ALD, spin-on or coating techniques, etc.) a second protection film; and patterning the second protection film according to a photolithographic mask (not shown), thereby defining the second protection layer **1702**. Subsequently, a patterning process is performed on the mandrel structures **1010** that are laterally offset from the second protection layer **1702** to remove them from the first and second device regions **502**, **504**, thereby defining third upper masking structures **1704** within the first and second device regions **502**, **504**. In some embodiments, the patterning process includes performing a wet etch process and/or dry etch process that etches the mandrel structures **1010** more quickly than the sidewall spacer structures **1202**. Thus, in various embodiments, the third upper masking structures **1704** each comprise the sidewall spacer structures **1202** disposed within the first and second device regions **502**, **504**. After performing the patterning process, a removal process is performed to remove the second protection layer **1702** (not shown).

In further embodiments, the second protection layer **1702** may comprise a single layer and/or may comprise a multi-layer structure. For example, the multi-layer stack may be or comprise one or more layers such as a first layer, a second layer, and a third layer (not shown). The second layer may be disposed between the first and third layers. In yet further embodiments, the first layer may comprise an allyl mono-

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mer, the second layer may comprise diazonaphthoquinone (DNQ), and the third layer may comprise an off-stoichiometry thiol-ene polymer.

As shown in cross-sectional view **1800** of FIG. **18**, a patterning process is performed on the second device region **504** to remove a first set of the third upper masking structures **1704** while leaving a second set of the third upper masking structures **1704** in place. In some embodiments, the patterning process may include performing a dry etch process according to a masking layer (not shown). Subsequently, an upper dielectric layer **1802** is formed over the masking structure **1002** and laterally across the first, second, and third device regions **502**, **504**, **506**. In some embodiments, the upper dielectric layer **1802** may, for example, be deposited by a CVD process, a PVD process, an ALD process, or another suitable growth or deposition process. In further embodiments, the upper dielectric layer **1802** may, for example, be or comprise silicon nitride, silicon carbide, or another suitable dielectric material and/or may be formed to a thickness of about 3 nanometers, or within a range of about 2.8 to 3.2 nanometers.

As shown in cross-sectional view **1900** of FIG. **19**, a patterning process is performed on the structure of FIG. **18** according to the upper dielectric layer **1802**, the first upper masking structures **1204**, the second upper masking structures **1502**, and the third upper masking structures **1704**. This, in part, forms a first stack of masking layers **1902**, a second stack of masking layers **1904**, a third stack of masking layers **1906**, and a fourth stack of masking layers **1908**. In some embodiments, the patterning process includes performing a dry etch process, a wet etch process, another suitable etch process, or any combination of the foregoing. Further, the first and fourth stack of masking layers **1902**, **1908** may respectively comprise the mandrel structure **1010** and the masking structure **1002**. Furthermore, the second and third stack of masking layers **1904**, **1906** may respectively comprise the masking structure **1002**.

Additionally, due to a difference in layers and/or structures of the first upper masking structures **1204**, the second upper masking structures **1502**, and/or the third upper masking structures **1704** the first, second, third, and fourth stack of masking layers **1902-1908** may have different widths from one another. For example, the first stack of masking layers **1902** has a first width **w1**, the second stack of masking layers **1904** has a second width **w2**, the third stack of masking layers **1906** has a third width **w3**, and the fourth stack of masking layers has a fourth width. In further embodiments, the first width **w1** is greater than the second width **w2**, and the second width **w2** is greater than the third width **w3**. In yet further embodiments, the fourth width **w4** is greater than or equal to the first width **w1**.

As shown in cross-sectional view **2000** of FIG. **20**, a bottom anti-reflectant coating (BARC) layer **2002** is formed over the semiconductor substrate **102**. In some embodiments, a process for forming the BARC layer **2002** may include: depositing a BARC film over the semiconductor substrate **102** and laterally between the stacks of masking layers **1902-1908**; and performing an etch back process on the BARC film, thereby forming the BARC layer **2002**. In some embodiments, the etch back process is performed in such a manner that a top surface of the upper masking layer **1008** is disposed vertically above a top surface of the BARC layer **2002**. In yet further embodiments, after performing the etch back process, a removal process (e.g., a wet etch process and/or a dry etch process) may be performed to remove the mandrel structures (**1010** of FIG. **19**) of the first and fourth stacks of masking layers **1902**, **1908**. In other

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embodiments, at least a portion of the mandrel structures (**1010** of FIG. **19**) may be removed by the etch back process.

As shown in cross-sectional view **2100** of FIG. **21**, a patterning process is performed on the semiconductor substrate **102** and the stack of semiconductor layers **708** according to the stacks of masking layers **1902-1908**. This, in turn, forms first fin structures **106** and second fin structures **108** within the first device region **502**, third fin structures **304** within the second device region **504**, and fourth fin structures **508** within the third device region **506** as illustrated in FIG. **21**. Further, the patterning process may remove the upper masking layer (**1008** of FIG. **20**). Furthermore, by virtue of the semiconductor substrate **102** being patterned according to the stacks of masking layers **1902-1908**, each of the aforementioned fin structures may have a thickness corresponding to an overlying stack of masking layers. For example, the first stack of masking layers **1902** overlies the first fin structures **106** and the first fin structures **106** are formed such that they have the first width **w1**. Thus, the patterning process is performed in such a manner that the first fin structures **106** each have the first width **w1**, the second fin structures **108** each have the second width **w2**, the third fin structures **304** each have the third width **w3**, and the fourth fin structures **508** each have the fourth width **w4**.

After performing the patterning process, the semiconductor substrate **102** comprises a top region **102a** that vertically extends from a bottom region **102b**. Further, the semiconductor substrate **102** has an upper surface **102us** that is disposed above a lower surface **102ls**. In some embodiments, the upper surface **102us** defines a topmost surface of the bottom region **102b** of the semiconductor substrate **102**. In some embodiments, a top surface **102ts** of the semiconductor substrate **102** defines a topmost surface of the top region **102a** of the semiconductor substrate **102**. In some embodiments, the top surface **102ts** of the semiconductor substrate **102** is aligned with a substantially straight line **301**.

A first height **h1** of the first fin structures **106** and the stack of semiconductor layers **708** is defined between the lower surface **102ls** and the substantially straight line **301**. Further, a second height **h2** of the second and third fin structures **108**, **304** is defined between the upper surface **102us** and the substantially straight line **301**. A third height **h3** of the fourth fin structures **508** and the stack of semiconductor layers **708** is defined between the lower surface **102ls** within the third device region **506** and the substantially straight line **301**. By virtue of the relatively large widths of the first and/or fourth fin structures **106**, **508** (e.g., greater than widths of the second and/or third fin structures **108**, **304**), the first and third heights **h1**, **h3** are respectively greater than the second height **h2**. This may be because of a removal of dielectric material (e.g., over-etching of the upper masking layer **1008** of FIG. **20**) in regions laterally adjacent to the first and/or fourth fin structures **106**, **508** (e.g., see FIGS. **30-34**). In some embodiments, in which the fourth width **w4** of the fourth fin structures **508** is less than the first width **w1** of the first fin structures **106**, the third height **h3** is less than the first height **h1**. In yet further embodiments, in which the fourth width **w4** of the fourth fin structures **508** is greater than the first width **w1** of the first fin structures **106**, the third height **h3** is greater than the first height **h1**. In some embodiments, a difference between the first height **h1** and the second height **h2** (e.g., **h1-h2**) may be within a range of about 5 to 30 nanometers.

As shown in the various views of FIGS. **22A-22C**, a removal process is performed to remove the lower masking layer (**1004** of FIG. **21**) and the middle masking layer (**1006** of FIG. **21**). Further, as illustrated in FIGS. **22B-22C**, an

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isolation patterning process is performed on the structure of FIG. 21 to define an isolation region 310 laterally within the first device region 502. Thereafter, an isolation structure 104 is formed over the semiconductor substrate 102. FIG. 22A illustrates some embodiments of a cross-sectional view 2200a taken along the line A-A' of top view 2200b of FIG. 22B. Further, FIG. 22C illustrates some embodiments of a cross-sectional view 2200c taken along the line B-B' of the top view 2200b of FIG. 22B.

In some embodiments, the removal process includes performing a wet etch process and/or a dry etch process to remove the lower masking layer (1004 of FIG. 21) and the middle masking layer (1006 of FIG. 21). Further, the removal process may include performing an etch process and/or a planarization process (e.g., a chemical mechanical polishing (CMP) process) into the semiconductor substrate 102 and/or the stack of semiconductor layers 708 to reduce a height of the second and third fin structures 108, 304 and/or to remove a topmost first semiconductor layer 704 of the stack of semiconductor layers 708.

Further, as illustrated in the top view 2200b of FIG. 22B and the cross-sectional view 2200c of FIG. 22C, the isolation patterning process defines the isolation region 310 laterally within the first device region 502. In some embodiments, the isolation patterning process may include forming a masking layer over the semiconductor substrate 102 and exposing unmasked regions of the semiconductor substrate 102 to one or more etchants (e.g., see FIGS. 35A-35B through 39A-39B). This, in part, removes the first and second fin structures 106, 108 from the isolation region 310. As illustrated by the cross-sectional view 2200c of FIG. 22C, a remnant 2202 of the second fin structures 108 remains within the isolation region 310 after performing the isolation patterning process. In some embodiments, the remnant 2202 of the second fin structures 108 remains due to a difference in widths and/or semiconductor materials of the first and second fin structures 106, 108.

Furthermore, as illustrated by the various views of FIGS. 22A-22C, after performing the isolation patterning process, the isolation structure 104 is formed over the semiconductor substrate 102. The isolation structure 104 may, for example, be formed by PVD, CVD, ALD, thermal oxidation, or another suitable growth or deposition process. In some embodiments, the isolation structure 104 may, for example, be a shallow trench isolation (STI) structure and/or may be or comprise silicon dioxide, silicon nitride, silicon carbide, another suitable dielectric material, or any combination of the foregoing.

As shown in the cross-sectional view 2300a of FIG. 23A and the top view 2300b of FIG. 23B, a dummy sidewall spacer layers 2302 and dummy gate structures 2304 are formed over the isolation structure 104, the first fin structures 106, the second fin structures 108, the third fin structures 304, the fourth fin structures 508, and the stack of semiconductor layers 708. FIG. 23A illustrates some embodiments of a cross-sectional view 2300a taken along the line A-A' of top view 2300b of FIG. 23B. The dummy sidewall spacer layers 2302 separate the dummy gate structures 2304 from the isolation structure 104, the first fin structures 106, the second fin structures 108, the third fin structures 304, the fourth fin structures 508, and the stack of semiconductor layers 708, as illustrated in FIG. 23A.

In some embodiments, a process for forming the dummy sidewall spacer layers 2302 and the dummy gate structures 2304 includes depositing a dummy dielectric layer (not shown) covering the isolation structure 104, the first fin structures 106, the second fin structures 108, the third fin

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structures 304, the fourth fin structures 508, and the stack of semiconductor layers 708. The dummy dielectric layer may be deposited as a conformal layer. The dummy dielectric layer may, for example, be or comprise an oxide (e.g., silicon dioxide), silicon nitride, silicon carbide, a high-k dielectric material, another suitable dielectric material, or any combination of the foregoing. The dummy dielectric layer be deposited by, for example, CVD, PVD, ALD, some other deposition process, or a combination of the foregoing. A dummy gate material layer (not shown) is then deposited on the dummy dielectric layer. The dummy gate material layer may be deposited by, for example, CVD, PVD, ALD, some other deposition process, or a combination of the foregoing.

Thereafter, a patterned masking layer (not shown) is formed over the dummy gate material layer. In some embodiments, the patterned masking layer may comprise a stack of dielectric materials. For example, the patterned masking layer may comprise a first dielectric layer (e.g., silicon dioxide), a second dielectric layer (e.g., silicon nitride) disposed over the first dielectric layer, and a third dielectric layer (e.g., silicon dioxide) disposed over the second dielectric layer. An etching process is then performed to remove unmasked portions of the dummy gate material layer and the dummy dielectric layer, thereby forming the dummy sidewall spacer layers 2302 and the dummy gate structures 2304.

Further, as illustrated in the cross-sectional view 2300a of FIG. 23A and the top view 2300b of FIG. 23B, sidewall spacer structures 2306 are formed along sidewalls of the dummy gate structures 2304. In some embodiments, the sidewall spacer structures 2306 may, for example, be or comprise silicon nitride, silicon carbide, another suitable dielectric material, or the like.

As shown in the various views of FIGS. 24A-24C, first source/drain regions 110, second source/drain regions 111, third source/drain regions 306, and fourth source/drain regions 514 are formed over the semiconductor substrate 102. FIG. 24A illustrates some embodiments of a cross-sectional view 2400a taken along the line A-A' of top view 2400b of FIG. 24B. Further, FIG. 24C illustrates some embodiments of a cross-sectional view 2400c taken along the line F-F' of the top view 2400b of FIG. 24B.

In some embodiments, a process for forming the first, second, third, and fourth source/drain regions 110, 111, 306, 514 includes first performing a removal process into the first, second, third, and fourth fin structures 106, 108, 304, 508 such that sections of the aforementioned fin structures are recessed below an upper surface of the isolation structure 104 (see, e.g., FIG. 24C). Further, the removal process removes the stack of semiconductor layers 708 from above the sections of the aforementioned fin structures. Secondly, a first epitaxial process is performed to form the first, second, third, and fourth source/drain regions 110, 111, 306, 514. In some embodiments, the first epitaxial process may include performing one or more epitaxial processes. For example, the one or more epitaxial processes may include performing an epitaxial processes for each of the first, second, third, and fourth source/drain regions 110, 111, 306, 514 (i.e., performing at least fourth epitaxial processes). In further embodiments, the one or more epitaxial processes may, for example, respectively be VPE, LPE, MBE, some other epitaxial process, or a combination of the foregoing. Further, the one or more epitaxial processes may in-situ dope the corresponding source/drain regions with first doping

type dopants (e.g., p-type dopants, such as boron atoms) or second doping type dopants (e.g., n-type dopants, such as phosphorus atoms).

As shown in the various views of FIGS. 25A-25C, a removal process is performed on the dummy gate structures 2304 and the dummy sidewall spacer layers 2302 to expose segments of the first fin structure 106 and the fourth fin structures 508, thereby exposing the stack of semiconductor layers 708. FIG. 25A illustrates some embodiments of a cross-sectional view 2500a taken along the line A-A' of top view 2500b of FIG. 25B. Further, FIG. 25C illustrates some embodiments of a cross-sectional view 2500c taken along the line F-F' of the top view 2500b of FIG. 25B.

In various embodiments, before performing the removal process, an inter-level dielectric (ILD) layer 2502 is formed over each of the first, second, third, and fourth source/drain regions 110, 111, 306, 514. The ILD layer 2502 may be formed by PVD, CVD, ALD, or another suitable deposition or growth process. Further, the ILD layer 2502 may, for example, be or comprise silicon dioxide, a low-k dielectric material, an extreme low-k dielectric material, another suitable dielectric material, or any combination of the foregoing. Subsequently, forming the ILD layer 2502 may include performing a planarization process into the ILD layer 2502 until an upper surface of the dummy gate structures 2304 is reached. In such embodiments, an upper surface of the ILD layer 2502 may be co-planar with an upper surface of the dummy gate structures 2304.

In some embodiments, the removal process illustrated in the various views of FIGS. 25A-25C includes performing a wet etch process and/or a dry etch process. In some embodiments, the removal process includes: forming a masking layer (not shown) over the semiconductor substrate 102; exposing unmasked regions of the dummy gate structures 2304 and the dummy sidewall spacer layers 2302 to one or more etchants; and removing the masking layer. Further, the removal process is performed in such a manner that segments of the dummy gate structures 2304 and the dummy sidewall spacer layers 2302 remain over the second and third fin structures 108, 304. In yet further embodiments, the removal process may entirely remove the dummy gate structures 2304 and the dummy sidewall spacer layers 2302, such that the second and third fin structures 108, 304 are exposed after performing the removal process (not shown).

As shown in the cross-sectional view 2600 of FIG. 26, a first plurality of nanostructures 117 are formed over the first fin structures 106 and a second plurality of nanostructures 512 are formed over the fourth fin structures 508. The first plurality of nanostructures 117 are vertically stacked over one another and the second plurality of nanostructures 512 are vertically stacked over one another. In some embodiments, the first plurality of nanostructures 117 is formed in such a manner that each nanostructure 117 comprises a same width as an underlying first fin structure 106. In further embodiments, the second plurality of nanostructures 512 is formed in such a manner that each nanostructure 512 comprises a same width as an underlying fourth fin structure 508.

In some embodiments, a process for forming the first and second plurality of nanostructures 117, 512 comprises selectively removing the first semiconductor layer (704 of FIGS. 25A-25C) of the stack of semiconductor layers (708 of FIGS. 25A-25C), thereby forming the first and second plurality of nanostructures 117, 512. The first semiconductor layer (704 of FIGS. 25A-25C) may be removed by performing an etching process (e.g., a wet etch) that selectively etches the first semiconductor layer (704 of FIGS. 25A-25C)

against the second semiconductor layer (706 of FIGS. 25A-25C). In some embodiments, the first semiconductor layer (704 of FIGS. 25A-25C) may be selectively removed by using a wet etchant, such as, ammonium hydroxide (NH₄OH), tetramethylammonium hydroxide (TMAH) solution, ethylenediamine pyrocatechol (EDP), potassium hydroxide (KOH) solution, or the like. In some embodiments, after forming the first and second plurality of nanostructures 117, 512, a removal process is performed to remove the dummy gate structures 2304 and the dummy sidewall spacer layers 2302 from over the second fin structures 108 and the third fin structures 304 (not shown).

As shown in the various views of FIGS. 27A-27C, a gate dielectric layer 112 and gate electrodes 114 are formed over the semiconductor substrate 102, thereby forming a first plurality of NSFETs 116, a first plurality of finFETs 118, a second plurality of finFETs 308, and a second plurality of NSFETs 510. FIG. 27A illustrates some embodiments of a cross-sectional view 2700a taken along the line A-A' of top view 2700b of FIG. 27B. Further, FIG. 27C illustrates some embodiments of a cross-sectional view 2700c taken along the line F-F' of the top view 2700b of FIG. 27B. The gate dielectric layer 112 separates the gate electrodes 114 from the first, second, third, and fourth fin structures 106, 108, 304, 508 and the first and second plurality of nanostructures 117, 512.

Further, it will be appreciated that in some instances, each NSFETs 116 may be also known as and/or referred to as, for example, a gate-all-around FET (GAAFET), a gate-surrounding FET, a multi-bridge channel FET (MBCFET), a nanowire FET, a nanoring FET, or the like.

In some embodiments, a process for forming the gate dielectric layer 112 and the gate electrodes 114 includes depositing (e.g., CVD, PVD, ALD, another suitable deposition process, etc.) a gate dielectric material (not shown) around each of the nanostructures 117, 512, and on exposed portions of the first, second, third, and fourth fin structures 106, 108, 304, 508 and the isolation structure 104. The gate dielectric material may be deposited as a conformal layer. The gate dielectric material may, for example, be or comprise an oxide (e.g., silicon dioxide), a high-k dielectric material, some other dielectric material, or any combination of the foregoing. The gate dielectric material may be multi-layered, for example, comprising one or more interfacial layers.

Thereafter, a gate electrode material (not shown) is deposited on the gate dielectric material and around each of the nanostructures 117, 512. The gate electrode material may be or comprise, for example, polysilicon (e.g., doped polysilicon), a metal (e.g., Al, W, etc.), some other conductive material, or a combination of the foregoing. The gate electrode material may be multi-layered, for example, comprising a work function metal layer (e.g., TiN, TaN, or the like), a metal fill layer (e.g., W), etc. In some embodiments, the gate electrode material may be deposited by, for example, CVD, PVD, ALD, electrochemical plating, electroless plating, some other deposition process, or a combination of the foregoing. A planarization process (e.g., CMP) is then performed on the gate electrode material and the gate dielectric material to remove upper portions of the gate electrode material and the gate dielectric material, thereby forming the gate electrodes 114 and the gate dielectric layer 112. The planarization process may co-planarize upper surfaces of the ILD layer 2502, the gate electrodes 114, the gate dielectric layer 112, and the sidewall spacer structures 2306.

As shown in the cross-sectional views 2800a and 2800b of FIGS. 28A and 28B, an upper ILD layer 2802 is formed

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over the gate electrodes **114** and the ILD layer **2502**. Subsequently, a plurality of conductive vias **212** are formed in the ILD layer **2502** over the gate electrodes **114** and over each of the first, second, third, and fourth source/drain regions **110**, **111**, **306**, **514**. In some embodiments, the conductive vias **212** may, for example, be or comprise aluminum, copper, tungsten, titanium nitride, tantalum nitride, another suitable conductive material, or any combination of the foregoing. In yet further embodiments, the conductive vias **212** may be formed, for example, be a single damascene process or another suitable process. The upper ILD layer **2802** may be deposited by, for example, PVD, CVD, ALD, or another suitable deposition or growth process. In yet further embodiments, the upper ILD layer **2802** may, for example, be or comprise silicon dioxide, a low-k dielectric material, an extreme low-k dielectric material, another suitable dielectric material, or any combination of the foregoing.

FIG. **29** illustrates a flowchart **2900** of some embodiments of a method for forming an IC comprising a plurality of fin structures having different widths that are laterally adjacent to one another on a same substrate according to the present disclosure. While the flowchart **2900** of FIG. **29** is illustrated and described herein as a series of acts or events, it will be appreciated that the illustrated ordering of such acts or events is not to be interpreted in a limiting sense. For example, some acts may occur in different orders and/or concurrently with other acts or events apart from those illustrated and/or described herein. Further, not all illustrated acts may be required to implement one or more aspects or embodiments of the description herein, and one or more of the acts depicted herein may be carried out in one or more separate acts and/or phases.

At act **2902**, a stack of semiconductor layers is formed over a base semiconductor substrate, where the stack of semiconductor layers comprises a plurality of first semiconductor layers and a plurality of second semiconductor layers. FIG. **7** illustrates a cross-sectional view **700** of some embodiments corresponding to act **2904**.

At act **2904**, an epitaxial structure is formed over the base semiconductor structure and along sides of the stack of semiconductor layers, thereby defining a semiconductor substrate. FIGS. **8** and **9** illustrate cross-sectional views **800** and **900** of some embodiments corresponding to act **2904**.

At act **2906**, a masking structure is formed over the semiconductor substrate. FIG. **10** illustrates a cross-sectional view **1000** of some embodiments corresponding to act **2906**.

At act **2908**, a plurality of mandrel structures is formed over the masking structure. FIG. **10** illustrates a cross-sectional view **1000** of some embodiments corresponding to act **2908**.

At act **2910**, a sidewall spacer structure is formed along sidewalls of each mandrel structure. FIGS. **11** and **12** illustrate cross-sectional views **1100** and **1200** of some embodiments corresponding to act **2910**.

At act **2912**, one or more patterning processes are performed on the mandrel structures and sidewall spacer structures to define first, second, and third upper masking structures. The first, second, and third upper masking structures each comprise a width different from one another. FIGS. **13-18** illustrate cross-sectional views **1300-1800** of some embodiments corresponding to act **2912**.

At act **2914**, the masking structure is patterned according to the first, second, and third upper masking structures, thereby defining first, second, third, and fourth stack of masking layers that each comprise a width different from

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one another. FIG. **19** illustrates a cross-sectional view **1900** of some embodiments corresponding to act **2914**.

At act **2916**, the semiconductor substrate and the stack of semiconductor layers are patterned to define first fin structures, second fin structures, third fin structures, and fourth fin structures. The first, second, third, and fourth fin structures each have different widths from one another. FIG. **21** illustrates a cross-sectional view **2100** of some embodiments corresponding to act **2916**.

At act **2918**, an isolation structure is formed over the first, second, third, and fourth fin structures. FIGS. **22A-22C** illustrate various views of some embodiments corresponding to act **2918**.

At act **2920**, dummy gate structures are formed over the isolation structure, the first, second, third, and fourth fin structures. FIGS. **23A** and **23B** illustrate various views of some embodiments corresponding to act **2920**.

At act **2922**, source/drain regions are formed over the first, second, third, and fourth fin structures. The source/drain regions are spaced laterally between the dummy gate structures. FIGS. **24A-24C** illustrate various views of some embodiments corresponding to act **2922**.

At act **2924**, a first plurality of nanostructures is formed over the first fin structures, and a second plurality of nanostructures is formed over the fourth fin structures by selectively removing the first semiconductor layers of the stack of semiconductor layers. FIGS. **25A-25C** and **26** illustrate various views of some embodiments corresponding to act **2924**.

At act **2926**, a gate dielectric layer and gate electrodes are formed over the isolation structure, the first, second, third, and fourth fin structures, and around the first and second plurality of nanostructures. FIGS. **27A-27C** illustrate various views of some embodiments corresponding to act **2926**.

At act **2928**, an inter-level dielectric (ILD) layer is formed over the isolation structure, the first, second, third, and fourth fin structures, the source/drain regions, and the gate electrodes. FIGS. **28A-28B** illustrate cross-sectional views **2800a** and **2800b** of some embodiments corresponding to act **2928**.

At act **2930**, conductive vias are formed within the ILD layer. FIGS. **28A-28B** illustrate cross-sectional views **2800a** and **2800b** of some embodiments corresponding to act **2930**.

FIGS. **30-34** illustrate cross-sectional views **3000-3400** of some embodiments of a second method for forming an IC comprising a plurality of fin structures having different widths that are laterally adjacent to one another on a same substrate according to the present disclosure. For example, FIGS. **30-34** illustrate alternative embodiments of acts that may be performed on the first device region **502** in the place of the acts at FIGS. **14A-21**. Thus, in some embodiments, the second method includes a method that alternately proceeds from FIGS. **7-13** to FIG. **30**, and then from FIG. **34** to FIGS. **22A-22C** to **28A-28B** (skipping FIGS. **14A-21**).

As illustrated by the cross-sectional view **3000** of FIG. **30**, a first protection layer **1402** is formed over a middle mandrel structure **1010**. In some embodiments, the middle mandrel structure **1010** is spaced laterally within a middle region **3002** of the first device region **502** and is laterally offset from the stack of semiconductor layers **708**. After forming the first protection layer **1402**, a removal process is performed to remove segments of the dielectric capping layer **1302** from regions laterally offset from the first protection layer **1402**. In some embodiments, the removal process includes performing a wet etch process and/or a dry etch process. As illustrated in FIG. **30**, the removal process may over-etch and remove portions of the upper masking layer

1008. The portions of the upper masking layer 1008 removed by the patterning process are laterally offset from the first protection layer 1402, such that an upper surface 1008_{us} of the upper masking layer 1008 is disposed vertically below a top surface 1008_{ts} of the upper masking layer 1008. After performing the removal process, an etching process (e.g., a wet etch and/or a dry etch) may be performed to remove the first protection layer 1402 (not shown).

As illustrated by the cross-sectional view 3100 of FIG. 31, a first patterning process is performed to remove the sidewall spacer structures 1202 from around the mandrel structures 1010 in areas laterally offset from the middle region 3002. In some embodiments, the dielectric capping layer (1302 of FIG. 30) acts as masking layer during the first patterning process and is configured to prevent damage to and/or removal of the sidewall spacer structure 1202 in the middle region 3002. Further, after performing the first patterning process, a second patterning process is performed to remove the dielectric capping layer 1302 from the middle region 3002. In further embodiments, the second patterning process may include performing a dry etching process and/or a wet etching process.

As illustrated by the cross-sectional view 3200 of FIG. 32, second protection layers 1702 are formed over the mandrel structures 1010 that are laterally offset from the middle region 3002. Subsequently, a patterning process is performed to remove the mandrel structure 1010 that is laterally spaced within the middle region 3002, thereby leaving the sidewall spacer structure 1202 in the middle region 3002. The patterning process may include performing a wet etch process and/or a dry etch process. Further, the patterning process may over-etch and remove portions of the upper masking layer 1008 laterally within the middle region 3002, such that an upper surface 1008_{us} of the upper masking layer 1008 is disposed vertically below a top surface 1008_{ts} of the upper masking layer 1008. After performing the patterning process, an etching process (e.g., a wet etch and/or a dry etch) may be performed to remove the second protection layer 1702 (not shown).

As illustrated by the cross-sectional view 3300 of FIG. 33, an upper dielectric layer 1802 is formed over the mandrel structures 1010 and the sidewall spacer structure 1202. The upper dielectric layer 1802 may, for example, be or comprise silicon nitride, silicon carbide, or another suitable dielectric material. The upper dielectric layer 1802 may, for example, be formed by CVD, PVD, ALD, or another suitable deposition or growth process. In further embodiments, after depositing the upper dielectric layer 1802, an etch process (e.g., a wet etch process and/or a dry etch process) may be performed on the upper dielectric layer 1802 to remove the upper dielectric layer 1802 from regions laterally offset from the mandrel structures 1010 and/or the sidewall spacer structure 1202.

Further, as illustrated in FIG. 33, due to the patterning and/or removal processes of FIGS. 30-32, dielectric material (e.g., silicon dioxide) from the upper masking layer 1008 is removed from the regions 3202, such that the upper surface 1008_{us} of the upper masking layer 1008 is disposed below the top surface 1008_{ts} of the upper masking layer 1008. Further, dielectric material from the upper masking layer 1008 may not be removed from a center region 3204 between opposing sidewalls of the sidewall spacer structure 1202. In addition, the upper dielectric layer 1802 extends continuously across the center region 3204 between the opposing sidewalls of the sidewall spacer structure 1202.

As illustrated by the cross-sectional view 3400 of FIG. 34, a patterning process is performed on the semiconductor

substrate 102, thereby defining the first fin structures 106 and the second fin structures 108. The patterning process is performed in such a manner that the first fin structures 106 have a first width w1 and the second fin structures 108 have a second width w2 different from the first width w1. In some embodiments, due to the removal of dielectric material (e.g., silicon dioxide) from the upper masking layer (1008 of FIG. 33) from the regions 3202, the patterning process may over-etch the semiconductor substrate 102 within the regions 3202, such that the semiconductor substrate 102 comprises a lower surface 102_{ls} disposed below an upper surface 102_{us}. A substantially straight line 301 is disposed along a top surface of the semiconductor substrate 102. Further, a first height h1 of the semiconductor substrate 102 is defined between the substantially straight line 301 and the lower surface 102_{ls} of the semiconductor substrate 102. A second height h2 of the semiconductor substrate 102 is defined between the substantially straight line 301 and the upper surface 102_{us} of the semiconductor substrate 102. In some embodiments, due to the over-etching, the second height h2 is less than the first height h1. In further embodiments, a difference between the first height h1 and the second height h2 (e.g., h1-h2) may be within a range of about 5 to 30 nanometers. In some embodiments, if the difference between the first height h1 and the second height h2 is relatively high (e.g., greater than about 30 nanometers), then a height of the second fin structures 108 may too small, thereby impeding formation of a selectively-conductive channel in the second fin structures 108.

FIGS. 35A-35B through 39A-39B illustrated various views of alternative embodiments of acts that may be performed on the first device region 502 to carry out the isolation patterning process illustrated and/or described in FIGS. 22A-22C.

As illustrated in the cross-sectional view 3500a of FIG. 35A and the top view 3500b of FIG. 35B, the semiconductor substrate 102 comprises the first plurality of fin structures 106, the second plurality of fin structures 108, and the third plurality of fin structures 304. In such embodiments, the third plurality of fin structures 304 are formed in such a manner that they are on opposing sides of the first fin structures 106. In some embodiments, FIG. 35A illustrates the cross-sectional view 3500a taken along the line A-A' of the top view 3500b of FIG. 35B. In some embodiments, the structure of FIGS. 35A-35B may be formed by the steps illustrated and/or described in FIGS. 7-21, FIGS. 30-34, another suitable method, or any combination of the aforementioned methods. Thus, in some embodiments, one or more patterning processes may be performed on the semiconductor substrate 102 to form the first, second, and third fin structures 106, 108, 304 (e.g., see FIG. 21 and/or FIG. 34). By virtue of the width and/or material composition of the first fin structures 106, remnants 1008_r of the upper masking layer (1008 of FIG. 33 and/or FIG. 20) may remain over a corresponding first fin structure 106 after performing the one or more patterning processes. In some embodiments, the remnants 1008_r may result in the semiconductor substrate 102 having the upper surface 102_{us} and the lower surface 102_{ls} (not shown), such that there is a difference in height between the first height h1 and the second height h2.

As illustrated in FIGS. 35A-35B, a bottom anti-reflectant coating (BARC) layer 3502 is formed over the semiconductor substrate 102. Further, a masking layer 3504 is formed over the BARC layer 3502. The BARC layer 3502 and/or the masking layer 3504 may be formed, for example, by CVD, PVD, ALD, or another suitable deposition or growth process.

As illustrated in the cross-sectional view **3600** of FIG. **36**, a patterning process is performed on the masking layer **3504** and the BARC layer **3502**, thereby defining an opening **3602** that exposes the remnants **1008r**.

As illustrated in the cross-sectional view **3700** of FIG. **37**, a patterning process is performed on the structure of FIG. **36** to remove the remnants (**1008r** of FIG. **36**). In some embodiments, the patterning process may include performing a wet etch process and/or a dry etch process. In further embodiments, the patterning process may expand the opening **3602**.

As illustrated in the cross-sectional view **3800** of FIG. **38**, a patterning process is performed on the structure of FIG. **37** to remove portions of the middle masking layer **1006** and the lower masking layer from over the first and second fin structures **106**, **108**. The patterning process may include, for example, performing a wet etch process, a dry etch process, or another suitable etch process.

As illustrated in the cross-sectional view **3900a** of FIG. **39A** and the top view **3900b** of FIG. **39B**, an isolation patterning process is performed on the structure of FIG. **38**, thereby defining an isolation region **310** (e.g., as illustrated in FIGS. **22A-22C**). In some embodiments, FIG. **39A** illustrates the cross-sectional view **3900a** taken along the line A-A' of the top view **3900b** of FIG. **39B**.

In some embodiments, the isolation patterning process may include exposing unmasked regions of the semiconductor substrate **102** to one or more etchants. This, in part, removes the first and second fin structures **106**, **108** from the isolation region **310**. As illustrated by the cross-sectional view **3900a** of FIG. **39A**, a remnant **3902** of the second fin structures **108** remains within the isolation region **310** after performing the isolation patterning process. In some embodiments, the remnant **3902** remains due to a difference in widths and/or semiconductor materials of the first and second fin structures **106**, **108**. Further, the isolation patterning process results in the semiconductor substrate **102** having an upper surface **102us** and a lower surface **102ls** laterally within the isolation region **310**, such that the upper surface **102us** is vertically offset from the lower surface **102ls**.

Accordingly, in some embodiments, the present disclosure provides an integrated chip (IC) having a first fin structure with a first width laterally adjacent to a second fin structure with a second width that is less than the first width.

In some embodiments, the present application provides an integrated chip (IC) including: a first fin structure vertically extending from a semiconductor substrate, wherein the first fin structure laterally extends along a first direction and has a first width; a second fin structure vertically extending from the semiconductor substrate, wherein the second fin structure laterally extends along the first direction and has a second width that is less than the first width; a first plurality of nanostructures directly overlying the first fin structure and vertically spaced from the first fin structure by a non-zero distance; and a gate electrode continuously laterally extending along a second direction that is substantially perpendicular to the first direction, wherein the gate electrode directly overlies the first and second fin structures, and wraps around the nanostructures.

In some embodiments, the present application provides an integrated chip (IC) including: an isolation structure overlying a semiconductor substrate; a plurality of first fin structures extending vertically from the semiconductor substrate through the isolation structure; wherein the first fin structures respectively have a first width; a plurality of second fin structures extending vertically from the semiconductor substrate through the isolation

structure, wherein the second fin structures respectively have a second width and are spaced laterally between the plurality of first fin structures, wherein the second width is less than the first width; a first plurality of nanosheet field-effect transistors (NSFETs) overlying the semiconductor substrate, wherein NSFETs in the first plurality of NSFETs respectively include: a pair of first source/drain regions disposed over a corresponding first fin structure; a first plurality of nanostructures directly overlying the corresponding first fin structure, wherein the first plurality of nanostructures extend laterally between the pair of first source/drain regions, wherein the first plurality of nanostructures have the first width; and a first plurality of fin field-effect transistors (finFETs) overlying the semiconductor substrate, wherein finFETs in the first plurality of finFETs respectively include: a pair of second source/drain regions disposed over a corresponding second fin structure; and an upper portion of the corresponding second fin structure that extends laterally between the pair of second source/drain regions.

In some embodiments, the present application provides a method for forming an integrated chip (IC), the method including: forming a stack of semiconductor layers over a semiconductor substrate, wherein the stack of semiconductor layers comprise a plurality of first semiconductor layers and a plurality of second semiconductor layers; forming a masking structure over the stack of semiconductor layers and the semiconductor substrate, wherein the masking structure comprises a plurality of masking layers; forming a plurality of mandrel structures over the masking structure, wherein the plurality of mandrel structures includes a first mandrel structure that overlies the stack of semiconductor layers and a second mandrel structure that is laterally offset from the stack of semiconductor layers by a non-zero distance; forming a first sidewall spacer structure along opposing sidewalls of the first mandrel structure and a second sidewall spacer structure along opposing sidewalls of the second mandrel structure; forming a first protection layer over the second mandrel structure; performing a first patterning process according to the first protection layer to remove the first sidewall spacer; forming a second protection layer over the first mandrel structure; performing a second patterning process according to the second protection layer to remove the second mandrel structure; performing a third patterning process according to the first mandrel structure and the second sidewall spacer to remove portions of the masking structure, thereby defining a first stack of masking layers over the stack of semiconductor layers with a first width and a second stack of masking layers laterally offset from the stack of semiconductor layers with a second width less than the first width; and performing a third patterning process on the semiconductor substrate and the stack of semiconductor layers according to the first and second stack of masking layers, thereby defining a first fin structure laterally adjacent to a second fin structure, wherein the first fin structure has the first width and the second fin structure has the second width, and wherein the stack of semiconductor layers directly overlie the first fin structure.

The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the

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spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. An integrated chip, comprising:
 - a first fin structure over a base region of a semiconductor substrate, wherein the first fin structure comprises a first sidewall opposite a second sidewall, wherein a first height of the first sidewall is greater than a second height of the second sidewall;
 - a first plurality of semiconductor channel structures stacked vertically with one another over the base region of the semiconductor substrate, wherein a first width of the first fin structure is different from a second width of the first plurality of semiconductor channel structures; and
 - a gate structure extending from the first fin structure to the first plurality of semiconductor channel structures.
2. The integrated chip of claim 1, further comprising:
 - a second fin structure over the base region of the semiconductor substrate and having a third width different from the second width, wherein a first lateral distance between the first fin structure and the second fin structure is less than a second lateral distance between the first fin structure and the first plurality of semiconductor channel structures.
3. The integrated chip of claim 2, wherein the first lateral distance is less than the second width.
4. The integrated chip of claim 2, further comprising:
 - a second plurality of semiconductor channel structures stacked vertically with one another over the base region of the semiconductor substrate, wherein the first and second fin structures are spaced laterally between the first plurality of semiconductor channel structures and the second plurality of semiconductor channel structures.
5. The integrated chip of claim 4, wherein a fourth width of the second plurality of semiconductor channel structures is different from the first and third widths.
6. The integrated chip of claim 1, further comprising:
 - a lower fin structure under the first plurality of semiconductor channel structures, wherein a first thickness of the base region directly adjacent to a side of the lower fin structure is greater than a second thickness of the base region directly adjacent to the first fin structure.
7. The integrated chip of claim 1, wherein the first fin structure is elongated in a first direction, and wherein the semiconductor substrate comprises a protrusion extending from the base region and laterally aligned with the first fin structure along the first direction.
8. The integrated chip of claim 7, wherein the protrusion directly underlies the gate structure.
9. The integrated chip of claim 1, further comprising:
 - a shallow trench isolation (STI) structure over the base region of the semiconductor substrate, wherein the STI structure contacts the first sidewall and the second sidewall.
10. An integrated chip, comprising:
 - a first fin structure over a base region of a semiconductor substrate;
 - a first plurality of semiconductor channel structures over the first fin structure;
 - a second fin structure over the base region;
 - a second plurality of semiconductor channel structures over the second fin structure;

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- a third fin structure over the base region of the semiconductor substrate, wherein the third fin structure is spaced laterally between the first fin structure and the second fin structure, and wherein a height of the third fin structure is greater than heights of the first and second fin structures; and
 - a gate structure laterally extending from the first fin structure to the second fin structure.
11. The integrated chip of claim 10, wherein the base region comprises a first horizontal surface and a second horizontal surface vertically offset the first horizontal surface, wherein the first horizontal surface abuts the first fin structure and the second horizontal surface abuts the third fin structure.
 12. The integrated chip of claim 10, wherein a first lateral distance between outer sidewalls of the first plurality of semiconductor channel structures is different from a second lateral distance between outer sidewalls of the third fin structure.
 13. The integrated chip of claim 10, further comprising:
 - a first gate-all-around field effect transistor (GAAFET) on the semiconductor substrate and comprising the first plurality of semiconductor channel structures and a first pair of source/drain structures disposed on opposing sides of the gate structure and abutting the first plurality of semiconductor channel structures; and
 - a first fin FET (finFET) on the semiconductor substrate and comprising at least a portion of the third fin structure and a second pair of source/drain structures disposed on opposing sides of the gate structure and abutting the third fin structure.
 14. The integrated chip of claim 13, wherein a width of the first pair of source/drain structures is less than a width of the second pair of source/drain structures.
 15. The integrated chip of claim 10, further comprising:
 - an isolation structure between the base region and the gate structure, wherein a height of the isolation structure is greater than the heights of the first and second fin structures and is less than the height of the third fin structure.
 16. The integrated chip of claim 10, further comprising:
 - a fourth fin structure over the base region of the semiconductor substrate and spaced laterally between the third fin structure and the second fin structure, wherein a first height of the base region between the third and fourth fin structures is less than a second height of the base region adjacent to the first fin structure and/or the second fin structure.
 17. The integrated chip of claim 10, wherein the height of the first fin structure is greater than a height of the first plurality of semiconductor channel structures.
 18. An integrated chip, comprising:
 - a semiconductor substrate comprising a first material and having a base region;
 - a first fin structure continuously vertically extending from the base region to a first height over the base region;
 - a plurality of semiconductor channel structures over the first fin structure;
 - a second fin structure laterally offset from the first fin structure and continuously vertically extending from the base region to a second height over the base region, wherein the second height is greater than the first height, wherein the first fin structure and the second fin structure comprise the first material, and wherein a first lateral distance between opposing sidewalls of an individual semiconductor channel structure in the plurality of semiconductor channel structures is different from a

second lateral distance between opposing sidewalls of the second fin structure; and
a gate structure over the plurality of semiconductor channel structures and the second fin structure.

19. The integrated chip of claim **18**, wherein the individual semiconductor channel structure directly underlies a topmost semiconductor channel structure in the plurality of semiconductor channel structures, wherein the individual semiconductor channel structure is vertically offset from the base region by a vertical distance that is less than the second height.

20. The integrated chip of claim **18**, wherein the opposing sidewalls of the second fin structure continuously extend from the base region to a top surface of the second fin structure.

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